



PCI Express® Base Specification Revision 6.0

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Status of this Document

This section describes the status of this document at the time of its publication. Other documents may supersede this document. A list of current PCISIG publications and the latest revision of this specification can be found at pcisig.com

This is the *PCI Express Base 6.0 Specification Revision 1.0*.

- The **NCB-PCI_Express_Base_6.0.pdf** is normative (i.e., the official specification). It contains no changebars.
- The **CB-PCI_Express_Base_6.0.pdf** is informative. It contains changebars relative to the *PCI Express Base 5.0 Specification*.
- The **CB-PCI_Express_Base_6.0-vs-r0.9.pdf** is informative. It contains changebars relative to the *PCI Express Base 6.0 Specification Revision 0.9*.

NOTE: Background on the new Document Process §

The new PCISIG document system is a variant of the w3c Respec tool (see <https://github.com/w3c/respec/wiki>). Respec is a widely used tool written to support the World Wide Web specifications. The PCISIG variant is <https://github.com/sglaser/respec>. Both Respec and the PCISIG variant are open source (MIT License) Javascript libraries. They operate in the author's browser and provide a rapid edit / review cycle without requiring any special tools be installed.

Respec is built on top of HTML5, the document format for the World Wide Web <http://www.w3.org/TR/html5/>. HTML is a text-based document format that allows us to deploy tools commonly used for software development (git, continuous integration, build scripts, etc.) to better manage and control the spec development process.

PCISIG enhancements to Respec support document formatting closer to existing PCISIG practice as well as automatic creation of register figures (eliminating about half of the manually drawn figures).

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Revision History

Revision	Revision History	Date
1.0	Initial release.	07/22/2002
1.0a	Incorporated Errata C1-C66 and E1-E4.17.	04/15/2003
1.1	Incorporated approved Errata and ECNs.	03/28/2005
2.0	Added 5.0 GT/s data rate and incorporated approved Errata and ECNs.	12/20/2006
2.1	Incorporated <i>Errata for the PCI Express Base Specification, Rev. 2.0</i> (February 27, 2009), and added the following ECNs: • Internal Error Reporting ECN (April 24, 2008) • Multicast ECN (December 14, 2007, approved by PWG May 8, 2008) • Atomic Operations ECN (January 15, 2008, approved by PWG April 17, 2008) • Resizable BAR Capability ECN (January 22, 2008, updated and approved by PWG April 24, 2008) • Dynamic Power Allocation ECN (May 24, 2008) • ID-Based Ordering ECN (January 16, 2008, updated 29 May 2008) • Latency Tolerance Reporting ECN (22 January 2008, updated 14 August 2008) • Alternative Routing-ID Interpretation (ARI) ECN (August 7, 2006, last updated June 4, 2007) • Extended Tag Enable Default ECN (September 5, 2008) • TLP Processing Hints ECN (September 11, 2008) • TLP Prefix ECN (December 15, 2008)	03/04/2009
3.0	Added 8.0 GT/s data rate, latest approved Errata, and the following ECNs: • Optimized Buffer Flush/Fill ECN (8 February 2008, updated 30 April 2009) • ASPM Optionality ECN (June 19, 2009, approved by the PWG August 20, 2009) • Incorporated End-End TLP Changes for RCs ECN (26 May 2010) and Protocol Multiplexing ECN (17 June 2010)	11/10/2010
3.1	Incorporated feedback from Member Review Incorporated Errata for the PCI Express® Base Specification Revision 3.0 Incorporated M-PCIe Errata (3p1_active_errata_list_mpcie_28Aug2014.doc and 3p1_active_errata_list_mpcie_part2_11Sept2014.doc) Incorporated the following ECNs: • ECN: Downstream Port containment (DPC) • ECN: Separate Refclk Independent SSC (SRIS) Architecture • ECN: Process Address Space ID (PASID) • ECN: Lightweight Notification (LN) Protocol	10/8/2014

Revision	Revision History	Date
	• ECN: Precision Time Measurement • ECN: Enhanced DPC (eDPC) • ECN: 8.0 GT/s Receiver Impedance • ECN: L1 PM Substates with CLKREQ • ECN: Change Root Complex Event Collector Class Code • ECN: M-PCIe • ECN: Readiness Notifications (RN) • ECN: Separate Refclk Independent SSC Architecture (SRIS) JTOL and SSC Profile Requirements	
3.1a	Minor update: Corrected: Equation 4.3.9 in Section 4.3.8.5., Separate Refclk With Independent SSC (SRIS) Architecture. Added missing square (exponent=2) in the definition of B. $B = 2.2 \times 10^{12} \times (2.\pi)^{\wedge 2}$ where $\wedge$= exponent.	12/5/2015
4.0	Version 0.3: Based on PCI Express® Base Specification Revision 3.1 (October 8, 2014) with some editorial feedback received in December 2013. • Added § Chapter 9., Electrical Sub-block: Added § Chapter 9. (Rev0.3-11-30-13_final.docx) • Changes related to Revision 0.3 release • Incorporated PCIe-relevant material from PCI Bus Power Management Interface Specification (Revision 1.2, dated March 3, 2004). This initial integration of the material will be updated as necessary and will supersede the standalone Power Management Interface specification. Version 0.5 (12/22/14, minor revisions on 1/26/15, minor corrections 2/6/15) • Added front matter with notes on expected discussions and changes. • Added ECN:Retimer (dated October 6, 2014) • Corrected § Chapter 4. title to, “Physical Layer Logical Block”. • Added Encoding subteam feedback on § Chapter 4. • Added Electrical work group changes from PCIe Electrical Specification Rev 0.5 RC1 into § Chapter 9.	2/6/2015
	Version 0.7: Based on PCI Express® Base Specification Version 4.0 Revision 0.5 (11/23/2015) • Added ECN_DVSEC-2015-08-04 • Applied ECN PASID-ATS dated 2011-03-31 • Applied PCIe Base Spec Errata: PCIe_Base_r3 1_Errata_2015-09-18 except: ◦ B216; RCIE ◦ B256; grammar is not clear • Changes to Chapter 7. Software Initialization and Configuration per PCIe_4.0_regs_0-3F_gord_7.docx • Added Chapter SR-IOV Spec Rev 1.2 (Rev 1.1 dated September 8, 2009 plus: ◦ SR-IOV_11_errata_table.doc	11/24/2015

Revision	Revision History	Date
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	2/18/2016 Changes from the Protocol Working Group • Applied changes from the following documents: ◦ FC Init/Revision scaled-flow-control-pcie-base40-2016-01-07.pdf (Steve.G) ◦ Register updates for integrated legacy specs PCIe_4.0_regs_0-3F_gord_8.docx (GordC) ◦ Tag Scaling PCIe 4_0 Tag Field scaling 2015-11-23 clean.docx (JoeC) ◦ MSI/MSI-X PCIe 4_0 MSI & MSI-X 2015-12-18 clean.docx (JoeC); register diagrams TBD on next draft. ◦ REPLAY_TIMER/Ack/FC Limits Ack_FC_Replay_Timers_ver8 (PeterJ)	2/18/16
	Chapter 10. SR-IOV related changes: • Incorporated “SR-IOV and Sharing Specification” Revision 1.1 dated January 20, 2010 (sr-iov1_1_20Jan10.pdf) as § Chapter 10. , with changes from the following documents ◦ Errata for the PCI Express® Base Specification Revision 3.1, Single Root I/O Virtualization and Sharing Revision 1.1, Address Translation and Sharing Revision 1.1, and M.2 Specification Revision 1.0: PCIe_Base_r3 1_Errata_2015-09-18_clean.pdf ◦ ECN__Integrated_Endpoints_and_IOV_updates__19 Nov 2015_Final.pdf ◦ Changes marked “editorial” only in marked PDF: sr-iov1_1_20Jan10-steve-manning-comments.pdf	4/26/16 [snapshot]
	Chapter 9. Electrical Sub-Block related changes: Source: WG approved word document from Dan Froelich (FileName: Electrical-PCI_Express_Base_4.0r0.7_April_7_wg_approved_redo_for_figure_corruption.docx.)	5/23/16[snapshot]
	Version 0.7 continued... Chapter 4. PHY Logical Changes based on: • Chapter4-PCI_Express_Base_4 0r0 7_May3_2016_draft.docx Chapter 7. . PHY Logical Changes based on: • PCI_Express_Base_4 0r0 7_Phy-Logical_Ch7_Delta_28_Apr_2016.docx	
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Revision	Revision History	Date
	PWG Feedback on 4.0 r0.7 Feb-Apr-May-2016 Drafts *EWG Feedback: -CB-PCI_Express_Base_4.0r0.7_May-2016 (Final).fdf -EWG f/b: Electrical-PCI_Express_Base_4.0r0.7_April_7_wg_approved_redo_for_figure_corruption_Broadco.docx *PWG Feedback: -PWG 0.7 fix list part1 and part 2.docx -PWG 0 7 fix list part3a.docx -PCI_Express_Base_4.0r0.7_pref_April-2016_chp5_PM_stuff_only_ver3.docx -PCI_Express_Base_4.0r0.7_pref_April-2016_chp5_PM_stuff_only_ver3.docx -scaled-flow-control-pcie-base40-2016-07-07.pdf -ECN_NOP_DLLP-2014-06-11_clean.pdf -ECN_RN_29_Aug_2013.pdf -3p1_active_errata_list_mpcie_28Aug2014.doc -3p1_active_errata_list_mpcie_part2_11Sept2014.doc -lane-margining-capability-snapshot-2016-06-16.pdf -Emergency Power Reduction Mechanism with PWRBRK Signal ECN -PWG 0 7 fix list part4.docx -ECN_Conventional_Adv_Caps_27Jul06.pdf -10-bit Tag related SR-IOV Updates *Other: -Merged Acknowledgements back pages from SR-IOV and ATS specifications into the main base spec. Acknowledgements page.	
	----- Changes since August 2016 for the September 2016 4.0 r0.7 Draft PDF----- Applied: PWG Feedback/Corrections on August draft ECN_SR-IOV_Table_Updates_16-June-2016.doc	9/28/16
	----- Changes since September 28 2016 for the October 2016 4.0 r0.7 Draft PDF---- EWG: Updates to § Chapter 9. - Electrical Sub-block (Sections: 9.4.1.4, 9.6.5.1, 9.6.5.2, 9.6.7) PWG: Updates to Sections: 3.2.1, 3.3, 3.5.1, 7.13, 7.13.3 (Figure: Data Link Status Register)	10/7/16
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	----- Changes to the November 3 2016 4.0 r0.7 Draft PDF----- § Section 2.6.1 Flow Control Rules: Updated Scaled Flow Control sub-bullet under FC initialization bullet (before Table 2-43)	11/3/16
	----- Changes to the November 11 2016 4.0 r0.7 Draft PDF----- Added M-PCIe statement to the Open Issues page Updated date to November 11, 2016	11/11/16
	----- Version 0.9: Based on PCI Express® Base Specification Version 4.0 Revision 0.7 (11/11/2016) Incorporated the following ECNs: -ECN-Hierarchy_ID-2017-02-23 -ECN_FPB_9_Feb_2017 -ECN Expanded Resizable BARs 2016-04-18 -ECN-VF-Resizable-BARs_6-July-2016 - § Chapter 7. reorganized: • New section 7.6 created per a PWG-approved reorganization to move sections 7.5, 7.6., and 7.10 to subsections 7.6.1 through 7.6.3 resp. • New section 7.7 created per a PWG-approved reorganization to move sections 7.7, 7.8., 7.12, 7.13, 7.40, 7.41 and 7.20 to subsections 7.7.1 through 7.7.7 resp. • New section 7.9 created per a PWG-approved reorganization to move sections 7.15, 7.22, 7.16, 7.23, 7.39, 7.24, 7.17, 7.18, 7.21, 7.25, 7.28, 7.30, 7.33, 7.34, 7.35, 7.38, and 7.42 to subsections 7.9.1 through 7.9.17 resp. -Removed § Chapter 8. : M-PCIe Logical Sub-Block -Updated § Chapter 9. (8 now), EWG Updates to § Chapter 9. - Electrical Sub-block per: Chapter9-PCI_Express_Base_4 0r09_March_30-2017_approved.docx -Updated § Chapter 4. : Physical Layer Logical Block per PCI_Express_Base_4 0_r0 9_Chapter4_Final_Draft.docx -Updated Figures in § Chapter 10. : ATS Specification -Removed § Appendix H. : M-PCIe timing Diagrams -Removed Appendix I: M-PCIe Compliance Patterns, pursuant to removing the M-PCIe Chapter this 0.9 version of the 4.0 Base Spec. -Added § Appendix H. : Flow Control Update Latency and ACK Update Latency Calculations -Added Appendix I: Vital Product Data (VPD)	April 28 2017

Revision	Revision History	Date
	-Updated editorial feedback on the Appendix section per: PCI_Express_Base_4.0r0.7_appendixes_November-11-2016_combined-editorial.docx -Deleted references to M-PCIe throughout the document. -Updated § Chapter 9. (8 now), EWG Updates to § Chapter 9. - Electrical Sub-block per: Chapter9-PCI_Express_Base_4.0r09_March_30-2017_approved.docx -Updated § Chapter 4. : Physical Layer Logical Block per PCI_Express_Base_4.0_r0 9_Chapter4_Final_Draft.docx -Updated Figures in § Chapter 10. : ATS Specification -Added § Appendix H. : Flow Control Update Latency and ACK Update Latency Calculations -Following items that were marked deleted in the Change Bar version of the April 28th snapshot have been “accepted” to no longer show up: pp 1070: Lane Equalization Control 2 Register (Offset TBD) Comment: Deleted per: PCI_Express_Base_4.0r0.7_Phy-Logical_Ch7_Delta_28_Apr_2016.docx pp 1074: Physical Layer 16.0 GT/s Margining Extended Capability section Comment: Deleted per: PCI_Express_Base_4.0r0.7_Phy-Logical_Ch7_Delta_28_Apr_2016.docx Comment: Replaced by Section Lane Margining at the Receiver Extended Capability per Fix3a #83 lane-margining-capability-snapshot-2016-06-16.pdf -Incorporated: PCIe 4_0 Tag Field scaling 2017-03-31.docx -Vital Product Data (VPD) -Added § Section 6.27 -Added § Section 7.9.4 -Incorporated feedback from April 28th snapshot.[source: 3 fdf files] -Completed editorial feedback on the Appendix section per: PCI_Express_Base_4.0r0.7_appendixes_November-11-2016_combined-editorial.docx -Incorporated ECN EMD for MSI 2016-05-10 -Updated per: PWG F2F changes from: PCI_Express_Base_4.0r0.7_pref_November-11-2016-F2F-2017-03-16-2017-03-30-sdg.docx -Updated figures per following lists (Gord Caruk): PCIe_4_0_fix_drawing_items.doc PCIe_4_0_fix_drawing_items_part2.doc	May 26, 2017
	Version 0.91 <i>***Note this version will be used as the base for the PCI Express® Base Specification Revision 5.0***</i> <i>Item numbers are with reference to PWG CheckList (https://members.pcisig.com/wg/PCIe-Protocol/document/10642)</i> -Moved Flattening Portal Bridge Section 7.10 to Section 7.8.10. PWG Checklist Items #12.1 -Fixed misc. feedback that needed clarification from the 0.9 version. Issues fall under the categories of figure updates, broken cross references. Also incorporated feedback received from member review of the 4.0 version rev. 0.9 Base Spec. -Updated to reconcile issues related to incorporating the Extended Message Data for MSI ECN. PWG Checklist Items #22	August 17, 2017

Revision	Revision History	Date
	-Completed incorporating all resolved editorial items from PWG Checklist Items #14, 14.1,15.1, 36, 42. TBD: Some minor editorial items from #13, #14 and #15 have been deferred to post 0.91 by reviewers. TBD: Errata and NPEM ECN	
	ECN: ECN_Native_PCl_e_Enclosure_Management_v10August2017.docx Deleted Section 5.11.1 through Section 5.14 Changes tracked by items 34.01 34.02 34.04 34.05 34.11 in the PWG checklist Errata: B265, C266, 267, 268, B269, A270, A271, B274, C275, B276, B277, B278, B279, B280, B281, B283, B284, B285, B286, B288, B289, B292, B293, B294, B295, B297, B299, B300, B301 Other minor edits per: NCB-PCI_Express_Base_4.0r0.91_August-17-2017__dh_sdg_Annot_2.fdf	August 28, 2017
	Applied fixes and corrections captured in NCB-PCI_Express_Base_4.0r1.0_August-28-2017.fdf (Revision 8): https://members.pcisig.com/wg/PCle-Protocol/document/10770 Updated contributor list in Appendix section.	September 20, 2017
	Updated contributor list in Appendix section. Inserted correct Figure 6-2. Applied minor fixes and corrections captured in: NCB-PCI_Express_Base_4.0r1.0_September-20-2017 https://members.pcisig.com/wg/PCle-Protocol/document/10770	September 27, 2017
	“-c” version: Changes to match -b version of the Final NCB PDF approved by PWG and EWG on September 29, 2017. See change bars. Details include: EWG Changes: -Typo in Equation 8-3; changed 1.6.0 GT/s to 16.0 GT/s - § Section 8.4.2.1 ; corrected references from Table 8-11 to Table 8-10 - § Section 8.5.1.3.3 & § Section 8.5.1.4.3 (Figure 8-47); changed “median” to “mean” PWG Changes: -Sub-Sub-Bullet before Figure 4-27. Added “or higher” after 8.0 GT/s - § Section 5.12 Power Management Events; deleted last two paragraphs and Implementation Note. -Updated Acknowledgements section with additional contacts.	September 29, 2017
5.0	Version 0.3 Summary of intended changes for 5.0. This was a short document, referencing the PCI Express Base Specification but not including it.	2017-06-01
	Version 0.5	2017-11-02

Revision	Revision History	Date
	Further details on intended changes for 5.0. This was a short document, referencing the PCI Express Base Specification but not including it.	
	Version 0.7 This was the first release of Base 5.0 based on the 4.0 Specification text. The 4.0 specification was converted into HTML format during this process. This conversion process was imperfect but does not impact the new 5.0 material.	2018-06-07
	Version 0.9 This includes: • Additional details regarding operating at 32.0 GT/s • Corrections to match published Base 4.0 • Redrawing of some figures • PCIe_Base_r4_0_Errata_2018-10-04a.pdf • ECN-Thermal-Reporting_2017May18.pdf • ECN-Link-Activation-07-Dec-2017.pdf	2018-10-18
	Version 1.0 This includes: • Corrections and clarification for support of the 32.0 GT/s operation • Editorial Changes: ◦ Rewrite misleading / confusing text ◦ Update terminology for consistency and accuracy ◦ Update grammar for readability ◦ Add many hotlinks / cross references • Implement all 4.0 Errata • Incorporate Expansion ROM Validation ECN Expansion ROM Validation ECN.pdf • Incorporate Enhanced PCIe Precision Time Measurement (ePTM) ECN ECN_ePTM_10_January_2019.pdf • Incorporate Root Complex Event Collector Bus Number Association ECN ECN_EventCollector_13Sept2018a.pdf • Incorporate PCIe Link Activation ECN ECN_Link_Activation_07_Dec_2017.pdf • Incorporate Advanced Capabilities for Conventional PCI ECN (updated for PCIe) ECN_Conventional_Adv_Caps_27Jul06.pdf • Incorporate Async Hot-Plug Updates ECN ECN_Async_Hot-Plug_Updates_2018-11-29.pdf • Incorporate ACS Enhanced Capability ECN ECN_ACS_25_Apr_2019_Clean.pdf	2019-05-16

Revision	Revision History	Date
	Incorporate the Subsystem ID and Subsystem Vendor ID Capability, from the PCI-to-PCI Bridge Architecture Specification, Revision 1.2 (updated for PCIe) ppb12.pdf	
6.0	Version 0.3 Initial Release of Base 6.0, Standalone Document	2019-10-04
	Version 0.5, Standalone Document - Add L0p - Add Shared Flow Control - Update Physical Layer / Logical Sublayer material - Add Deprecation items: - MR-IOV - Lightweight Notification (LN) - Update new TLP Header material - Update Electrical Layer material	2020-01-30
	Version 0.7, Integrated Document First version relative to Base 5.0 Specification text. - Incorporate Base 5.0 Errata Matching Errata document to be published. - Incorporate Approved Base 5.0 ECNs: - ACS Enhanced Capability - ATS Memory Attributes Shadow Functions - CMA - DOE - PTM Byte Order Adaptation - DMWr - PASID for Untranslated Addr - Support Flit Mode changes in Chapter 2 14 bit Tag support - Flit Mode TLP Format changes, including translation rules - Support Shared Flow Control, L0p, NULL2 DLLP in Chapter 3 - Integrate Flit Mode material into Chapter 4 - Flit Mode changes to Error Handling in Chapter 6 - TLP Translation Blocked error - Integrate PAM4 Electrical changes into Chapter 8 - Refactor SR-IOV Registers from Chapter 9 into Chapter 7 Moves “VFs do things this way” material next to the original.	

Revision	Revision History	Date
	- Define Shared Flow Control Supported and Shared Flow Control Enable bits in Chapter 7 (removed in Version 0.9) <u>Virtual Channel Extended Capability</u> <u>Multi-Function Virtual Channel Extended Capability</u> - Define new capabilities associated with 64.0 GT/s and Flit Mode in Chapter 7 <u>Physical Layer 64.0 GT/s Extended Capability</u> <u>Flit Logging Extended Capability</u> - Device 3 Extended Capability Structure <u>Flit Performance Measurement Extended Capability</u> <u>Flit Error Injection Extended Capability</u> - Deprecate material: - MR-IOV - Lightweight Notification (LN)	
	Version 0.71, Integrated Document - Incorporate Combined Power ECN - Incorporate IDE ECN, plus Partial Header Encryption - Incorporate Errata: B90a, B90b, B90c, B90d, B90g, B90h, B90j, B90k, B90l, B90m, B90n, B90o, B90w, B90x, B90y, B90z, B113, B114 - L0p updates - Shared Flow control Updates - PAM4, 64.0 GT/s electrical updates - Physical Layer, Logical Sub-block updates - Rework Flit Ack/Nak protocol - Training Set changes - Max Payload Supported changes - TLP Layout changes 14 bit tag updates - SR-IOV updates - Data Link Feature DLLP updates - DRS is <i>MUST@FLIT</i>, explain how software should use - Specify additional error behavior	2021-06-24
	Version 0.9, Integrated Document - Implement Errata B23, B64, B66-67, B74-75, B77, B78, B80-81, B85-88, B89a, B90f, B93-B95, B98-101, B103-105, B107-111, B112, B115-118, B120-121, B123-125, B127-128, B132, B133a-B133c, B133e - Simplified Shared Flow Control (always on in Flit Mode) and improvements to text - Numerous Phy Logical updates and issue fixes - Updated and Improved Flit Mode TLP type earmarking	2021-09-23

Revision	Revision History	Date
	- Updated and Improved OHC content - Improve Segment-related content - Improve Max Payload Size content <i>MUST@FLIT</i> changes, esp for Completion Timeout mechanism - Numerous editorial improvements	
	Version 1.0, Published Document - Incorporate Relaxed Detect Timing ECN - Numerous editorial improvements - Update errata B117 - Add missing artwork - Fix and add cross references - Electrical section clarifications - Add Remote L0p Supported bit in Device Status 3 - Define Null Capability and Null Extended Capability - Update Reference Documents - Update encoding of Flit Mode Local TLP Prefix (was inconsistent in 0.9)	2021-12-16

Objective of the PCI Express® Architecture

This document defines the “base” specification for the PCI Express architecture, including the electrical, protocol, platform architecture and programming interface elements required to design and build devices and systems. A key goal of the PCI Express architecture is to enable devices from different vendors to inter-operate in an open architecture, spanning multiple market segments including clients, servers, embedded, and communication devices. The architecture provides a flexible framework for product versatility and market differentiation.

This specification describes the PCI Express® architecture, interconnect attributes, fabric management, and the programming interface required to design and build systems and peripherals that are compliant with the PCI Express Specification.

The goal is to enable such devices from different vendors to inter-operate in an open architecture. The specification is intended as an enhancement to the PCI™ architecture spanning multiple market segments; clients (desktops and mobile), servers (standard and enterprise), and embedded and communication devices. The specification allows system OEMs and peripheral developers adequate room for product versatility and market differentiation without the burden of carrying obsolete interfaces or losing compatibility.

PCI Express Architecture Specification Organization

The PCI Express specifications are organized as a base specification and a set of companion documents.

The *PCI Express Base Specification* contains the technical details of the architecture, protocol, Data Link Layer, Physical Layer, and software interface. The *PCI Express Base Specification* (this document) is applicable to all variants of PCI Express.

The companion specifications define a variety of form factors, including mechanical and electrical chapters covering topics including auxiliary signals, power delivery, and the Adapter interconnect electrical budget.

Documentation Conventions

Capitalization

Some terms are capitalized to distinguish their definition in the context of this document from their common English meaning. Words not capitalized have their common English meaning. When terms such as “memory write” or “memory read” appear completely in lower case, they include all transactions of that type.

Register names and the names of fields and bits in registers and headers are presented with the first letter capitalized and a mixture of capitalization for the remainder.

Numbers and Number Bases

Hexadecimal numbers are written with a lower case “h” suffix, e.g., FFFh and 80h. Hexadecimal numbers larger than four digits are represented with a space dividing each group of four digits, as in 1E FFFF FFFFh. Binary numbers are written with a lower case “b” suffix, e.g., 1001b and 10b. Binary numbers larger than four digits are written with a space dividing each group of four digits, as in 1000 0101 0010b.

All other numbers are decimal.

Implementation Notes

IMPLEMENTATION NOTE:

§ Implementation Notes should not be considered to be part of this specification. They are included for clarification and illustration only.

Notes

NOTE §

Notes pertain to the specification itself as opposed to implementations of the specification. For example, they are used to describe the document process. They are also used for forward looking information describing anticipated changes for a future version of this specification.

Issues

ISSUE 1

Issues are outstanding items in the specification. They indicate things like missing, outdated, or lower quality artwork, anticipated changes that are being deferred to a subsequent version of the specification, potential errata items noticed during the editing process, etc.

PCI-SIG's goal is to resolve all notes before the 1.0 published release of the specification (where “resolving” includes deferring an item to later, or determining that the item is not needed or is incorrect).

Implementation Notes, Notes, and Issues can also be inline.

Terms and Acronyms

8b/10b

The data encoding scheme¹ used in the PCI Express Physical Layer for 5.0 GT/s and below.

10-Bit Tags

A Tag's capability that provides a total of 10 bits for the Tag field. See Tag.

14-Bit Tags

A Tag's capability that provides a total of 14 bits for the Tag field. See Tag.

Access Control Services, ACS

A set of capabilities and control registers used to implement access control over routing within a PCI Express component.

ACS Violation

An error that applies to a Posted or Non-Posted Request when the Completer detects an access control violation.

Adapter

Used generically to refer to an add-in card or module.

Advanced Error Reporting, AER

Advanced Error Reporting (see § [Section 7.8.4](#)).

Alternative Routing-ID, ARI

Alternative Routing-ID Interpretation. Applicable to Requester IDs and Completer IDs as well as Routing IDs.

ARI Device

A Device associated with an Upstream Port, whose Functions each contain an [ARI Extended Capability](#) structure.

ARI Downstream Port

A Switch Downstream Port or Root Port that supports ARI Forwarding.

ARI Forwarding

Functionality that enables the Downstream Port immediately above an ARI Device to access the Devices extended Functions. Enabling ARI Forwarding ensures the logic that determines when to turn a Type 1 Configuration Request into a Type 0 Configuration Request no longer enforces a restriction on the traditional Device Number field being 0.

Asserted

The active logical state of a conceptual or actual signal.

Async Removal

Removal of an adapter or cable from a slot without lock-step synchronization with the operating system (e.g., in an asynchronous manner without button presses).

Atomic Operation, AtomicOp

One of three architected Atomic Operations where a single PCI Express transaction targeting a location in Memory Space reads the location's value, potentially writes a new value to the location, and returns the original value. This read-modify-write sequence to the location is performed atomically. AtomicOps include [FetchAdd](#), [Swap](#), and [CAS](#).

1. IBM Journal of Research and Development, Vol. 27, #5, September 1983 "A DC-Balanced, Partitioned-Block 8B/10B Transmission Code" by Widmer and Franaszek.

Attribute

Transaction handling preferences indicated by specified Packet header bits and fields (e.g., non-snoop).

Authentication

A process for determining that an entity is what it appears to be (its identity) using defined data objects and digital signatures.

Base Address Register, BAR

Base Address Registers exist within Configuration Space and are used to determine the amount of system memory space needed by a Function and to provide the base address for a mapping to Function memory space. A Base Address Register may map to memory space or I/O space.

Beacon

An optional 30 kHz to 500 MHz in-band signal used to exit the L2 Link Power Management state. One of two defined mechanisms for waking up a Link in L2 (see Wakeup).

Bridge

One of several defined System Elements. A Function that virtually or actually connects a PCI/PCI-X segment or PCI Express Port with an internal component interconnect or with another PCI/PCI-X bus segment or PCI Express Port. A virtual Bridge in a Root Complex or Switch must use the software configuration interface described in this specification.

by-1, x1

A Link or Port with one Physical Lane.

by-8, x8

A Link or Port with eight Physical Lanes.

by-N, xN

A Link or Port with “N” Physical Lanes.

Compare and Swap, CAS

An AtomicOp where the value of a target location is compared to a specified value and, if they match, another specified value is written back to the location. Regardless, the original value of the location is returned.

Character

An 8-bit quantity treated as an atomic entity; a byte.

Clear

A bit is Clear when its value is 0b.

Cold Reset

A Fundamental Reset following the application of main power.

Completer

The Function that terminates or “completes” a given Request, and generates a Completion if appropriate. Generally, the Function targeted by the Request serves as the Completer. For cases when an uncorrectable error prevents the Request from reaching its targeted Function, the Function that detects and handles the error serves as the Completer.

Completer Abort, CA

1. A status that applies to a posted or non-posted Request that the Completer is permanently unable to complete successfully, due to a violation of the Completer’s programming model or to an unrecoverable error associated with the Completer.
2. A status indication returned with a Completion for a non-posted Request that suffered a Completer Abort at the Completer.

Completer ID

The combination of a Completer's Bus Number, Device Number, and Function Number that uniquely identifies the Completer of the Request within a Hierarchy. With an ARI Completer ID, bits traditionally used for the Device Number field are used instead to expand the Function Number field, and the Device Number is implied to be 0.

Completion

A Packet used to terminate, or to partially terminate, a transaction sequence. A Completion always corresponds to a preceding Request, and, in some cases, includes data.

component

A physical device (a single package).

Configuration Software

The component of system software responsible for accessing Configuration Space and configuring the PCI/PCIe bus.

Configuration Space

One of the four address spaces within the PCI Express architecture. Packets with a Configuration Space address are used to configure Functions.

Configuration-Ready

A Function is Configuration-Ready when it is guaranteed that the Function will respond to a valid Configuration Request targeting the Function with a Completion indicating Successful Completion status.

Containment Error Recovery, CER

A general error containment and recovery approach supported by Downstream Port Containment (DPC), where with suitable software/firmware support, many uncorrectable errors can be handled without disrupting applications.

Conventional PCI

Behaviors or features originally defined in the PCI Local Bus Specification. The PCI Express Base 4.0 and subsequent specifications incorporate the relevant requirements from the PCI Local Bus Specification.

Conventional Reset

A Hot, Warm, or Cold Reset. Distinct from Function Level Reset (FLR).

Data Link Layer

The intermediate Layer that is between the Transaction Layer and the Physical Layer.

Data Link Layer Packet, DLLP

A Packet generated in the Data Link Layer to support Link management functions.

data payload

Information following the header in some packets that is destined for consumption by the targeted Function receiving the Packet (for example, Write Requests or Read Completions).

deasserted

The inactive logical state of a conceptual or actual signal.

Deferrable Memory Write, DMWr

A Memory Write where the Requester attempts to write to a given location in Memory Space using the non-posted DMWr TLP Type. A Completer that supports this TLP Type can accept or decline the Request, indicating this by means of the Completion status returned. See § Section 6.32 .

Design for Testability, DFT

Design for Testability.

Device (uppercase 'D')

A collection of one or more Functions within a single hierarchy identified by common Bus Number and Device Number. An SR-IOV Device may have additional Functions accessed via additional Bus Numbers and/or Device Numbers configured through one or more SR-IOV Extended Capability structures.

device (lowercase 'd')

1. A physical or logical entity that performs a specific type of I/O.
2. A component on either end of a PCI Express Link.
3. A common imprecise synonym for Function, particularly when a device has a single Function.

Device Readiness Status, DRS

A mechanism for indicating that a Device is Configuration-Ready (see § Section 6.22.1).

DLP

In Flit Mode, the Data Link Layer Payload within a Flit.

Downstream

1. The relative position of an interconnect/System Element (Port/component) that is farther from the Root Complex. The Ports on a Switch that are not the Upstream Port are Downstream Ports. All Ports on a Root Complex are Downstream Ports. The Downstream component on a Link is the component farther from the Root Complex.
2. A direction of information flow where the information is flowing away from the Root Complex.

Downstream Path

The flow of data through a Retimer from the Upstream Pseudo Port Receiver to the Downstream Pseudo Port Transmitter.

Downstream Port Containment, DPC

The automatic disabling of the Link below a Downstream Port following an uncorrectable error, which prevents TLPs subsequent to the error from propagating Upstream or Downstream.

DWORD, DW

Four bytes. Used in the context of a data payload, the 4 bytes of data must be on a naturally aligned 4-byte boundary (the least significant 2 bits of the byte address are 00b).

Egress Port

The transmitting Port; that is, the Port that sends outgoing traffic.

Electrical Idle

A Link state used in a variety of defined cases, with specific requirements defined for the Transmitter and Receiver.

End-End TLP Prefix

A TLP Prefix that is carried along with a TLP from source to destination. See § Section 2.2.10.4.

Endpoint

One of several defined System Elements. A Function that has a Type 00h Configuration Space header.

error detection

Mechanisms that determine that an error exists, either by the first agent to discover the error (e.g., Malformed TLP) or by the recipient of a signaled error (e.g., receiver of a poisoned TLP).

error logging

A detector setting one or more bits in architected registers based on the detection of an error. The detector might be the original discoverer of an error or a recipient of a signaled error.

error reporting

In a broad context, the general notification of errors. In the context of the Device Control register, sending an error Message. In the context of the Root Error Command register, signaling an interrupt as a result of receiving an error Message.

error signaling

One agent notifying another agent of an error either by (1) sending an error Message, (2) sending a Completion with UR/CA Status, or (3) poisoning a TLP.

Extension Device

A component whose purpose is to extend the physical length of a Link.

Extended Function

Within an ARI Device, a Function whose Function Number is greater than 7. Extended Functions are accessible only after ARI-aware software has enabled ARI Forwarding in the Downstream Port immediately above the ARI Device.

FetchAdd, Fetch and Add

An AtomicOp where the value of a target location is incremented by a specified value using two's complement arithmetic ignoring any carry or overflow, and the result is written back to the location. The original value of the location is returned.

Flow Control

The method for communicating receive buffer status from a Receiver to a Transmitter to prevent receive buffer overflow and allow Transmitter compliance with ordering rules.

Flow Control Packet, FCP

A DLLP used to send Flow Control information from the Transaction Layer in one component to the Transaction Layer in another component.

Flow-Through

Refers to the behavior, by a Switch or Root Complex supporting peer-to-peer, of passing an IDE TLP associated with a Selective IDE Stream from Ingress Port to Egress Port without modification.

Function

Within a Device, an addressable entity in Configuration Space associated with a single Function Number. Used to refer to one Function of a Multi-Function Device, or to the only Function in a Single-Function Device. Specifically included are special types of Functions defined in § Chapter 9, notably Physical Functions and Virtual Functions.

Function Group

Within an ARI Device, a configurable set of Functions that are associated with a single Function Group Number. Function Groups can optionally serve as the basis for VC arbitration or access control between multiple Functions within the ARI Device.

Function Level Reset, FLR

A mechanism for resetting a specific Endpoint Function (see § Section 6.6.2).

Function Readiness Status, FRS

A mechanism for indicating that a Function is Configuration-Ready (see § Section 6.22.2)

Fundamental Reset

A hardware mechanism for setting or returning all Port states to the initial conditions specified in this document (see § Section 6.6).

GT/s

The number of encoded bits transferred in a second on a direction of a Lane. Short for Giga Transfers per Second.

header

A set of fields that appear at or near the front of a Packet that contain the information required to determine the characteristics and purpose of the Packet.

Hierarchy

A PCI Express I/O interconnect topology, wherein the Configuration Space addresses, referred to as the tuple of Bus/Device/Function Numbers (or just Bus/Function Numbers, for ARI cases), are unique. These addresses are used for Configuration Request routing, Completion routing, some Message routing, and for other purposes. In some contexts a Hierarchy is also called a Segment, and in Flit Mode, the Segment number is sometimes also included in the ID of a Function.

hierarchy domain

The part of a Hierarchy originating from a single Root Port.

Host Bridge

Part of a Root Complex that connects a host CPU or CPUs to a Hierarchy.

Hot Reset

A reset propagated in-band across a Link using a Physical Layer mechanism.

IDE Partner Port

The remote IDE Terminus for an IDE Stream.

IDE Stream

A Port to Port connection established using the mechanisms defined by Integrity and Data Encryption (IDE) to secure TLP traffic between the two Ports. The connection may be in the form of a Selective IDE Stream, in which case it is possible for IDE TLPs to flow through Switches without affecting their security, or in the form of a Link IDE Stream, in which case the two Ports must be connected without intervening Switches.

IDE Terminus

A Port acting as the originator or ultimate destination for IDE TLPs associated with one or more IDE Streams.

IDE TLP

A TLP associated with an IDE Stream and secured using Integrity and Data Encryption (IDE).

in-band signaling

A method for signaling events and conditions using the Link between two components, as opposed to the use of separate physical (sideband) signals. All mechanisms defined in this document can be implemented using in-band signaling, although in some form factors sideband signaling may be used instead.

Ingress Port

Receiving Port; that is, the Port that accepts incoming traffic.

Internal Error

An error associated with a PCI Express interface that occurs within a component and which may not be attributable to a packet or event on the PCI Express interface itself or on behalf of transactions initiated on PCI Express.

I/O Space

One of the four address spaces of the PCI Express architecture.

isochronous

Data associated with time-sensitive applications, such as audio or video applications.

invariant

A field of a TLP header or TLP Prefix that contains a value that cannot legally be modified as the TLP flows through the PCI Express fabric.

Lane

A set of differential signal pairs, one pair for transmission and one pair for reception. A by-N Link is composed of N Lanes.

Layer

A unit of distinction applied to this specification to help clarify the behavior of key elements. The use of the term Layer does not imply a specific implementation.

Link

The collection of two Ports and their interconnecting Lanes. A Link is a dual-simplex communications path between two components.

Link IDE Stream

An IDE Stream applied to all TLPs, except those associated with Selective IDE Stream(s), where the two Ports are connected without intervening Switches, although extension devices may be present on the Link.

Link Segment

The collection of a Port and a Pseudo Port or two Pseudo Ports and their interconnecting Lanes. A Link Segment is a dual simplex communications path between a Component and a Retimer or between two Retimers (two Pseudo Ports).

Lightweight Notification, LN

This protocol is now deprecated. It was a lightweight protocol that supported notifications to Endpoints via a hardware mechanism when cachelines of interest were updated.

Local TLP Prefix

A TLP Prefix that is carried along with a TLP on a single Link. See [§ Section 2.2.10.2](#).

Logical Bus

The logical connection among a collection of Devices that have the same Bus Number in Configuration Space.

Logical Idle

A period of one or more Symbol Times when no information (TLPs, PMUX Packets, DLLPs, or any special Symbol) is being transmitted or received. Unlike Electrical Idle, during Logical Idle the Idle data Symbol is being transmitted and received.

LTR

Abbreviation for Latency Tolerance Reporting

Malformed Packet

A TLP that violates specific TLP formation rules as defined in this specification.

Measurement

A process for calculating a cryptographic hash value of firmware or other configuration state, applying a digital signature, and returning this information.

Memory Space

One of the four address spaces of the PCI Express architecture.

Message

A TLP used to communicate information outside of the Memory, I/O, and Configuration Spaces.

Message Signaled Interrupt, MSI/MSI-X

Two similar but separate mechanisms that enable a Function to request service by writing a system-specified DWORD of data to a system-specified address using a Memory Write Request. Compared to MSI, MSI-X supports a larger maximum number of vectors and independent message address and data for each vector.

Message Space

One of the four address spaces of the PCI Express architecture.

MPS

Abbreviation for Max_Payload_Size.

Multicast, MC

A feature and associated mechanisms that enable a single Posted Request TLP sent by a source to be distributed to multiple targets.

Multicast Group, MCG

A set of Endpoints that are the target of Multicast TLPs in a particular address range.

Multicast Hit

The determination by a Receiver that a TLP will be handled as a Multicast TLP.

Multicast TLP

A TLP that is potentially distributed to multiple targets, as controlled by Multicast Capability structures in the components through which the TLP travels.

Multicast Window

A region of Memory Space where Posted Request TLPs that target it will be handled as Multicast TLPs.

Multi-Function Device, MFD

A Device that has multiple Functions.

Multi-Root I/O Virtualization, MR-IOV

A Function that supports the MR-IOV capability. See [MR-IOV] for additional information.

MUST@FLIT

MUST@FLIT features are mandatory for components that support Flit Mode (Flit Mode Supported is Set). *MUST@FLIT* features are strongly recommended for all other components.

naturally aligned

A data payload with a starting address equal to an integer multiple of a power of two, usually a specific power of two. For example, 64-byte naturally aligned means the least significant 6 bits of the byte address are 00 0000b.

NPEM

Native PCIe Enclosure Management

OBFF

Optimized Buffer Flush/Fill

Operating System

Throughout this specification, the terms operating system and system software refer to the combination of power management services, device drivers, user-mode services, and/or kernel mode services.

orderly removal

A hot-plug removal model where the OS is notified when a user/operator wishes to remove an adapter, and the OS has the opportunity to prepare for the event (e.g., quiescing adapter activity) before granting permission for removal.

P2P

Peer-to-peer.

Path

The flow of data through a Retimer, in either the Upstream Path or the Downstream Path.

Packet

A fundamental unit of information transfer consisting of an optional TLP Prefix, followed by a header and, in some cases, followed by a data payload.

Parts per Million, ppm

Applied to frequency, the difference, in millionths of a Hertz, between a stated ideal frequency, and the measured long-term average of a frequency.

PCIe®

PCI Express®

PCI Bridge

See Type 1 Function.

PCI Software Model

The software model necessary to initialize, discover, configure, and use a PCI-compatible device, as specified in [PCI-3.0], [PCI-X], and [Firmware].

Phantom Function Number, PFN

An unclaimed Function Number that may be used to expand the number of outstanding transaction identifiers by logically combining the PFN with the Tag identifier to create a unique transaction identifier.

Physical Function, PF

A PCI Function that contains an SR-IOV Extended Capability structure and supports the SR-IOV capabilities defined in § Chapter 9. .

Physical Lane

See Lane.

Physical Layer

The Layer that directly interacts with the communication medium between two components.

Port

1. Logically, an interface between a component and a PCI Express Link.
2. Physically, a group of Transmitters and Receivers located on the same chip that define a Link.

Power Management

Software or Hardware mechanisms used to minimize system power consumption, manage system thermal limits, and maximize system battery life. Power management involves tradeoffs among system speed, noise, battery life, and AC power consumption.

PMUX Channel

A multiplexed channel on a PMUX Link that is configured to transport a specific multiplexed protocol. See § Appendix G. .

PMUX Link

A Link where Protocol Multiplexing is supported and enabled. See § Appendix G. .

PMUX Packet

A non-PCI Express Packet transported over a PCI Express Link. See § Appendix G. .

Precision Time Measurement, PTM

An optional capability for communicating precise timing information between components.

Process Address Space ID, PASID

The Process Address Space ID, in conjunction with the Requester ID, uniquely identifies the address space associated with a transaction.

Programmed I/O, PIO

A transaction sequence that's initiated by a host processor, often as the result of executing a single load or store instruction that targets a special address range, but can be generated by other mechanisms such as the PCI-Compatible Configuration Mechanism. Notably, host processor loads or stores targeting an ECAM address range generate Configuration Space transactions. Other memory-mapped ranges typically exist to generate Memory Space and I/O Space transactions.

Pseudo Port

1. Logically, an interface between a Retimer and a PCI Express Link Segment.
2. Physically, a group of Transmitters and Receivers located on the same Retimer chip that define a Link Segment.

Quality of Service, QoS

Attributes affecting the bandwidth, latency, jitter, relative priority, etc., for differentiated classes of traffic.

QWORD, QW

Eight bytes. Used in the context of a data payload, the 8 bytes of data must be on a naturally aligned 8-byte boundary (the least significant 3 bits of the address are 000b).

RCiEP

Root Complex Integrated Endpoint.

Receiver, Rx

The component that receives Packet information across a Link.

Receiving Port

In the context of a specific TLP, PMUX Packet, or DLLP, the Port that receives the Packet on a given Link.

Re-driver

A non-protocol aware, software transparent, Extension Device.

repeater

An imprecise term for Extension Device.

Reported Error

An error subject to the logging and signaling requirements architecturally defined in this document.

Request

A Packet used to initiate a transaction sequence. A *Request* includes operation code and, in some cases, address and length, data, or other information.

Requester

The Function that first introduces a transaction sequence into the PCI Express domain.

Requester ID

The combination of a Requester's Bus Number, Device Number, and Function Number that uniquely identifies the Requester within a Hierarchy. With an ARI Requester ID, bits traditionally used for the Device Number field are used instead to expand the Function Number field, and the Device Number is implied to be 0.

Reserved

The contents, states, or information are not defined at this time. Using any Reserved area (for example, packet header bit-fields, configuration register bits) is not permitted. Reserved register fields must be read only and must

return 0 (all 0's for multi-bit fields) when read. Reserved encodings for register and packet fields must not be used. Any implementation dependence on a Reserved field value or encoding will result in an implementation that is not PCI Express-compliant. The functionality of such an implementation cannot be guaranteed in this or any future revision of this specification.

Refclk

An abbreviation for Reference Clock.

Retimer

A Physical Layer protocol aware, software transparent, Extension Device that forms two separate electrical Link Segments.

Root Complex, RC

A defined System Element that includes at least one Host Bridge, Root Port, or Root Complex Integrated Endpoint.

Root Complex Component

A logical aggregation of Root Ports, Root Complex Register Blocks, Root Complex Integrated Endpoints, and Root Complex Event Collectors.

Root Port, RP

A PCI Express Port on a Root Complex that maps a portion of a Hierarchy through an associated virtual PCI-PCI Bridge.

Routing Element

A term referring to a Root Complex, Switch, or Bridge in regard to its ability to route, multicast, or block TLPs.

Routing ID, RID

Either the Requester ID or Completer ID that identifies a PCI Express Function.

RP PIO

Root Port Programmed I/O. See § Section 6.2.11.3 .

Rx_MPS_Limit

The computed data payload size limit for a Function receiving a TLP, which is determined by the Rx_MPS_Fixed bit value and Max_Payload_Size setting in one or more Functions. See § Section 2.2.2 for details.

Segment

See Hierarchy

Selective IDE Stream

An IDE Stream applied selectively to TLPs based on ranges of Memory Addresses and RIDs, and where it is possible for secured TLPs to flow through Switches without affecting their security.

Set

A bit is Set when its value is 1b.

Shadow Function

An otherwise unimplemented Function, where its Transaction ID space is used by a Function that implements the Shadow Functions Extended Capability structure.

sideband signaling

A method for signaling events and conditions using physical signals separate from the signals forming the Link between two components. All mechanisms defined in this document can be implemented using in-band signaling, although in some form factors sideband signaling may be used instead.

Single-Function Device, SFD

A device that has a single Function.

Single Root I/O Virtualization, SR-IOV

A Function that supports the SR-IOV Extended Capability defined in this specification.

Single Root PCI Manager, SR-PCIM

Software responsible for configuration and management of the SR-IOV Extended Capability and PF/VF as well as dealing with associated error handling. Multiple implementation options exist; therefore, SR-PCIM implementation is outside the scope of this specification.

SR-IOV Device

A Device containing one or more Functions that have an SR-IOV Extended Capability structure.

SSD

Solid State Drive

Swap, Unconditional Swap

An AtomicOp where a specified value is written to a target location, and the original value of the location is returned.

Switch

A defined System Element that connects two or more Ports to allow Packets to be routed from one Port to another. To configuration software, a Switch appears as a collection of virtual PCI-to-PCI Bridges [PCI-to-PCI-Bridge].

Symbol

A 10-bit quantity when using 8b/10b encoding. An 8-bit quantity when using 128b/130b encoding.

Symbol Time

The period of time required to place a Symbol on a Lane (10 times the Unit Interval when using 8b/10b encoding and 8 times the Unit Interval when using 128b/130b encoding).

System Element

A defined Device or collection of Devices that operate according to distinct sets of rules. The following System Elements are defined: Root Complex, Endpoint, Switch, and Bridge.

System Image, SI

A software component running on a virtual system to which specific Functions, PFs, and VFs can be assigned. Specification of the behavior and architecture of an SI is outside the scope of this specification. Examples of SIs include guest operating systems and shared/non-shared protected domain device drivers.

System Software

Includes System Firmware (BIOS, UEFI), Operating System, VMM, management software, platform vendor's add-on to the Operating System.

Tag

A number assigned to a given Non-Posted Request to distinguish Completions for that Request from other Requests.

TLP Prefix

Additional information that may be optionally prepended to a TLP. TLP Prefixes are either Local or End-End. A TLP can have multiple TLP Prefixes. See § Section 2.2.10 .

TPH

Abbreviation for TLP Processing Hints

Transaction Descriptor

An element of a Packet header that, in addition to Address, Length, and Type, describes the properties of the Transaction.

Transaction ID

A component of the Transaction Descriptor including Requester ID and Tag.

Transaction Layer

The Layer that operates at the level of transactions (for example, read, write).

Transaction Layer Packet, TLP

A Packet generated in the Transaction Layer to convey a Request or Completion.

transaction sequence

A single Request and zero or more Completions associated with carrying out a single logical transfer by a Requester.

Transceiver

The physical Transmitter and Receiver pair on a single chip.

Translated Request

A Request using a Translated Memory Address, as indicated by the AT field.

Transmitter, Tx

The component sending Packet information across a Link.

Transmitting Port

In the context of a specific TLP, PMUX Packet, or DLLP, the Port that transmits the Packet on a given Link.

Trusted Execution Environment, TEE

Refers to an environment, which may include only a portion of a device, the whole of a device, or a composition of multiple devices, within which some level of “trust” is established, such that operations (including code execution) that occur within this environment are considered “trustworthy”. It is generally the case that one TEE is isolated from other TEEs that are intended to be distinct, and that all TEEs are isolated from untrusted environments.

Tx_MPS_Limit

The computed data payload size limit for a Function transmitting a TLP, which is determined by the Max_Payload_Size setting in one or more Functions. See § Section 2.2.2 for details.

Type 0 Function

Function with a Type 0 Configuration Space Header (see § Section 7.5.1.2).

Type 1 Function

Function with a Type 1 Configuration Space Header (see § Section 7.5.1.3).

Unconditional Swap, Swap

An AtomicOp where a specified value is written to a target location, and the original value of the location is returned.

Unit Interval, UI

Given a data stream of a repeating pattern of alternating 1 and 0 values, the Unit Interval is the value measured by averaging the time interval between voltage transitions, over a time interval long enough to make all intentional frequency modulation of the source clock negligible (see RX: UI and TX: UI).

Unsupported Request, UR

1. A status that applies to a posted or non-posted Request that specifies some action or access to some space that is not supported by the Completer.
2. A status indication returned with a Completion for a non-posted Request that suffered an Unsupported Request at the Completer.

Upstream

1. The relative position of an interconnect/System Element (Port/component) that is closer to the Root Complex. The Port on a Switch that is closest topologically to the Root Complex is the Upstream Port. The Port on a component that contains only Endpoint or Bridge Functions is an Upstream Port. The Upstream component on a Link is the component closer to the Root Complex.
2. A direction of information flow where the information is flowing towards the Root Complex.

Upstream Path

The flow of data through a Retimer from the Downstream Pseudo Port Receiver to the Upstream Pseudo Port Transmitter.

variant

A field of a TLP header that contains a value that is subject to possible modification according to the rules of this specification as the TLP flows through the PCI Express fabric.

Virtual Function, VF

A Function that is associated with a Physical Function. A VF shares one or more physical resources, such as a Link, with the Physical Function and other VFs that are associated with the same PF.

Virtualization Intermediary, VI

A software component supporting one or more SIs-colloquially known as a hypervisor or virtual machine monitor. Specification of the behavior and architecture of the VI is outside the scope of this specification.

wakeup

An optional mechanism used by a component to request the reapplication of main power when in the L2 Link state. Two such mechanisms are defined: Beacon (using in-band signaling) and WAKE# (using sideband signaling).

Warm Reset

A Fundamental Reset without cycling main power.

Zero

The numerical value of zero in a bit, field, or register, of appropriate width for that bit, field, or register.

Reference Documents

PCI

PCI-3.0

PCI Local Bus Specification, Revision 3.0

PCIe

PCIe-6.0

PCI Express Base Specification, Revision 6.0

PCIe-5.0

PCI Express Base Specification, Revision 5.0

PCIe-4.0

PCI Express Base Specification, Revision 4.0

PCIe-3.1

PCIe-3.1a

PCI Express Base Specification, Revision 3.1a

PCIe-3.0

PCI Express Base Specification, Revision 3.0

PCIe-2.1

PCI Express Base Specification, Revision 2.1

PCIe-2.0

PCI Express Base Specification, Revision 2.0

PCIe-1.1

PCI Express Base Specification, Revision 1.1

PCIe-1.0

PCIe-1.0a

PCI Express Base Specification, Revision 1.0a

CEM

CEM-5.0

PCI Express Card Electromechanical Specification, Revision 5.0

CEM-4.0

PCI Express Card Electromechanical Specification, Revision 4.0

CEM-3.0

PCI Express Card Electromechanical Specification, Revision 3.0

CEM-2.0

PCI Express Card Electromechanical Specification, Revision 2.0

PCIe-to-PCI-PCI-X-Bridge

PCI Express to PCI/PCI-X Bridge Specification, Revision 1.0

PCI-to-PCI-Bridge

PCI-to-PCI Bridge Architecture Specification Revision 1.2

PCI-X

PCI-X Addendum to the PCI Local Bus Specification, Revision 2.0a

Mini-Card

PCI Express Mini Card Electromechanical Specification, Revision 2.1

OCuLink

PCI Express OCuLink Specification, Revision 1.1

M.2

PCI Express M.2 Specification, Revision 4.0

MR-IOV

MR-IOV Specification, Revision 1.0

U.2**SFF-8639**

PCI Express SFF-8639 Module Specification, Revision 4.0, Version 1.0

Ext-Cabling

PCI Express External Cabling Specification, Revision 3.0a

ExpressModule

PCI Express ExpressModule Electromechanical Specification, Revision 1.0

PCI-Hot-Plug**PCI-Hot-Plug-1.1**

PCI Hot-Plug Specification, Revision 1.1

PCI-PM

PCI Bus Power Management Interface Specification, Revision 1.2

PCI-Code-and-ID

PCI Code and ID Assignment Specification, Revision 1.11 (or later)

Firmware

PCI Firmware Specification, Revision 3.2

ACPI

Advanced Configuration and Power Interface Specification, Revision 6.2

UEFI

Unified Extensible Firmware Interface (UEFI) Specification, Version 2.8

EUI-48**EUI-64**

Guidelines for Use of Extended Unique Identifier (EUI), Organizationally Unique Identifier (OUI), and Company ID (CID)

JEDEC-JESD22-C101

JEDEC JESD22-C101F: Field-Induced Charged-Device Model Test Method for Electrostatic Discharge Withstand Thresholds of Microelectronic Components

JEDEC-JEP155-JEP157

JEDEC JEP155: Recommended ESD Target Levels for HBM/MM Qualification and JEP157 Recommended ESD-CDM Target Levels

ESDA-JEDEC-JS-001-2010

ESDA/JEDEC JS-001-2010: Joint JEDEC/ESDA Standard for Electrostatic Discharge Sensitivity Test - Human Body Model (HBM) - Component Level

ITU-T-Rec-X-667

ITU T-Rec. X.667: Information technology - Procedures for the operation of object identifier registration authorities: Generation of universally unique identifiers and their use in object identifiers

ISO-IEC-9834-8

ISO/IEC 9834-8: Information technology -- Procedures for the operation of object identifier registration authorities -- Part 8: Generation of universally unique identifiers (UUIDs) and their use in object identifiers

RFC-4122

IETF RFC-4122: A Universally Unique Identifier (UUID) URN Namespace

DNS**RFC-1034**

IETF RFC-1034: DOMAIN NAMES - CONCEPTS AND FACILITIES

PICMG

PICMG

PLUG-PLAY-ISA-1.0a

Plug and Play ISA Specification, Version 1.0a, May 5, 1994

PC-Card

PC-Card

MCTP-VDM

Management Component Transport Protocol (MCTP) PCIe VDM Transport Binding Specification - <https://www.dmtf.org/dsp/DSP0238>.

SPDM

DMTF Security Protocol & Data Model (SPDM) Specification - <https://www.dmtf.org/dsp/DSP0274>. CMA requires SPDM Version 1.0 or above. IDE requires SPDM Version 1.1 or above.

SPDM-MCTP

Security Protocol and Data Model (SPDM) over MCTP Binding Specification – <https://www.dmtf.org/dsp/DSP0275>.

AES-GCM

NIST Special Publication 800-38D Recommendation for Block Cipher Modes of Operation: Galois/Counter Mode (GCM) and GMAC <https://nvlpubs.nist.gov/nistpubs/Legacy/SP/nistspecialpublication800-38d.pdf>

Secured SPDM

Secured Messages using SPDM Specification (IDE requires version 1.0 or above) - <https://www.dmtf.org/dsp/DSP0277>

Secured MCTP

Secured Messages using SPDM over MCTP Binding Specification (version 1.0 or above) - <https://www.dmtf.org/dsp/DSP0276>

Compute Express Link <https://www.computeexpresslink.org>

1. Introduction §

This chapter presents an overview of the PCI Express architecture and key concepts. PCI Express is a high performance, general purpose I/O interconnect defined for a wide variety of future computing and communication platforms. Key attributes, such as usage model, load-store architecture, and software interfaces, are maintained from PCI Local Bus, whereas PCI Local Bus's parallel bus implementation is replaced by a highly scalable, fully serial interface. PCI Express takes advantage of advances in point-to-point interconnects, Switch-based technology, and packetized protocol to deliver new levels of performance and features. Power Management, Quality of Service (QoS), Hot-Plug/hot-swap support, data integrity, and error handling are among some of the advanced features supported by PCI Express.

1.1 An Evolving I/O Interconnect §

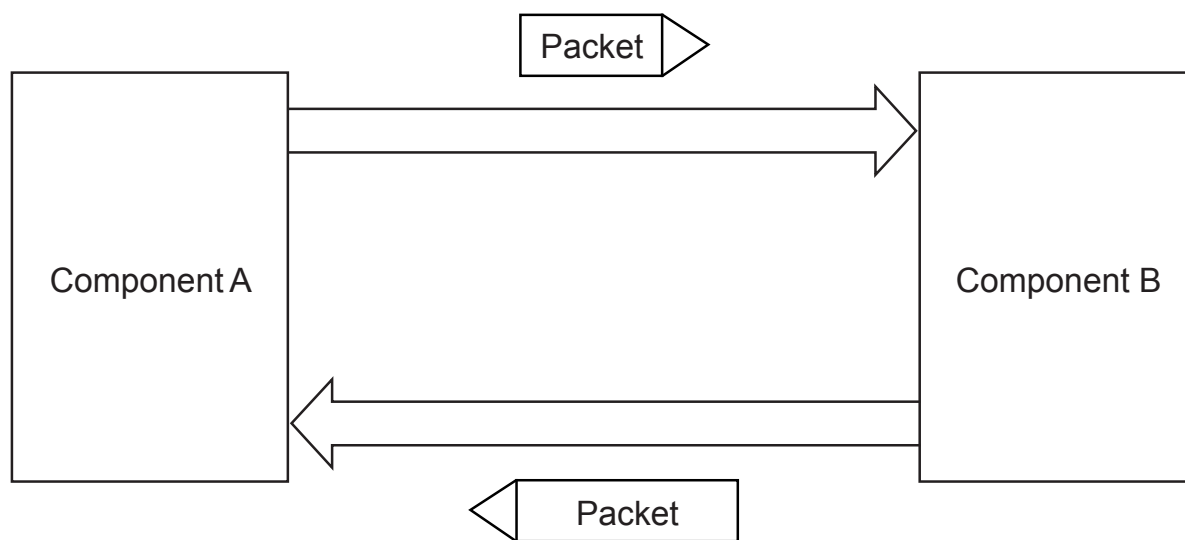
The high-level requirements for this evolving I/O interconnect are as follows:

- **Supports multiple market segments and emerging applications:**
 - Unifying I/O architecture for desktop, mobile, workstation, server, communications platforms, and embedded devices
- **Ability to deliver low cost, high volume solutions:**
 - Cost at or below PCI cost structure at the system level
- **Support multiple platform interconnect usages:**
 - Chip-to-chip, board-to-board via connector or cabling
- **A variety of mechanical form factors:**
 - [M.2], [CEM] (Card Electro-Mechanical), [U.2], [OCuLink]
- **PCI-compatible software model:**
 - Ability to enumerate and configure PCI Express hardware using PCI system configuration software implementations with no modifications
 - Ability to boot existing operating systems with no modifications
 - Ability to support existing I/O device drivers with no modifications
 - Ability to configure/enable new PCI Express functionality by adopting the PCI configuration paradigm
- **Performance:**
 - Low-overhead, low-latency communications to maximize application payload bandwidth and Link efficiency
 - High-bandwidth per pin to minimize pin count per device and connector interface
 - Scalable performance via aggregated Lanes and signaling frequency
- **Advanced features:**
 - Comprehend different data types and ordering rules
 - Power management and budgeting
 - Ability to identify power management capabilities of a Device of a specific Function
 - Ability to transition a Device or Function into a specific power state
 - Ability to receive notification of the current power state of a Device of Function
 - Ability to generate a request to wakeup from a power-off state of the main power supply

- Ability to sequence Device power-up to allow graceful platform policy in power budgeting
- Ability to support differentiated services, i.e., different (QoS)
 - Ability to have dedicated Link resources per QoS data flow to improve fabric efficiency and effective application-level performance in the face of head-of-line blocking
 - Ability to configure fabric QoS arbitration policies within every component
 - Ability to tag end-to-end QoS with each packet
 - Ability to create end-to-end isochronous (time-based, injection rate control) solutions
- Hot-Plug support
 - Ability to support existing PCI Hot-Plug solutions
 - Ability to support native Hot-Plug solutions (no sideband signals required)
 - Ability to support async removal
 - Ability to support a unified software model for all form factors
- Data Integrity
 - Ability to support Link-level data integrity for all types of transaction and Data Link packets
 - Ability to support end-to-end data integrity for high availability solutions
- Error handling
 - Ability to support PCI-Compatible error handling
 - Ability to support advanced error reporting and handling to improve fault isolation and recovery solutions
- Process Technology Independence
 - Ability to support different DC common mode voltages at Transmitter and Receiver
- Ease of Testing
 - Ability to test electrical compliance via simple connection to test equipment

1.2 PCI Express Link §

A Link represents a dual-simplex communications channel between two components. The fundamental PCI Express Link consists of two, low-voltage, differentially driven signal pairs: a Transmit pair and a Receive pair as shown in [§ Figure 1-1](#). A PCI Express Link consists of a PCIe PHY as defined in [§ Chapter 4](#).



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Figure 1-1 PCI Express Link

The primary Link attributes for PCI Express Link are:

- The basic Link - PCI Express Link consists of dual unidirectional differential Links, implemented as a Transmit pair and a Receive pair. A data clock is embedded using an encoding scheme (see § Chapter 4.) to achieve very high data rates.
- The Signaling method – Each major revision of PCI Express signaling has evolved one (or more) characteristics to increase bandwidth. Throughout this specification, the term GT/s is used to refer to the number of encoded bits transferred in a second on a direction of a Lane. The actual effective data rate is dependent on a combination of modulation method, encoding method, and data rate. § Table 1-1 provides a summary of Max Data Rate, Modulation Scheme, Encoding Method, and Effective Max Data Rate with the accounting of only encoding overhead for all the six major revisions of PCI Express.² See § Chapter 4. for more information about the combined signaling method and § Chapter 8. for electrical specification details for each major PCI Express revision.

Table 1-1 PCIe Signaling Characteristics

Data Rate	Modulation	Encoding	Effective Data Rate (after removing Encoding overhead)	Base Specification Revision					
				6.x	5.x	4.x	3.0	2.0	1.0
2.5 GT/s	NRZ	8b/10b	2 Gbit/s	✓	✓	✓	✓	✓	✓
5.0 GT/s	NRZ	8b/10b	4 Gbit/s	✓	✓	✓	✓	✓	
8.0 GT/s	NRZ	128b/130b	~8 Gbit/s	✓	✓	✓	✓		
16.0 GT/s	NRZ	128b/130b	~16 Gbit/s	✓	✓	✓			
32.0 GT/s	NRZ	128b/130b	~32 Gbit/s	✓	✓				
64.0 GT/s	PAM4	1b/1b	64 Gbit/s	✓					

2. Terms like “PCIe Gen3” are ambiguous and should be avoided. For example, “gen3” could mean (1) compliant with Base 3.0, (2) compliant with Base 3.1 (last revision of 3.x), (3) compliant with Base 3.0 and supporting 8.0 GT/s, (4) compliant with Base 3.0 or later and supporting 8.0 GT/s,

- Lanes - A Link must support at least one Lane - each Lane represents a set of differential signal pairs (one pair for transmission, one pair for reception). To scale bandwidth, a Link may aggregate multiple Lanes denoted by xN where N may be any of the supported Link widths. A x8 Link operating at the 2.5 GT/s data rate represents an aggregate bandwidth of 20 Gigabits/second of raw bandwidth in each direction. This specification describes operations for x1, x2, x4, x8, and x16 Lane widths.
- Initialization - During hardware initialization, each PCI Express Link is set up following a negotiation of Lane widths and frequency of operation by the two agents at each end of the Link. No firmware or operating system software is involved.
- Symmetry - Each Link must support a symmetric number of Lanes in each direction, i.e., a x16 Link indicates there are 16 differential signal pairs in each direction.

1.3 PCI Express Fabric Topology §

A fabric is composed of point-to-point Links that interconnect a set of components - an example fabric topology is shown in § Figure 1-2. This figure illustrates a single fabric instance with two Hierarchies composed of a Root Complex (RC), multiple Endpoints, and multiple Switches, interconnected via PCI Express Links.

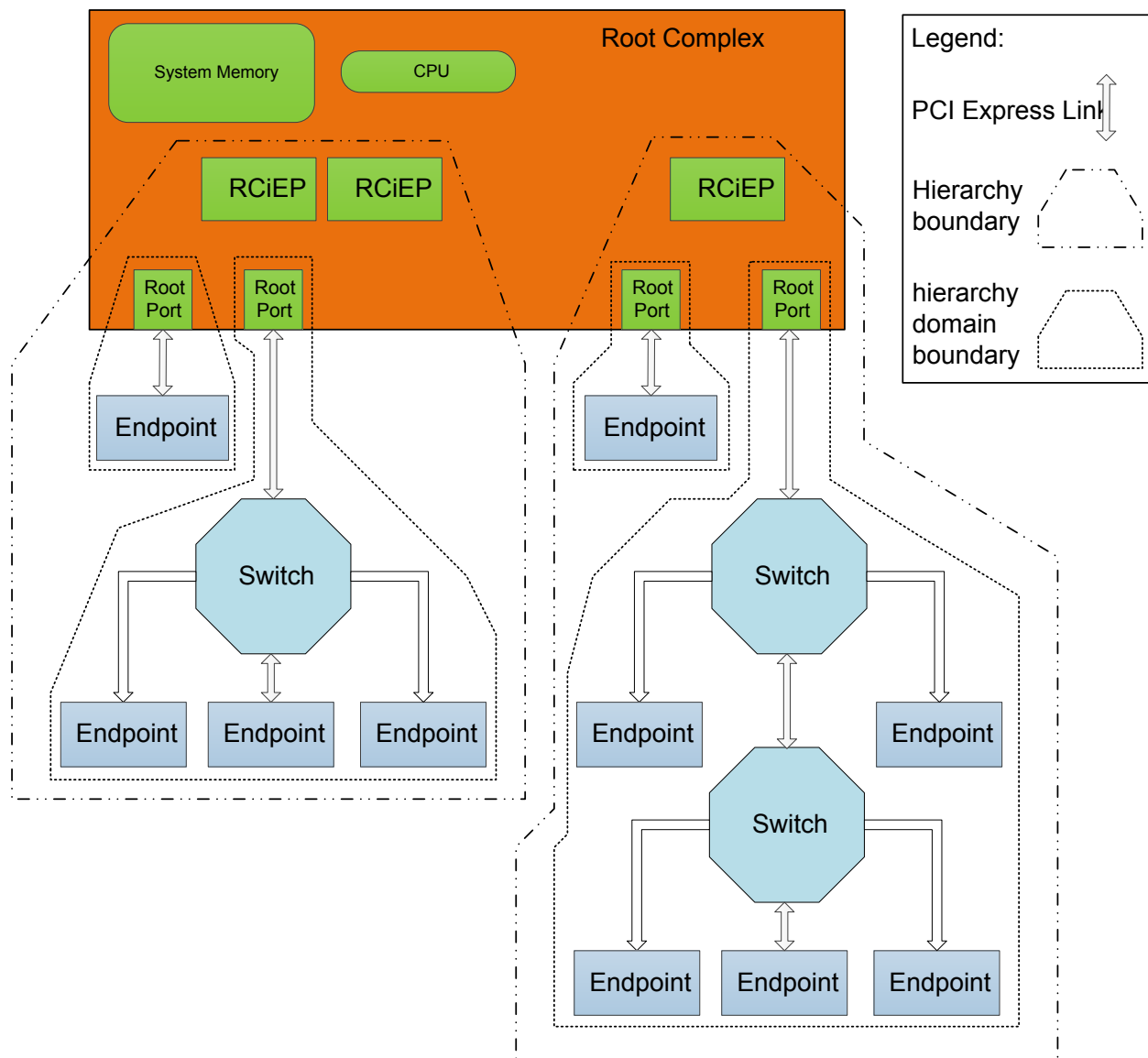


Figure 1-2 Example PCI Express Topology

1.3.1 Root Complex §

- An RC denotes the root of an I/O hierarchy that connects the CPU/memory subsystem to the I/O.
- As illustrated in § Figure 1-2, an RC may support one or more PCI Express Ports. Each interface defines a separate hierarchy domain. Each hierarchy domain may be composed of a single Endpoint or a sub-hierarchy containing one or more Switch components and Endpoints.
- The capability to route peer-to-peer transactions between hierarchy domains through an RC is optional and implementation dependent. An RC is permitted to "take ownership" of Requests that pass peer-to-peer between Root Ports, reforming and potentially splitting a Request such that it may appear to the ultimate Completer that the RC was the origin of the Request, and subsequently the RC must reform the Completion(s)

being returned to the original Requester. Alternately, an RC implementation may incorporate a real or virtual Switch internally within the Root Complex to enable full peer-to-peer support in a software transparent way. Unlike the rules for a Switch, an RC is generally permitted to split a packet into smaller packets when routing transactions peer-to-peer between hierarchy domains (except as noted below), e.g., split a single packet with a 256-byte payload into two packets of 128 bytes payload each. The resulting packets are subject to the normal packet formation rules contained in this specification (e.g., Max_Payload_Size, Read Completion Boundary (RCB), etc.). Component designers should note that splitting a packet into smaller packets may have negative performance consequences, especially for a transaction addressing a device behind a PCI Express to PCI/PCI-X bridge.

Exception: An RC that supports peer-to-peer routing of Deferrable Memory Write Requests is not permitted to split a Deferrable Memory Write Request packet into smaller packets (see § Section 6.32).

Exception: An RC that supports peer-to-peer routing of Vendor_Defined Messages is not permitted to split a Vendor_Defined Message packet into smaller packets except at 128-byte boundaries (i.e., all resulting packets except the last must be an integral multiple of 128 bytes in length) in order to retain the ability to forward the Message across a PCI Express to PCI/PCI-X Bridge.

- An RC must support generation of Configuration Requests as a Requester.
- An RC is permitted to support the generation of I/O Requests as a Requester.
 - An RC is permitted to generate I/O Requests to either or both of locations 80h and 84h to a selected Root Port, without regard to that Root Port's PCI Bridge I/O decode configuration; it is recommended that this mechanism only be enabled when specifically needed.
- An RC must not support Lock semantics as a Completer.
- An RC is permitted to support generation of Locked Requests as a Requester.

1.3.2 Endpoints §

Endpoint refers to a type of Function that can be the Requester or Completer of a PCI Express transaction either on its own behalf or on behalf of a distinct non-PCI Express device (other than a PCI device or host CPU), e.g., a PCI Express attached graphics controller or a PCI Express-USB host controller. Endpoints are classified as either legacy, PCI Express, or Root Complex Integrated Endpoints (RCiEPs).

1.3.2.1 Legacy Endpoint Rules §

- A Legacy Endpoint must be a Function with a Type 00h Configuration Space header.
- A Legacy Endpoint must support Configuration Requests as a Completer.
- A Legacy Endpoint may support I/O Requests as a Completer.
 - A Legacy Endpoint is permitted to accept I/O Requests to either or both of locations 80h and 84h, without regard to that Endpoint's I/O decode configuration.
- A Legacy Endpoint may generate I/O Requests.
- A Legacy Endpoint may support Lock memory semantics as a Completer if that is required by the device's legacy software support requirements.
- A Legacy Endpoint must not issue a Locked Request.
- A Legacy Endpoint may implement Extended Configuration Space Capabilities, but such Capabilities may be ignored by software.

- A Legacy Endpoint operating as the Requester of a Memory Transaction is not required to be capable of generating addresses 4 GB or greater.
- A Legacy Endpoint is required to support MSI or MSI-X or both if an interrupt resource is requested. If MSI is implemented, a Legacy Endpoint is permitted to support either the 32-bit or 64-bit Message Address version of the MSI Capability structure.
- A Legacy Endpoint is permitted to support 32-bit addressing for Base Address Registers that request memory resources.
- A Legacy Endpoint must appear within one of the hierarchy domains originated by the Root Complex.

1.3.2.2 PCI Express Endpoint Rules §

- A PCI Express Endpoint must be a Function with a Type 00h Configuration Space header.
- A PCI Express Endpoint must support Configuration Requests as a Completer.
- A PCI Express Endpoint must not depend on operating system allocation of I/O resources claimed through BAR(s).
- A PCI Express Endpoint must not generate I/O Requests.
- A PCI Express Endpoint must not support Locked Requests as a Completer or generate them as a Requester. PCI Express-compliant device drivers and applications must be written to prevent the use of lock semantics when accessing a PCI Express Endpoint.
- A PCI Express Endpoint operating as the Requester of a Memory Transaction is required to be capable of generating addresses greater than 4 GB.
- A PCI Express Endpoint is required to support MSI or MSI-X or both if an interrupt resource is requested. If MSI is implemented, a PCI Express Endpoint must support the 64-bit Message Address version of the MSI Capability structure.
- A PCI Express Endpoint requesting memory resources through a BAR must set the BAR's Prefetchable bit unless the range contains locations with read side-effects or locations in which the Function does not tolerate write merging. See § Section 7.5.1.2.1 for additional guidance on having the Prefetchable bit Set.
- For a PCI Express Endpoint, 64-bit addressing must be supported for all BARs that have the Prefetchable bit Set. 32-bit addressing is permitted for all BARs that do not have the Prefetchable bit Set.
- The minimum memory address range requested by a BAR is 128 bytes.
- A PCI Express Endpoint must appear within one of the hierarchy domains originated by the Root Complex.

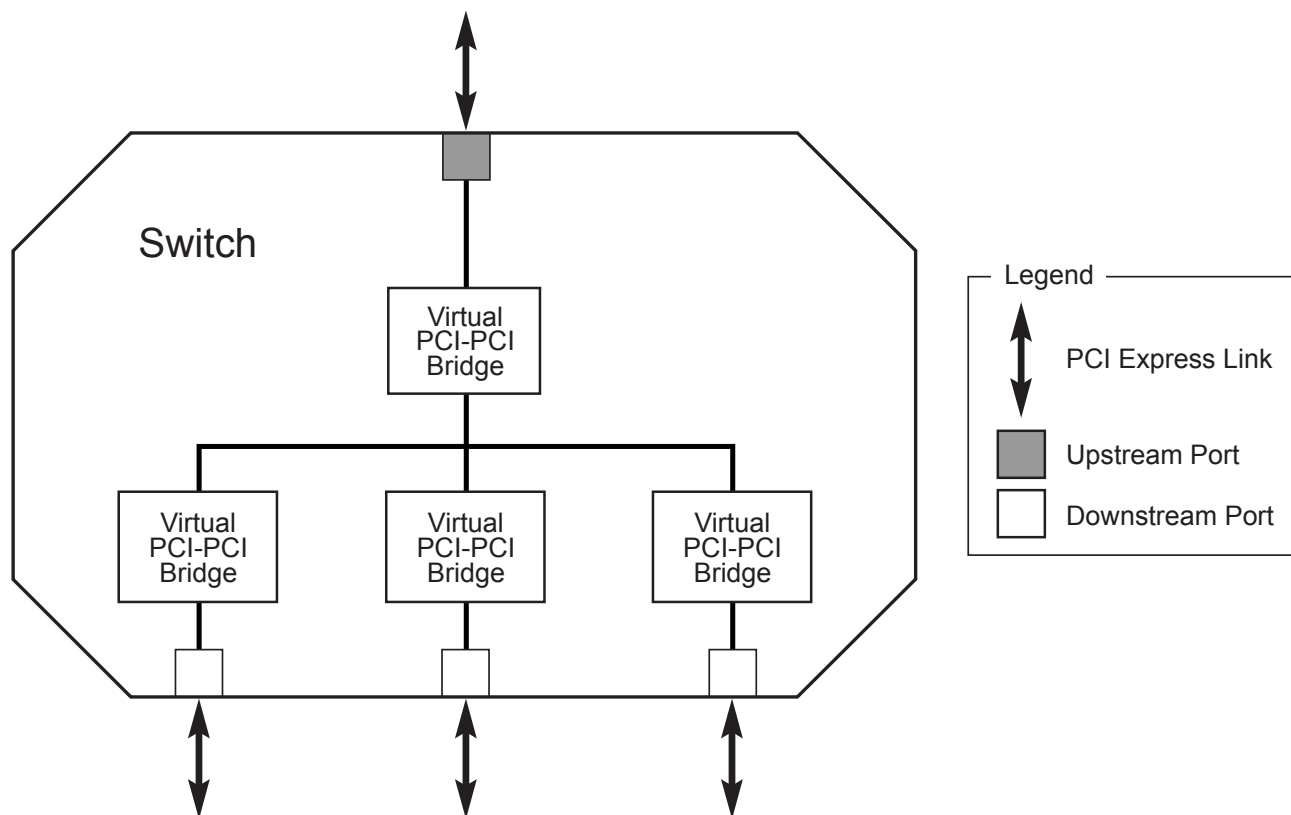
1.3.2.3 Root Complex Integrated Endpoint Rules §

- A Root Complex Integrated Endpoint (RCiEP) is implemented on internal logic of Root Complexes that contains the Root Ports.
- An RCiEP must be a Function with a Type 00h Configuration Space header.
- An RCiEP must support Configuration Requests as a Completer.
- An RCiEP must not require I/O resources claimed through BAR(s).
- An RCiEP must not generate I/O Requests.
- An RCiEP must not support Locked Requests as a Completer or generate them as a Requester. PCI Express-compliant device drivers and applications must be written to prevent the use of lock semantics when accessing an RCiEP.

- An RCiEP operating as the Requester of a Memory Transaction is required to be capable of generating addresses equal to or greater than the Host is capable of handling as a Completer.
- An RCiEP is required to support MSI or MSI-X or both if an interrupt resource is requested. If MSI is implemented, an RCiEP is permitted to support either the 32-bit or 64-bit Message Address version of the MSI Capability structure.
- An RCiEP is permitted to support 32-bit addressing for Base Address Registers that request memory resources.
- An RCiEP must not implement Link Capabilities, Link Status, Link Control, Link Capabilities 2, Link Status 2, and Link Control 2 registers in the PCI Express Extended Capability.
- If an RCiEP is associated with an optional Root Complex Event Collector it must signal PME and error conditions through the Root Complex Event Collector.
- An RCiEP must not be associated with more than one Root Complex Event Collector.
- An RCiEP does not implement Active State Power Management.
- An RCiEP may not be hot-plugged independent of the Root Complex as a whole.
- An RCiEP must not appear in any of the hierarchy domains exposed by the Root Complex.
- An RCiEP must not appear in Switches.

1.3.3 Switch §

A Switch is defined as a logical assembly of multiple virtual PCI-to-PCI Bridge devices as illustrated in § Figure 1-3. All Switches are governed by the following base rules.



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Figure 1-3 Logical Block Diagram of a Switch

- Switches appear to configuration software as two or more logical PCI-to-PCI Bridges.
- A Switch forwards transactions using PCI Bridge mechanisms; e.g., address-based routing except when engaged in a Multicast, as defined in § Section 6.14 .
- Except as noted in this document, a Switch must forward all types of Transaction Layer Packets (TLPs) between any set of Ports.
- Locked Requests must be supported as specified in § Section 6.5 . Switches are not required to support Downstream Ports as initiating Ports for Locked Requests.
- Each enabled Switch Port must comply with the Flow Control specification within this document.
- A Switch is not allowed to split a packet into smaller packets, e.g., a single packet with a 256-byte payload must not be divided into two packets of 128 bytes payload each.
- Arbitration between Ingress Ports (inbound Link) of a Switch may be implemented using round robin or weighted round robin when contention occurs on the same Virtual Channel. This is described in more detail later within the specification.
- Endpoints (represented by Type 00h Configuration Space headers) must not appear to configuration software on the Switch's internal bus as peers of the virtual PCI-to-PCI Bridges representing the Switch Downstream Ports.

1.3.4 Root Complex Event Collector §

- A Root Complex Event Collector (RCEC) provides support for terminating error and PME messages from RCiEPs.
- A Root Complex Event Collector must follow all rules for an RCiEP (unless otherwise specified).
- A Root Complex Event Collector is not required to decode any memory or I/O resources.
- A Root Complex Event Collector is identified by its Device/Port Type value (see § Section 7.5.3.2).
- A Root Complex Event Collector has the Base Class 08h, Sub-Class 07h and Programming Interface 00h. ³
- A Root Complex Event Collector resides on a Bus in the Root Complex. Multiple Root Complex Event Collectors are permitted to reside on a single Bus.
- A Root Complex Event Collector explicitly declares supported RCiEPs through the Root Complex Event Collector Endpoint Association Extended Capability.
- Root Complex Event Collectors are optional.

1.3.5 PCI Express to PCI/PCI-X Bridge §

- A PCI Express to PCI/PCI-X Bridge provides a connection between a PCI Express fabric and a PCI/PCI-X hierarchy.

1.4 Hardware/Software Model for Discovery, Configuration and Operation §

The PCI/PCIe hardware/software model includes architectural constructs necessary to discover, configure, and use a Function, without needing Function-specific knowledge. Key elements include:

- A configuration model which provides system software the means to discover hardware Functions available in a system
- Mechanisms to perform basic resource allocation for addressable resources such as memory space and interrupts
- Enable/disable controls for Function response to received Requests, and initiation of Requests
- Well-defined ordering and flow control models to support the consistent and robust implementation of hardware/software interfaces

The PCI Express configuration model supports two mechanisms:

- PCI-compatible configuration mechanism: The PCI-compatible mechanism supports 100% binary compatibility with Conventional PCI aware operating systems and their corresponding bus enumeration and configuration software.
- PCI Express enhanced configuration mechanism: The enhanced mechanism is provided to increase the size of available Configuration Space and to optimize access mechanisms.

Each PCI Express Link is mapped through a virtual PCI-to-PCI Bridge structure and has a Logical Bus associated with it. The virtual PCI-to-PCI Bridge structure may be part of a PCI Express Root Complex Port, a Switch Upstream Port, or a

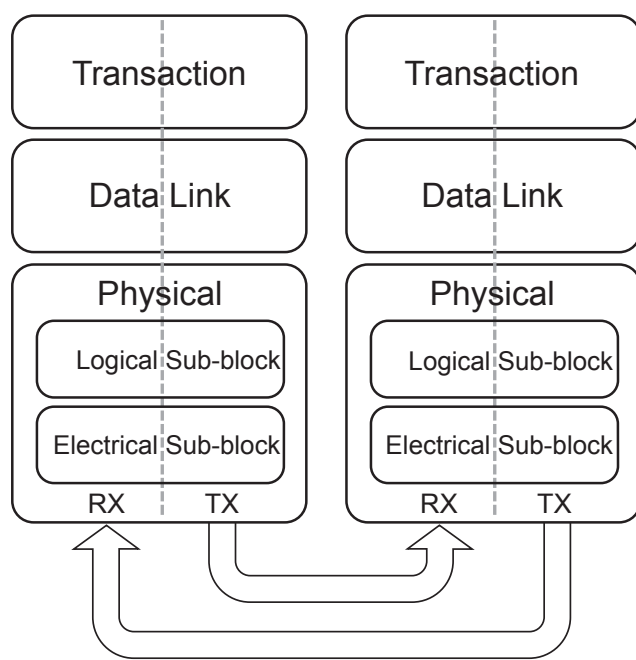
3. Since an earlier version of this specification used Sub-Class 06h for this purpose, an implementation is still permitted to use Sub-Class 06h, but this is strongly discouraged.

Switch Downstream Port. A Root Port is a virtual PCI-to-PCI Bridge structure that originates a PCI Express hierarchy domain from a PCI Express Root Complex. Devices are mapped into Configuration Space such that each will respond to a particular Device Number.

1.5 PCI Express Layering Overview §

This document specifies the architecture in terms of three discrete logical layers: the Transaction Layer, the Data Link Layer, and the Physical Layer. Each of these layers is divided into two sections: one that processes outbound (to be transmitted) information and one that processes inbound (received) information, as shown in § Figure 1-4.

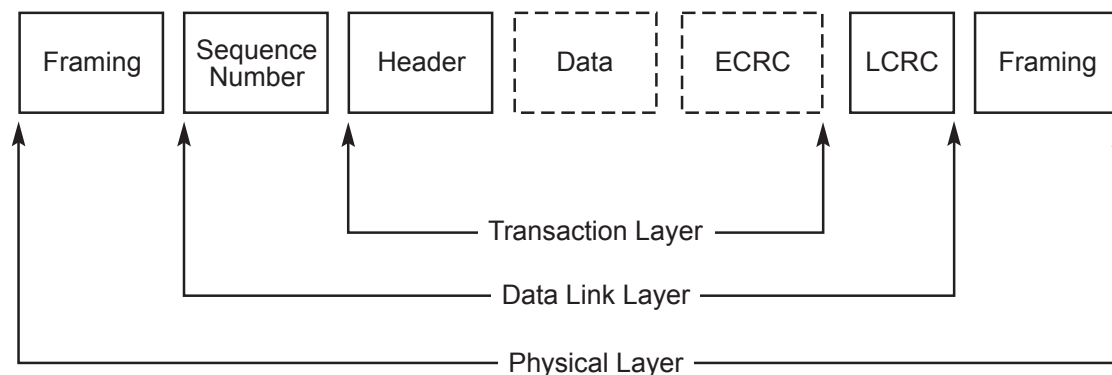
The fundamental goal of this layering definition is to facilitate the reader's understanding of the specification. Note that this layering does not imply a particular PCI Express implementation.



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§Figure 1-4 High-Level Layering Diagram

PCI Express uses packets to communicate information between components. Packets are formed in the Transaction and Data Link Layers to carry the information from the transmitting component to the receiving component. As the transmitted packets flow through the other layers, they are extended with additional information necessary to handle packets at those layers. At the receiving side the reverse process occurs and packets get transformed from their Physical Layer representation to the Data Link Layer representation and finally (for Transaction Layer Packets) to the form that can be processed by the Transaction Layer of the receiving device. § Figure 1-5 shows the conceptual flow of transaction level packet information through the layers.



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Figure 1-5 Packet Flow Through the Layers

Note that a simpler form of packet communication is supported between two Data Link Layers (connected to the same Link) for the purpose of Link management.

1.5.1 Transaction Layer §

The upper Layer of the architecture is the Transaction Layer. The Transaction Layer's primary responsibility is the assembly and disassembly of TLPs. TLPs are used to communicate transactions, such as read and write, as well as certain types of events. The Transaction Layer is also responsible for managing credit-based flow control for TLPs.

Every request packet requiring a response packet is implemented as a Split Transaction. Each packet has a unique identifier that enables response packets to be directed to the correct originator. The packet format supports different forms of addressing depending on the type of the transaction (Memory, I/O, Configuration, and Message). The Packets may also have attributes such as No Snoop, Relaxed Ordering, and ID-Based Ordering (IDO).

The Transaction Layer supports four address spaces: it includes the three PCI address spaces (memory, I/O, and configuration) and adds Message Space. This specification uses Message Space to support all prior sideband signals, such as interrupts, power-management requests, and so on, as in-band Message transactions. You could think of PCI Express Message transactions as “virtual wires” since their effect is to eliminate the wide array of sideband signals currently used in a platform implementation.

1.5.2 Data Link Layer §

The middle Layer in the stack, the Data Link Layer, serves as an intermediate stage between the Transaction Layer and the Physical Layer. The primary responsibilities of the Data Link Layer include Link management and data integrity, including error detection and error correction.

The transmission side of the Data Link Layer accepts TLPs assembled by the Transaction Layer, calculates and applies a data protection code and TLP sequence number, and submits them to Physical Layer for transmission across the Link. The receiving Data Link Layer is responsible for checking the integrity of received TLPs and for submitting them to the Transaction Layer for further processing. On detection of TLP error(s), this Layer is responsible for requesting retransmission of TLPs until information is correctly received, or the Link is determined to have failed.

The Data Link Layer also generates and consumes packets that are used for Link management functions. To differentiate these packets from those used by the Transaction Layer (TLP), the term Data Link Layer Packet (DLLP) will be used when referring to packets that are generated and consumed at the Data Link Layer.

1.5.3 Physical Layer §

The Physical Layer includes all circuitry for interface operation, including driver and input buffers, parallel-to-serial and serial-to-parallel conversion, PLL(s), and impedance matching circuitry. It also includes logical functions related to interface initialization and maintenance. The Physical Layer exchanges information with the Data Link Layer in an implementation specific format. This Layer is responsible for converting information received from the Data Link Layer into an appropriate serialized format and transmitting it across the PCI Express Link at a frequency and width compatible with the component connected to the other side of the Link.

The PCI Express architecture has “hooks” to support future performance enhancements via speed upgrades and advanced encoding techniques. The future speeds, encoding techniques or media may only impact the Physical Layer definition.

1.5.4 Layer Functions and Services §

1.5.4.1 Transaction Layer Services §

The Transaction Layer, in the process of generating and receiving TLPs, exchanges Flow Control information with its complementary Transaction Layer on the other side of the Link. It is also responsible for supporting both software and hardware-initiated power management.

Initialization and configuration functions require the Transaction Layer to:

- Store Link configuration information generated by the processor or management device,
- Store Link capabilities generated by Physical Layer hardware negotiation of width and operational frequency.

A Transaction Layer's Packet generation and processing services require it to:

- Generate TLPs from device core Requests
- Convert received Request TLPs into Requests for the device core,
- Convert received Completion Packets into a payload, or status information, deliverable to the core,
- Detect unsupported TLPs and invoke appropriate mechanisms for handling them,
- If end-to-end data integrity is supported, generate the end-to-end data integrity CRC and update the TLP header accordingly.

Flow Control services:

- The Transaction Layer tracks Flow Control credits for TLPs across the Link.
- Transaction credit status is periodically transmitted to the remote Transaction Layer using transport services of the Data Link Layer.
- Remote Flow Control information is used to throttle TLP transmission.

Ordering rules:

- PCI/PCI-X compliant producer/consumer ordering model,
- Extensions to support Relaxed Ordering,
- Extensions to support ID-Based Ordering.

Power management services:

- Software-controlled power management through mechanisms, as dictated by system software.
- Hardware-controlled autonomous power management minimizes power during full-on power states.

Virtual Channels and Traffic Class:

- The combination of Virtual Channel mechanism and Traffic Class identification is provided to support differentiated services and QoS support for certain classes of applications.
- Virtual Channels: Virtual Channels provide a means to support multiple independent logical data flows over given common physical resources of the Link. Conceptually this involves multiplexing different data flows onto a single physical Link.
- Traffic Class: The Traffic Class is a Transaction Layer Packet label that is transmitted unmodified end-to-end through the fabric. At every service point (e.g., Switch) within the fabric, Traffic Class labels are used to apply appropriate servicing policies. Each Traffic Class label defines a unique ordering domain - no ordering guarantees are provided for packets that contain different Traffic Class labels.

1.5.4.2 Data Link Layer Services §

The Data Link Layer is responsible for reliably exchanging information with its counterpart on the opposite side of the Link.

Initialization and power management services:

- Accept power state Requests from the Transaction Layer and convey to the Physical Layer
- Convey active/reset/disconnected/power managed state to the Transaction Layer

Data protection, error checking, and retry services:

- CRC generation
- Transmitted TLP storage for Data Link level retry
- Error checking
- TLP acknowledgement and retry Messages
- Error indication for error reporting and logging

1.5.4.3 Physical Layer Services §

Interface initialization, maintenance control, and status tracking:

- Reset/Hot-Plug control/status
- Interconnect power management
- Width and Lane mapping negotiation
- Lane polarity inversion

Symbol and special Ordered Set generation:

- 8b/10b encoding/decoding

- Embedded clock tuning and alignment

Block and special Ordered Set generation:

- 128b/130b encoding/decoding
- 1b/1b encoding/decoding
- Link Equalization

Symbol transmission and alignment:

- Transmission circuits
- Reception circuits
- Elastic buffer at receiving side
- Multi-Lane de-skew (for widths > x1) at receiving side

System Design For Testability (DFT) support features:

- Compliance Pattern (see § Section 4.2.9 , § Section 4.2.11 , and § Section 4.2.14)
- Modified Compliance Pattern (see § Section 4.2.10 , § Section 4.2.12 , and § Section 4.2.15)
- Jitter Measurement Pattern (see § Section 4.2.13 and § Section 4.2.16)
- Flit Error Injection

1.5.4.4 Inter-Layer Interfaces §

1.5.4.4.1 Transaction/Data Link Interface §

The Transaction to Data Link interface provides:

- Byte or multi-byte data to be sent across the Link
 - Local TLP-transfer handshake mechanism
 - TLP boundary information
- Requested power state for the Link

The Data Link to Transaction interface provides:

- Byte or multi-byte data received from the PCI Express Link
- TLP framing information for the received byte
- Actual power state for the Link
- Link status information

1.5.4.4.2 Data Link/Physical Interface §

The Data Link to Physical interface provides:

- Byte or multi-byte wide data to be sent across the Link

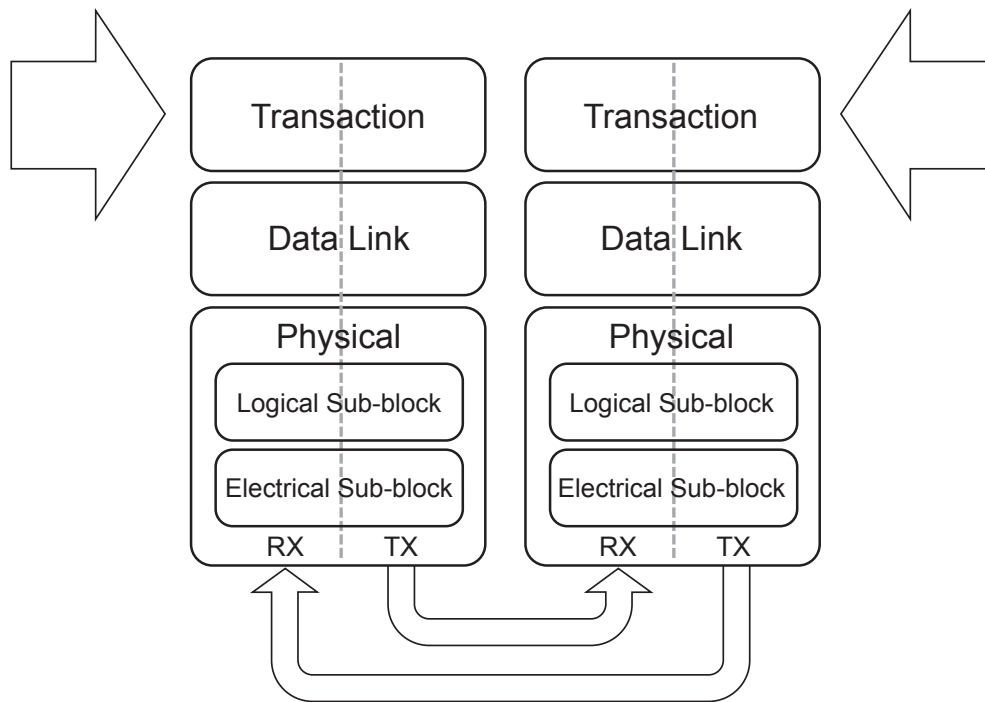
- Data transfer handshake mechanism
- TLP and DLLP boundary information for bytes
- Requested power state for the Link

The Physical to Data Link interface provides:

- Byte or multi-byte wide data received from the PCI Express Link
- TLP and DLLP framing information for data
- Indication of errors detected by the Physical Layer
- Actual power state for the Link
- Connection status information

2. Transaction Layer Specification §

2.1 Transaction Layer Overview §



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Figure 2-1 Layering Diagram Highlighting the Transaction Layer

At a high level, the key aspects of the Transaction Layer are:

- A pipelined full Split-Transaction protocol
- Mechanisms for differentiating the ordering and processing requirements of Transaction Layer Packets (TLPs)
- Credit-based flow control
- Optional support for data poisoning and end-to-end data integrity detection.

The Transaction Layer comprehends the following:

- TLP construction and processing
- Association of transaction-level mechanisms with device resources including:
 - Flow Control
 - Virtual Channel management
- Rules for ordering and management of TLPs
 - PCI/PCI-X compatible ordering

- Including Traffic Class differentiation

This chapter specifies the behaviors associated with the Transaction Layer.

2.1.1 Address Spaces, Transaction Types, and Usage §

Transactions form the basis for information transfer between a Requester and Completer. Four address spaces are defined, and different Transaction types are defined, each with its own unique intended usage, as shown in § Table 2-1.

Table 2-1 Transaction Types for Different Address Spaces

Address Space	Transaction Types	Basic Usage
Memory	Read Write	Transfer data to/from a memory-mapped location
I/O	Read Write	Transfer data to/from an I/O-mapped location
Configuration	Read Write	Device Function configuration/setup
Message	Baseline (including Vendor-Defined)	From event signaling mechanism to general purpose messaging

Details about the rules associated with usage of these address formats and the associated TLP formats are described later in this chapter.

2.1.1.1 Memory Transactions §

Memory Transactions include the following types:

- Read Request/Completion
- Write Request
- Deferrable Memory Write Request/Completion
- AtomicOp Request/Completion

Memory Transactions use two different address formats:

- Short Address Format: 32-bit address
- Long Address Format: 64-bit address

Certain Memory Transactions can optionally include a PASID TLP Prefix (Non-Flit Mode) or OHC (Flit Mode) containing the Process Address Space ID (PASID). See § Section 6.20 for details.

2.1.1.2 I/O Transactions §

PCI Express supports I/O Space for compatibility with legacy devices that require their use. Future revisions of this specification may deprecate the use of I/O Space. I/O Transactions include the following types:

- Read Request/Completion

- Write Request/Completion

I/O Transactions use a single address format:

- Short Address Format: 32-bit address

2.1.1.3 Configuration Transactions §

Configuration Transactions are used to access configuration registers of Functions within devices.

Configuration Transactions include the following types:

- Read Request/Completion
- Write Request/Completion

2.1.1.4 Message Transactions §

The Message Transactions, or simply Messages, are used to support in-band communication of events between devices.

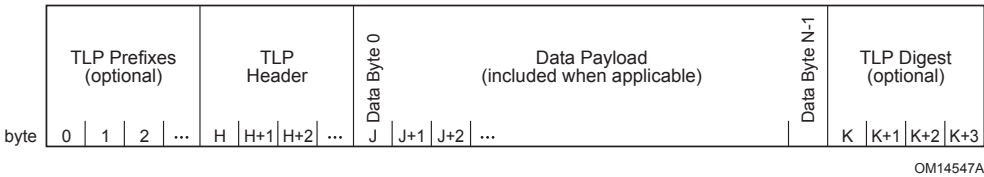
In addition to specific Messages defined in this document, PCI Express provides support for Vendor-Defined Messages using specified Message codes. Except for Vendor-Defined Messages that use the PCI-SIG® Vendor ID (0001h), the definition of specific Vendor-Defined Messages is outside the scope of this document.

This specification establishes a standard framework within which vendors can specify their own Vendor-Defined Messages tailored to fit the specific requirements of their platforms (see § Section 2.2.8.6).

Note that these Vendor-Defined Messages are not guaranteed to be interoperable with components from different vendors.

2.1.2 Packet Format Overview §

Transactions consist of Requests and Completions, which are communicated using packets. § Figure 2-2 shows a high level serialized view of a TLP, consisting of one or more optional TLP Prefixes, a TLP header, a data payload (for some types of packets), and an optional TLP Digest. § Figure 2-3 shows a more detailed view of the TLP. The following sections of this chapter define the detailed structure of the packet headers and digest.



§ Figure 2-2 Serial View of a TLP

PCI Express conceptually transfers information as a serialized stream of bytes as shown in § Figure 2-2. Note that at the byte level, information is transmitted/received over the interconnect with the left-most byte of the TLP as shown in § Figure 2-2 being transmitted/received first (byte 0 if one or more optional TLP Prefixes are present else byte H). Refer to § Section 4.2 for details on how individual bytes of the packet are encoded and transmitted over the physical media.

Detailed layouts of the TLP Prefix, TLP Header and TLP Digest (presented in generic form in § Figure 2-3) are drawn with the lower numbered bytes on the left rather than on the right as has traditionally been depicted in other PCI specifications. The header layout is optimized for performance on a serialized interconnect, driven by the requirement that the most time critical information be transferred first. For example, within the TLP header, the most significant byte of the address field is transferred first so that it may be used for early address decode.

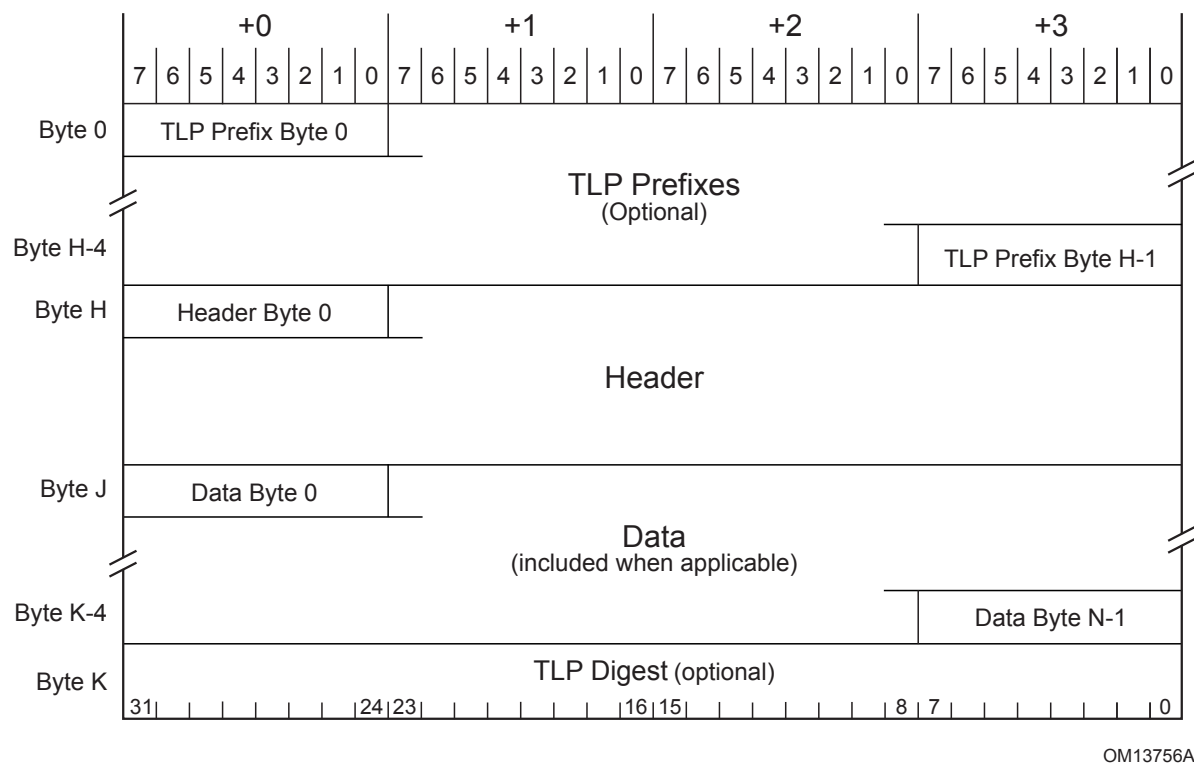


Figure 2-3 Generic TLP Format - Non-Flit Mode

The data payload within a TLP is depicted with the lowest addressed byte (byte J in § Figure 2-3) shown to the upper left. Detailed layouts depicting data structure organization (such as the Configuration Space depictions in § Chapter 7.) retain the traditional PCI byte layout with the lowest addressed byte shown on the right. Regardless of depiction, all bytes are conceptually transmitted over the Link in increasing byte number order.

Depending on the type of a packet, the header for that packet will include some of the following types of fields:

- Format of the packet
- Type of the packet
- Length for any associated data
- Transaction Descriptor, including:
 - Transaction ID
 - Attributes
 - Traffic Class
- Address/routing information
- Byte Enables
- Message encoding

- Completion status

2.2 Transaction Layer Protocol - Packet Definition §

PCI Express uses a packet based protocol to exchange information between the Transaction Layers of the two components communicating with each other over the Link. PCI Express supports the following basic transaction types: Memory, I/O, Configuration, and Messages. Two addressing formats for Memory Requests are supported: 32 bit and 64 bit.

Transactions are carried using Requests and Completions. Completions are used only where required, for example, to return read data, or to acknowledge Completion of I/O and Configuration Write Transactions. Completions are associated with their corresponding Requests by the value in the Transaction ID field of the Packet header.

All TLP fields marked Reserved (sometimes abbreviated as R) must be filled with all 0's when a TLP is formed. Values in such fields must be ignored by Receivers and forwarded unmodified by Switches. Note that for certain fields there are both specified and Reserved values - the handling of Reserved values in these cases is specified separately for each case.

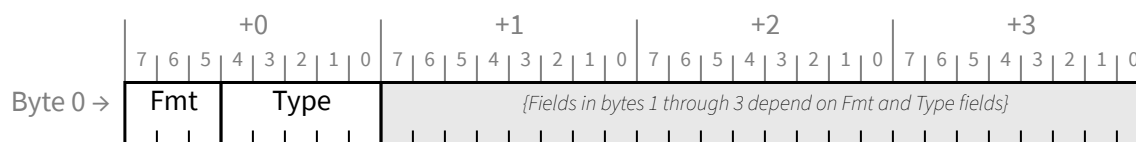
There are different header formats for Non-Flit Mode (NFM) and Flit Mode (FM). Routing elements must translate between the FM and NFM TLP formats when the Ingress Port and Egress Port are in different modes. In some cases translation is not possible, and the handling of such cases is also defined in this Chapter.

2.2.1 Common Packet Header Fields §

2.2.1.1 Common Packet Header Fields for Non-Flit Mode §

All TLP prefixes and headers contain the following fields (see § Figure 2-4):

- Fmt[2:0] - Format of TLP (see § Table 2-2) - bits 7:5 of byte 0
- Type[4:0] - Type of TLP - bits 4:0 of byte 0



§ Figure 2-4 Fields Present in All TLPs

The Fmt field(s) indicates the presence of one or more TLP Prefixes and the Type field(s) indicates the associated TLP Prefix type(s).

The Fmt and Type fields of the TLP Header provide the information required to determine the size of the remaining part of the TLP Header, and if the packet contains a data payload following the header.

The Fmt, Type, TD, and Length fields of the TLP Header contain all information necessary to determine the overall size of the non-prefix portion of the TLP. The Type field, in addition to defining the type of the TLP also determines how the TLP is routed by a Switch. Different types of TLPs are discussed in more detail in the following sections.

- Permitted Fmt[2:0] and Type[4:0] field values are shown in § Table 2-3.

- All other encodings are Reserved (see § Section 2.3).
- TC[2:0] - Traffic Class (see § Section 2.2.6.6) - bits [6:4] of byte 1
- R (byte 1 bit 1) - Reserved; formerly was the Lightweight Notification (LN) bit, but is now available for reassignment.
- TLP Hints (TH) - 1b indicates the presence of TLP Processing Hints (TPH) in the TLP header and optional TPH TLP Prefix (if present) - bit 0 of byte 1 (see § Section 2.2.7.1.1)
- Attr[1:0] - Attributes (see § Section 2.2.6.3) - bits [5:4] of byte 2
- Attr[2] - Attribute (see § Section 2.2.6.3) - bit 2 of byte 1 (shown as A2 in figures)
- TD - 1b indicates presence of TLP Digest in the form of a single Double Word (DW) at the end of the TLP (see § Section 2.2.3) - bit 7 of byte 2
- Error Poisoned (EP) - indicates the TLP is poisoned (see § Section 2.7) - bit 6 of byte 2
- Length[9:0] - Length of data payload in DW (see § Table 2-4) - bits 1:0 of byte 2 concatenated with bits 7:0 of byte 3
 - TLP data must be 4-byte naturally aligned and in increments of 4-byte DW.
 - Reserved for TLPs that do not contain or refer to data payloads, including Cpl, CplLk, and Messages (except as specified)

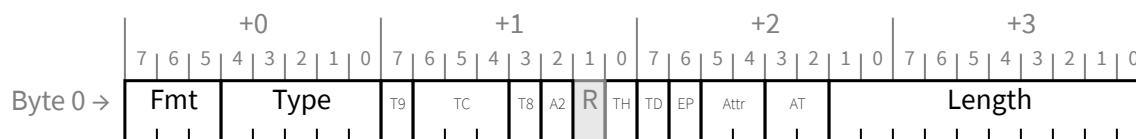


Figure 2-5 Fields Present in All Non-Flit Mode TLP Headers

Table 2-2 Fmt[2:0] Field Values

Fmt[2:0]	Corresponding TLP Format
000b	3 DW header, no data
001b	4 DW header, no data
010b	3 DW header, with data
011b	4 DW header, with data
100b	TLP Prefix
	All encodings not shown above are Reserved (see § Section 2.3).

Table 2-3 Fmt[2:0] and Type[4:0] Field Encodings

TLP Type	Fmt [2:0] 4 (b)	Type [4:0] (b)	Description
MRd	000 001	0 0000	Memory Read Request

TLP Type	Fmt [2:0] (b)	Type [4:0] (b)	Description
MRdLk	000 001	0 0001	Memory Read Request-Locked
MWr	010 011	0 0000	Memory Write Request
IORead	000	0 0010	I/O Read Request
IOWr	010	0 0010	I/O Write Request
CfgRd0	000	0 0100	Configuration Read Type 0
CfgWr0	010	0 0100	Configuration Write Type 0
CfgRd1	000	0 0101	Configuration Read Type 1
CfgWr1	010	0 0101	Configuration Write Type 1
TCfgRd	000	1 1011	Deprecated TLP Type ⁵
DMWr	010 011	1 1011	Deferrable Memory Write Request ⁶
Msg	001	1 0r ₂ r ₁ r ₀	Message Request - The sub-field r[2:0] specifies the Message routing mechanism (see § Table 2-20).
MsgD	011	1 0r ₂ r ₁ r ₀	Message Request with data payload - The sub-field r[2:0] specifies the Message routing mechanism (see § Table 2-20).
Cpl	000	0 1010	Completion without Data - Used for I/O, Configuration Write, and Deferrable Memory Write Completions with any Completion Status. Also used for AtomicOp Completions and Read Completions (I/O, Configuration, or Memory) with Completion Status other than Successful Completion.
CplD	010	0 1010	Completion with Data - Used for Memory, I/O, and Configuration Read Completions. Also used for AtomicOp Completions.
CplLk	000	0 1011	Completion for Locked Memory Read without Data - Used only in error case.
CplDLk	010	0 1011	Completion for Locked Memory Read - Otherwise like CplD.
FetchAdd	010 011	0 1100	Fetch and Add AtomicOp Request
Swap	010 011	0 1101	Unconditional Swap AtomicOp Request
CAS	010 011	0 1110	Compare and Swap AtomicOp Request
LPrfx	100	0 L ₃ L ₂ L ₁ L ₀	Local TLP Prefix - The sub-field L[3:0] specifies the Local TLP Prefix type (see § Table 2-38).
EPrfx	100	1 E ₃ E ₂ E ₁ E ₀	End-End TLP Prefix - The sub-field E[3:0] specifies the End-End TLP Prefix type (see § Table 2-39).

5. Deprecated TLP Types: previously used for Trusted Configuration Space (TCS), which is no longer supported by this specification. If a Receiver does not implement TCS, the Receiver must treat such Requests as Malformed Packets.

6. This TLP Type value was previously used for Trusted Configuration Space (TCS) Writes, which are no longer supported by this specification.

TLP Type	Fmt [2:0] (b)	Type [4:0] (b)	Description
			All encodings not shown above are Reserved (see § Section 2.3).

Table 2-4 Length[9:0] Field Encoding

Length[9:0]	Corresponding TLP Data Payload Size
00 0000 0001b	1 DW
00 0000 0010b	2 DW
...	...
11 1111 1111b	1023 DW
00 0000 0000b	1024 DW

2.2.1.2 Common Packet Header Fields for Flit Mode §

The TLP grammar is defined as:

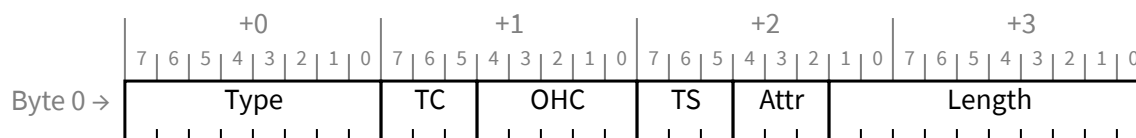
- zero or more 1DW Local Vendor-Defined TLP prefixes
- TLP Header Base with size indicated by Type[7:0] field, followed by zero to 7 DW of Orthogonal Header Content (OHC) as indicated by OHC[4:0] field
- TLP data payload of 0 to 1024DW
- TLP Trailer, if present as indicated by TS[2:0] field
- zero or more 1DW end-to-end suffixes ⁷

It is required to transmit NOP TLPs while TLP transmission is active if there are no other TLPs to transmit. NOP TLPs must be discarded without effect by the Receiver. All header fields other than the Type field are Reserved for NOP TLPs.

Other notable differences between Flit Mode and Non-Flit Mode TLPs include the following:

- Content that in Non-Flit Mode is included in End-to-End TLP prefixes is now incorporated within the header, as OHC.
- In Flit Mode, Steering Tags are not overloaded with the Byte Enables. The PH, Steering Tags, and AMA/AV fields are consolidated in OHC.

All Flit Mode TLPs contain the same fields in the first DW of the Header Base (see § Figure 2-6)



§ Figure 2-6 First DW of Header Base

7. Suffixes are defined as a placeholder for future use.

§ Table 2-5 defines the values for the Type[7:0] field for Flit Mode.

- The Type[7:0] field must be fully decoded by all Receivers regardless of which specific encodings are supported.
- All Receivers must handle Flow Control for all Type[7:0] field encodings as specified.
 - Switch Ports must buffer and route TLPs, including Reserved entries, as specified.
 - Endpoint Upstream Ports and Root Ports are required to buffer, including for Header Logging, up to the largest Header Base size plus all OHC content supported by the Port, but are permitted, after accounting for Flow Control, to discard Header Base and OHC content that is not supported by the Port.
- For all Reserved entries, TLP routing must be handled as indicated in the Description field, and the Header Base fields used for routing are at the same location within the Header as with the non-Reserved Header Base formats.
- Endpoint Upstream Ports and Root Ports must handle received TLPs using Reserved Type[7:0] encodings of FC Type PR and NPR as Unsupported Requests, and of FC Type CPL as Unexpected Completions.

In the Translation Rule column, an entry of “1:1” indicates that there is no change in meaning or behavior when translating between Non-Flit Mode and Flit Mode in either direction. TLPs that cannot be translated must result in a TLP Translation Egress Blocked error being reported by the Egress Port.

Table 2-5 Flit Mode TLP Header Type Encodings

#	Type								Description	Name	FC Type	Data Payload?	Header Base Size (DW)	New for Flit Mode	Translation Rule
	7	6	5	4	3	2	1	0							
0	0	0	0	0	0	0	0	0	No Operation	NOP	none	n	1	y	NFM uses this Type code for MRd (see #3)
1	0	0	0	0	0	0	0	1	Memory Read Request Locked, 32b address routed	MRdLk	NPR	n	3	n	1:1
2	0	0	0	0	0	0	1	0	IO Read Request	IORd	NPR	n	3	n	1:1
3	0	0	0	0	0	0	1	1	Memory Read Request, 32b address routed	MRd	NPR	n	3	y/n	Requires change of Type field value
4	0	0	0	0	0	1	0	0	Configuration Read Request, Type 0	CfgRd0	NPR	n	3	n	1:1
5	0	0	0	0	0	1	0	1	Configuration Read Request, Type 1	CfgRd1	NPR	n	3	n	1:1
6	0	0	0	0	0	1	1	0	Reserved – ID routed		CPL	Length	4	y	Block at NFM Egress – Fatal
7	0	0	0	0	0	1	1	1			CPL	Length	4	y	
8	0	0	0	0	1	0	0	0	Reserved – ID routed		PR	n	3	y	Block at NFM Egress – Fatal
9	0	0	0	0	1	0	0	1			PR	n	3	y	

#	Type								Description	Name	FC Type	Data Payload?	Header Base Size (DW)	New for Flit Mode	Translation Rule
	7	6	5	4	3	2	1	0							
10	0	0	0	0	1	0	1	0	Completion without Data	Cpl	CPL	n	3	n	1:1
11	0	0	0	0	1	0	1	1	Completion without Data, Locked (only for error cases)	CplLk	CPL	n	3	n	1:1
12-15	0	0	0	0	1	1	X	X	Reserved – ID routed		CPL	n	3	y	Block at NFM Egress – Fatal
16-19	0	0	0	1	0	0	X	X	Reserved – 64b address routed		NPR	Length	5	y	Block at NFM Egress – Fatal
20-21	0	0	0	1	0	1	0	X	Reserved – 64b address routed		NPR	Length	5	y	Block at NFM Egress – Fatal
22-23	0	0	0	1	0	1	1	X	Reserved – 64b address routed		NPR	Length	7	y	Block at NFM Egress – Fatal
24	0	0	0	1	1	0	0	0	Reserved – ID routed		CPL	n	7	y	Block at NFM Egress – Fatal
25	0	0	0	1	1	0	0	1			CPL	n	7	y	
26	0	0	0	1	1	0	1	0			CPL	Length	7	y	
27	0	0	0	1	1	0	1	1	Reserved – ID routed <i>{was: Trusted Configuration Read (deprecated)}</i>		CPL	Length	7	y	
28-29	0	0	0	1	1	1	0	X	Reserved – ID routed		NPR	n	3	y	Block at NFM Egress – Fatal
30-31	0	0	0	1	1	1	1	X	Reserved – ID routed		NPR	n	6	y	Block at NFM Egress – Fatal
32	0	0	1	0	0	0	0	0	Memory Read Request, 64b address routed	MRd	NPR	n	4	n	1:1
33	0	0	1	0	0	0	0	1	Memory Read Request Locked, 64b address routed	MRdLk	NPR	n	4	n	1:1
34-35	0	0	1	0	0	0	1	X	Reserved – 64b address routed		NPR	n	4	y	Block at NFM Egress – Fatal
36-39	0	0	1	0	0	1	X	X			NPR	n	4	y	
40-43	0	0	1	0	1	0	X	X	Reserved – ID routed		CPL	n	4	y	Block at NFM Egress Port – Fatal

#	Type								Description	Name	FC Type	Data Payload?	Header Base Size (DW)	New for Flit Mode	Translation Rule
	7	6	5	4	3	2	1	0							
44-45	0	0	1	0	1	1	0	X	Reserved – ID routed		PR	n	4	y	Block at NFM Egress Port – Fatal
46-47	0	0	1	0	1	1	1	X	Reserved – ID routed		PR	n	5	y	Block at NFM Egress Port – Fatal
48	0	0	1	1	0	0	0	0	Message w/o Data, Routed to Root Complex	Msg	PR	n	4	n	1:1
49	0	0	1	1	0	0	0	1	Message w/o Data, Routed by Address (64b) - NONE DEFINED	Msg	PR	n	4	n	1:1
50	0	0	1	1	0	0	1	0	Message w/o Data, Routed by ID	Msg	PR	n	4	n	1:1
51	0	0	1	1	0	0	1	1	Message w/o Data, Broadcast from Root Complex	Msg	PR	n	4	n	1:1
52	0	0	1	1	0	1	0	0	Message w/o Data, Local - terminate at Receiver	Msg	PR	n	4	n	1:1
53	0	0	1	1	0	1	0	1	Message w/o Data, Gathered and routed to RC (PME_TO_Ack)	Msg	PR	n	4	n	1:1
54	0	0	1	1	0	1	1	0	Message w/o Data -- RESERVED	Msg	PR	n	4	n	Terminate at FM Ingress Port
55	0	0	1	1	0	1	1	1		Msg	PR	n	4	n	
56-59	0	0	1	1	1	0	X	X	Reserved – 64b address routed		NPR	n	4	y	Block at NFM Egress – Fatal
60-61	0	0	1	1	1	1	0	X	Reserved – ID routed		NPR	n	4	y	
62-63	0	0	1	1	1	1	1	X	Reserved – ID routed		NPR	n	5	y	
64	0	1	0	0	0	0	0	0	Memory Write Request, 32b address routed	MWr	PR	Length	3	n	1:1
65	0	1	0	0	0	0	0	1	Reserved – ID routed		PR	Length	6	y	Block at NFM Egress – Fatal

#	Type								Description	Name	FC Type	Data Payload?	Header Base Size (DW)	New for Flit Mode	Translation Rule
	7	6	5	4	3	2	1	0							
66	0	1	0	0	0	0	1	0	IO Write Request	IOWr	NPR	Length	3	n	1:1
67	0	1	0	0	0	0	1	1	Reserved – ID routed		PR	Length	6	y	Block at NFM Egress – Fatal
68	0	1	0	0	0	1	0	0	Configuration Write Request Type 0,	CfgWr0	NPR	Length	3	n	1:1
69	0	1	0	0	0	1	0	1	Configuration Write Request Type 1,	CfgWr1	NPR	Length	3	n	1:1
70	0	1	0	0	0	1	1	0	Reserved – ID routed		NPR	Length	3	y	Block at NFM Egress – Fatal
71	0	1	0	0	0	1	1	1			NPR	Length	3	y	
72	0	1	0	0	1	0	0	0	Reserved – ID routed		CPL	Length	3	y	Block at NFM Egress – Fatal
73	0	1	0	0	1	0	0	1			CPL	Length	3	y	
74	0	1	0	0	1	0	1	0	Completion with Data	CplD	CPL	Length	3	n	1:1
75	0	1	0	0	1	0	1	1	Completion with Data, Locked	CplDLk	CPL	Length	3	n	1:1
76	0	1	0	0	1	1	0	0	Fetch and Add AtomicOp Request, 32b address routed	FetchAdd	NPR	Length	3	n	1:1
77	0	1	0	0	1	1	0	1	Unconditional Swap AtomicOp Request, 32b address routed	Swap	NPR	Length	3	n	1:1
78	0	1	0	0	1	1	1	0	Compare and Swap AtomicOp Request, 32b address routed	CAS	NPR	Length	3	n	1:1
79	0	1	0	0	1	1	1	1	Reserved – 64b address routed		PR	n	4	y	Block at NFM Egress – Fatal
80-83	0	1	0	1	0	0	X	X	Reserved – 64b address routed		NPR	Length	6	y	Block at NFM Egress Port – Fatal
84-85	0	1	0	1	0	1	0	X	Reserved – 64b address routed		NPR	Length	6	y	
86-87	0	1	0	1	0	1	1	X	Reserved – 64b address routed		NPR	Length	7	y	
88-89	0	1	0	1	1	0	0	X	Reserved – ID routed		PR	Length	3	y	Block at NFM Egress – Fatal

#	Type								Description	Name	FC Type	Data Payload?	Header Base Size (DW)	New for Flit Mode	Translation Rule
	7	6	5	4	3	2	1	0							
90	0	1	0	1	1	0	1	0	Reserved – 64b address routed		PR	Length	4	y	
91	0	1	0	1	1	0	1	1	Deferrable Memory Write Request, 32b address routed {was: Trusted Configuration Write (deprecated)}	DMWr	NPR	Length	3	n	1:1
92-93	0	1	0	1	1	1	0	X	Reserved – ID routed		PR	Length	4	y	Block at NFM Egress – Fatal
94-95	0	1	0	1	1	1	1	X	Reserved – ID routed		PR	Length	5	y	
96	0	1	1	0	0	0	0	0	Memory Write Request, 64b address routed	MWr	PR	Length	4	n	1:1
97	0	1	1	0	0	0	0	1	Reserved – 64b address routed		PR	Length	4	y	Block at NFM Egress – Fatal
98-99	0	1	1	0	0	0	1	X			PR	Length	4	y	
100-103	0	1	1	0	0	1	X	X			PR	Length	4	y	
104-107	0	1	1	0	1	0	X	X	Reserved – 64b address routed		PR	Length	4	y	Block at NFM Egress – Fatal
108	0	1	1	0	1	1	0	0	Fetch and Add AtomicOp Request, 64b address routed	FetchAdd	NPR	Length	4	n	1:1
109	0	1	1	0	1	1	0	1	Unconditional Swap AtomicOp Request, 64b address routed	Swap	NPR	Length	4	n	1:1
110	0	1	1	0	1	1	1	0	Compare and Swap AtomicOp Request, 64b address routed	CAS	NPR	Length	4	n	1:1
111	0	1	1	0	1	1	1	1	Reserved – 64b address routed		NPR	Length	4	y	Block at NFM Egress – Fatal
112	0	1	1	1	0	0	0	0	Message with Data, Routed to Root Complex	MsgD	PR	Length	4	n	1:1
113	0	1	1	1	0	0	0	1	Message with Data, Routed by	MsgD	PR	Length	4	n	1:1

#	Type								Description	Name	FC Type	Data Payload?	Header Base Size (DW)	New for Flit Mode	Translation Rule
	7	6	5	4	3	2	1	0							
									Address (64b) - NONE DEFINED						
114	0	1	1	1	0	0	1	0	Message with Data, Routed by ID	MsgD	PR	Length	4	n	1:1
115	0	1	1	1	0	0	1	1	Message with Data, Broadcast from Root Complex	MsgD	PR	Length	4	n	1:1
116	0	1	1	1	0	1	0	0	Message with Data, Local - terminate at Receiver	MsgD	PR	Length	4	n	1:1
117	0	1	1	1	0	1	0	1	Message with Data, Gathered and routed to RC (MsgD NOT USED)	MsgD	PR	Length	4	n	1:1
118	0	1	1	1	0	1	1	0	Message with Data -- RESERVED	MsgD	PR	Length	4	n	Terminate at FM Ingress Port
119	0	1	1	1	0	1	1	1		MsgD	PR	Length	4	n	
120	0	1	1	1	1	0	0	0	Reserved – 64b address routed		NPR	Length	4	y	Block at NFM Egress – Fatal
121	0	1	1	1	1	0	0	1			NPR	Length	4	y	
122	0	1	1	1	1	0	1	0			NPR	Length	4	y	
123	0	1	1	1	1	0	1	1	Deferrable Memory Write Request, 64b address routed	DMWr	NPR	Length	4	n	1:1
124-127	0	1	1	1	1	1	X	X	Reserved – 64b address routed		NPR	Length	4	y	Block at NFM Egress – Fatal
128-135	1	0	0	0	0	X	X	X	Reserved – Local Suffix		none	n	1	n	Terminate at FM Ingress Port Error if follows NOP
136-139	1	0	0	0	1	0	X	X	Reserved – Local		none	n	1	n	Terminate at FM Ingress Port
140	1	0	0	0	1	1	0	0			none	n	1	n	
141	1	0	0	0	1	1	0	1	Flit Mode Local TLP Prefix	FlitModePrefix	none	n	1	n	

#	Type								Description	Name	FC Type	Data Payload?	Header Base Size (DW)	New for Flit Mode	Translation Rule
	7	6	5	4	3	2	1	0							
142	1	0	0	0	1	1	1	0	1 DW Prefix-Vendor Defined Local 0	VendPrefixL0	none	n	1	n	Terminate at FM Ingress Port Vendor-Defined behavior
143	1	0	0	0	1	1	1	1	1 DW Prefix - Vendor Defined Local 1	VendPrefixL1	none	n	1	n	Terminate at FM Ingress Port Vendor-Defined behavior
144-147	1	0	0	1	0	0	X	X	Reserved – 64b address routed		PR	n	4	y	Terminate at FM Ingress Port
148-151	1	0	0	1	0	1	X	X	Reserved – 64b address routed		PR	n	5	y	
152-155	1	0	0	1	1	0	X	X	Reserved – 64b address routed		PR	n	6	y	Terminate at FM Ingress Port
156-159	1	0	0	1	1	1	X	X	Reserved – 64b address routed		PR	n	7	y	
160-167	1	0	1	0	0	X	X	X	Reserved – 64b address routed		NPR	n	5	y	Block at NFM Egress – Fatal
168-169	1	0	1	0	1	0	0	X	Reserved – ID routed		PR	n	6	y	Block at NFM Egress – Fatal
170-171	1	0	1	0	1	0	1	X	Reserved – ID routed		PR	n	7	y	Block at NFM Egress – Fatal
172-173	1	0	1	0	1	1	0	X	Reserved – ID routed		CPL	n	5	y	Block at NFM Egress – Fatal
174-175	1	0	1	0	1	1	1	X	Reserved – ID routed		CPL	n	6	y	Block at NFM Egress – Fatal
176-183	1	0	1	1	0	X	X	X	Reserved – 64b address routed		PR	Length	5	y	Block at NFM Egress – Fatal
184-191	1	0	1	1	1	X	X	X	Reserved – 64b address routed		PR	Length	5	y	Block at NFM Egress – Fatal
192-199	1	1	0	0	0	X	X	X	Reserved – 64b address routed		NPR	n	6	y	Block at NFM Egress – Fatal
200-201	1	1	0	0	1	0	0	X	Reserved – ID routed		NPR	n	7	y	Block at NFM Egress – Fatal
202-203	1	1	0	0	1	0	1	X	Reserved – ID routed		CPL	Length	5	y	Block at NFM Egress – Fatal
204-205	1	1	0	0	1	1	0	X	Reserved – ID routed		CPL	Length	6	y	Block at NFM Egress – Fatal

#	Type								Description	Name	FC Type	Data Payload?	Header Base Size (DW)	New for Flit Mode	Translation Rule
	7	6	5	4	3	2	1	0							
206-207	1	1	0	0	1	1	1	X	Reserved – ID routed		PR	Length	7	y	Block at NFM Egress – Fatal
208-215	1	1	0	1	0	X	X	X	Reserved – 64b address routed		PR	Length	6	y	Block at NFM Egress – Fatal
216-223	1	1	0	1	1	X	X	X	Reserved – 64b address routed		PR	Length	6	y	Block at NFM Egress – Fatal
224-225	1	1	1	0	0	0	0	X	Reserved – Local routed (terminate at receiver)		none	n	4	y	Terminate at FM Ingress Port
226-227	1	1	1	0	0	0	1	X	Reserved – Local routed (terminate at receiver)		none	n	6	y	Terminate at FM Ingress Port
228-229	1	1	1	0	0	1	0	X	Reserved – Local routed (terminate at receiver)		none	Length	4	y	Terminate at FM Ingress Port
230-231	1	1	1	0	0	1	1	X	Reserved – Local routed (terminate at receiver)		none	Length	6	y	Terminate at FM Ingress Port
232-239	1	1	1	0	1	X	X	X	Reserved – 64b address routed		NPR	n	7	y	Block at NFM Egress – Fatal
240-241	1	1	1	1	0	0	0	X	Reserved – ID routed		NPR	Length	4	y	Block at NFM Egress – Fatal
242-243	1	1	1	1	0	0	1	X	Reserved – ID routed		NPR	Length	5	y	Block at NFM Egress – Fatal
244-245	1	1	1	1	0	1	0	X	Reserved – ID routed		NPR	Length	6	y	Block at NFM Egress – Fatal
246-247	1	1	1	1	0	1	1	X	Reserved – ID routed		NPR	Length	7	y	Block at NFM Egress – Fatal
248-255	1	1	1	1	1	X	X	X	Reserved – 64b address routed		PR	Length	7	y	Block at NFM Egress – Fatal

The TS[2:0] field indicates Trailer Size and use encoded as:

- 000b – No Trailer
- 001b – 1DW Trailer containing ECRC

- 010b – 1DW Trailer – Content Reserved
- 011b – 2DW Trailer – Content Reserved
- 100b – 2DW Trailer – Content Reserved
- 101b – 3DW Trailer with IDE MAC if and only if OHC-C present and indicates IDE TLP; Else 3DW Trailer – Content Reserved
- 110b – 4DW Trailer with IDE MAC and PCRC if and only if OHC-C present and indicates IDE TLP; Else 4DW Trailer – Content Reserved
- 111b – 5DW Trailer – Content Reserved

The definitions of the TC, Attr and Length fields in Flit Mode are the same as in Non-Flit Mode.

Bit 1 in byte 1 of Non-Flit Mode is now Reserved, but it was the LN bit associated with the now deprecated Lightweight Notification (LN) protocol. This bit is not supported in Flit Mode. Thus, it must be ignored when translating from Non-Flit Mode to Flit Mode, and it must be set to 0b when translating from Flit Mode to Non-Flit Mode.

The OHC[4:0] field indicates the presence of “Orthogonal Header Content” (OHC) encoded as:

- 0 0000b = No OHC present
- x xxx1b = OHC-A present
- x xx1xb = OHC-B present
- x x1xxb = OHC-C present
- 0 0xxxb = No OHC-Ex present
- 0 1xxxb = OHC-E1 present
- 1 0xxxb = OHC-E2 present
- 1 1xxxb = OHC-E4 present

When present, OHC must follow the Header Base. It is permitted for any combination of OHC content to be present, but, when present, must follow the Header Base, in A-B-C-E order. The contents of the OHC in some cases varies depending on the TLP type.

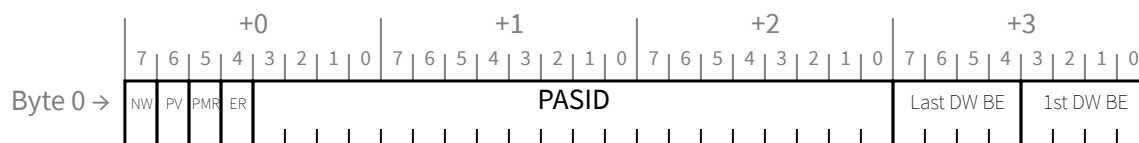
For specific TLP types, as defined in this specification, specific OHC content must be included by the Transmitter. Receivers must check for violations of these rules. If a Receiver determines that a Request violates a rule requiring specific OHC content, the Request must be handled as an Unsupported Request. If a Receiver determines that a Completion violates a rule requiring specific OHC content, the Completion must be handled as an Unexpected Completion.

*Table 2-6 **OHC-A** Included Fields for OHC-A1 through OHC-A5 (see § Figure 2-7 through § Figure 2-11)*

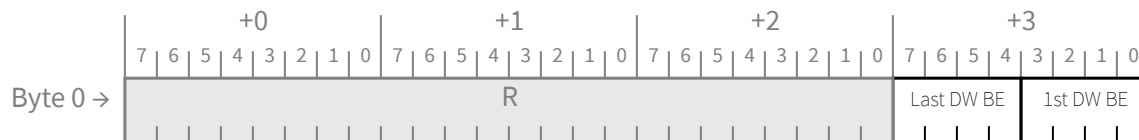
Name	Required for	Byte Enables	PASID, ER, PMR	Destination Segment, DSV	Completer Segment	Completion Status	Lower Address[1:0]	NW Flag
<u>OHC-A1</u>	Memory Requests with explicit Byte Enables and/or PASID, Translation Requests and Address Routed Messages with PASID	Y	Y					Y
<u>OHC-A2</u>	IO Requests	Y						
<u>OHC-A3</u>	Configuration Requests	Y		Y				

Name	Required for	Byte Enables	PASID, ER, PMR	Destination Segment, DSV	Completer Segment	Completion Status	Lower Address[1:0]	NW Flag
<u>OHC-A4</u>	ID-Routed Messages that require Destination Segment and/or PASID		Y	Y				
<u>OHC-A5</u>	Completions when required as defined in § Section 2.2.9.2.			Y	Y	Y	Y	

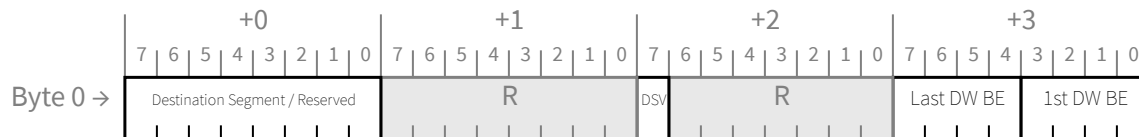
In OHC-A1, the ER bit is Execute Requested, and the PMR bit is Privileged Mode Requested (see § Section 6.20). When OHC-A1 is included with a TLP, if the PASID is not known or has not been assigned, then the PV ("PASID Valid") bit must be Clear. The ER and PMR bits are Reserved if PV is Clear. The NW bit is No Write (NW), for Translation Requests only, and for otherwise Reserved.



§

Figure 2-7 **OHC-A1**

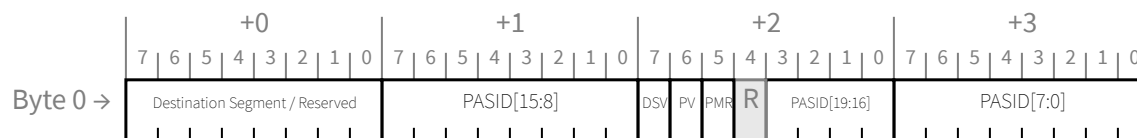
§

Figure 2-8 **OHC-A2**

§

Figure 2-9 **OHC-A3**

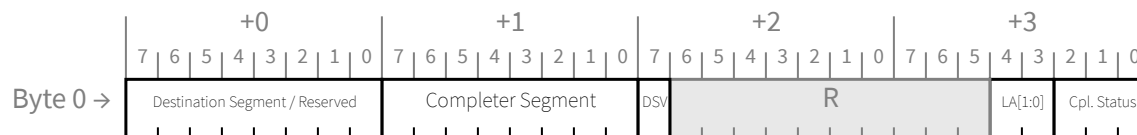
In OHC-A4, the PMR bit is Privileged Mode Requested (see § Section 6.20). When OHC-A4 is included with a TLP, if the PASID is not known or has not been assigned, then the PV ("PASID Valid") bit must be Clear. The PMR bit is Reserved if PV is Clear.



§

Figure 2-10 OHC-A4

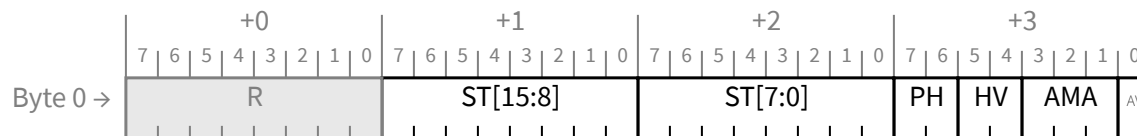
In OHC-A5, LA[1:0] is Lower Address[1:0].



§

Figure 2-11 OHC-A5

When TLP Processing Hints (TPH) are used (applicable only for Memory Address Routed Request TLPs) OHC-B must be included with the appropriate PH, ST, AMA and AV values.



§

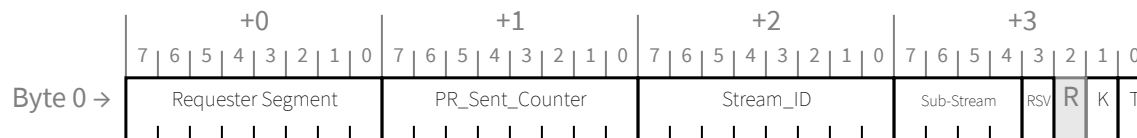
Figure 2-12 OHC-B

OHC-B includes the PH and ST fields, qualified by the HV[1:0] ("Hints Valid") field, defined as:

- 00b: PH[1:0], ST[15:0] not valid – fields Reserved
- 01b: PH[1:0] & ST[7:0] valid, ST[15:8] not valid & Reserved
- 10b: Reserved
- 11b: PH[1:0] & ST[15:0] valid

The AV qualifies AMA[2:0] independently of HV[1:0].

In Flit Mode, IDE TLPs must include OHC-C as shown in § Figure 2-13. Non-IDE TLPs will, in some cases, also include OHC-C to indicate the Requester Segment (see Segment Rules below). Compared with the IDE Prefix used in Non-Flit Mode, OHC-C does not include the M and P bits because the presence of a MAC/PCRC is indicated using the TS field. When the Requester Segment field is valid the RSV bit must be Set.



§

Figure 2-13 OHC-C

Because IDE TLPs cannot be modified between the two Partner Ports, the IDE Partner Ports and the path between them must operate entirely in Non-Flit Mode or in Flit mode. Root Complexes that support peer-to-peer and Switches cannot modify IDE TLPs associated with Flow-Through Selective IDE Streams, making TLP Translation impossible. If an IDE TLP is directed out an Egress Port operating in a different mode from the Ingress Port, the IDE TLP must be dropped and the result must be reported as a Misrouted IDE TLP error.

It is permitted to configure a Root Complex or Switch such that the Ingress Port is a terminus for an IDE connection and the Egress Port another terminus, such that the TLP is passed through the RC/Switch unprotected by IDE. Doing this requires that the RC/Switch to be trusted, and requires the Root/Switch Ports to have the ability to act as an IDE Terminus, not simply to support Flow-Through IDE.

For IDE Completion TLPs, the Requester Segment field is Reserved and the RSV bit must be Clear.

In Flit Mode, NOP TLPs must never be transmitted as IDE TLPs. Receivers are not required to check for violations of this rule, but, if checked, Receivers must handle NOP TLPs received as IDE TLPs as Malformed TLPs.

Segment Rules:

In Flit Mode, it is possible, and in some cases required, to include Segment fields in TLPs. One benefit of the Segment fields is to enable routing Route-by-ID TLPs between Hierarchies, which are, by definition, in different Segments. Root Complexes are the only place where peer-to-peer Requests will traverse from one Hierarchy to another. Peer-to-peer Route-by-ID Message Requests can traverse Hierarchies when the Requester includes a valid Destination Segment field. Memory Requests are address routed between Hierarchies, but the associated Completions are ID routed. To aid in Root Complex routing of Completions between Hierarchies, FM Completions can include the Destination Segment field which reflects the value of the Requester Segment field from the associated NP Memory Request. This allows a Root Complex to route Non-Posted Memory Requests between Hierarchies without the need to assume ownership of each outstanding transaction for the purpose of routing the associated Completions back to the Hierarchy of the original Requester. This can lead to performance improvements for peer-to-peer transfers between Hierarchies.

A second use of the Segment fields is to improve error logging. When FM TLP headers are logged in the AER Capability structure the Segment fields will be included. The Segment fields improve traceability when identical Requester/Completer IDs exist in different Hierarchies. The rules in this section allow the Segment fields to be omitted in some cases to reduce FM TLP overhead. It should be noted that omitting the Segment fields in these cases could forfeit the improved error-logging traceability benefit. It is permitted to use implementation specific mechanisms to select when optional Segment fields are included (e.g., during debug) while still achieving optimal performance during normal operation by omitting non-required Segment fields.

These fields, which exist only in FM, are used to communicate Segment information:

- The Requester Segment field indicates the Hierarchy in which the Requester is located. This field exists in OHC-C and is sometimes included in Memory and Message Requests.
 - The Requester Segment Valid (RSV) bit, when Set, indicates that the Requester Segment field is valid.
 - When Requester Segment Valid (RSV) is Clear then the Requester Segment field is Reserved.
 - For TLPs with OHC-C that are not IDE TLPs, the Sub-Stream[3:0] field must be 0111b, and the Stream ID, PR_Sent_Counter, K and T fields/bits are Reserved.

- In Flit Mode, IDE Requests (see § Section 6.33) other than Configuration Requests must include Requester Segment in OHC-C.
- The Completer Segment field indicates the Hierarchy in which the Completer is located. This field exists in OHC-A5 and is sometimes included in Completions.
- The Destination Segment field indicates the Hierarchy to which the TLP should be routed for ID Based Routing. In Configuration Write Requests this field is also used to configure the Segment of the completing Function. Configuration Requests in FM always include this field in OHC-A3 unless the Request had previously traversed a NFM Link. Route-by-ID Message Requests sometimes include this field in OHC-A4. Completions sometimes include this field in OHC-A5.
 - The Destination Segment Valid (DSV) bit, when Set, indicates that the Destination Segment field is valid.
 - When Destination Segment Valid (DSV) is Clear then the Destination Segment field is Reserved.

In addition to the following rules that apply specifically to Root Complexes, Requesters and Completers within Root Complexes must also follow the rules later in this section that apply to Requesters and Completers.

- All Configuration Requests transmitted by a Root Port in Flit Mode, including those initiated through the SFI Configuration Access Method, must include OHC-A3 with the DSV bit set and a valid Destination Segment. The Destination Segment is necessary for the Completer to capture its Segment as described in § Section 2.2.6.2
 - The Root Complex must indicate the correct Segment value in the Destination Segment field, even if only one Segment is implemented.
- Completions associated with Configuration Requests must be identifiable solely by Transaction ID when received at a RP. Such Completions will not include a Destination Segment field because Configuration Requests do not include a Requester Segment field.

IMPLEMENTATION NOTE: ROOT COMPLEX SUPPORT FOR PEER-TO-PEER NON-POSTED MEMORY TRANSACTIONS THAT TRAVERSE HIERARCHIES §

Because Segment fields aren't communicated across Links in NFM, Root Complexes take on additional burden for peer-to-peer NP Memory Requests that cross from one Hierarchy to another. With the loss of the Requester Segment field when a Request is translated to NFM, the Requester ID that remains in the original NP Memory Request might be indistinguishable from that of other Requesters within the hierarchy domain. Unless all Links along the path from the egress RP to the Completer are known to be in FM, Root Complexes must replace the Requester ID in peer-to-peer NP Memory Requests that cross from one Hierarchy to another. The Requester ID supplied by the Root Complex must be an ID associated with the Root Complex itself. This action is sometimes called "taking ownership" of the NP Request. It is necessary for the Root Complex to take ownership of such Requests to ensure that the Requester ID remains unique within the hierarchy domain of the egress RP, and that the associated Completions can be routed correctly by any Switches within that hierarchy domain. The egress RP also must track all such outstanding NP Memory Requests in order to route the associated Completion(s) to the Hierarchy of the original Requester within the Root Complex, as well as to restore the original Requester ID (Destination BDF/BF in FM) within the Completion(s).

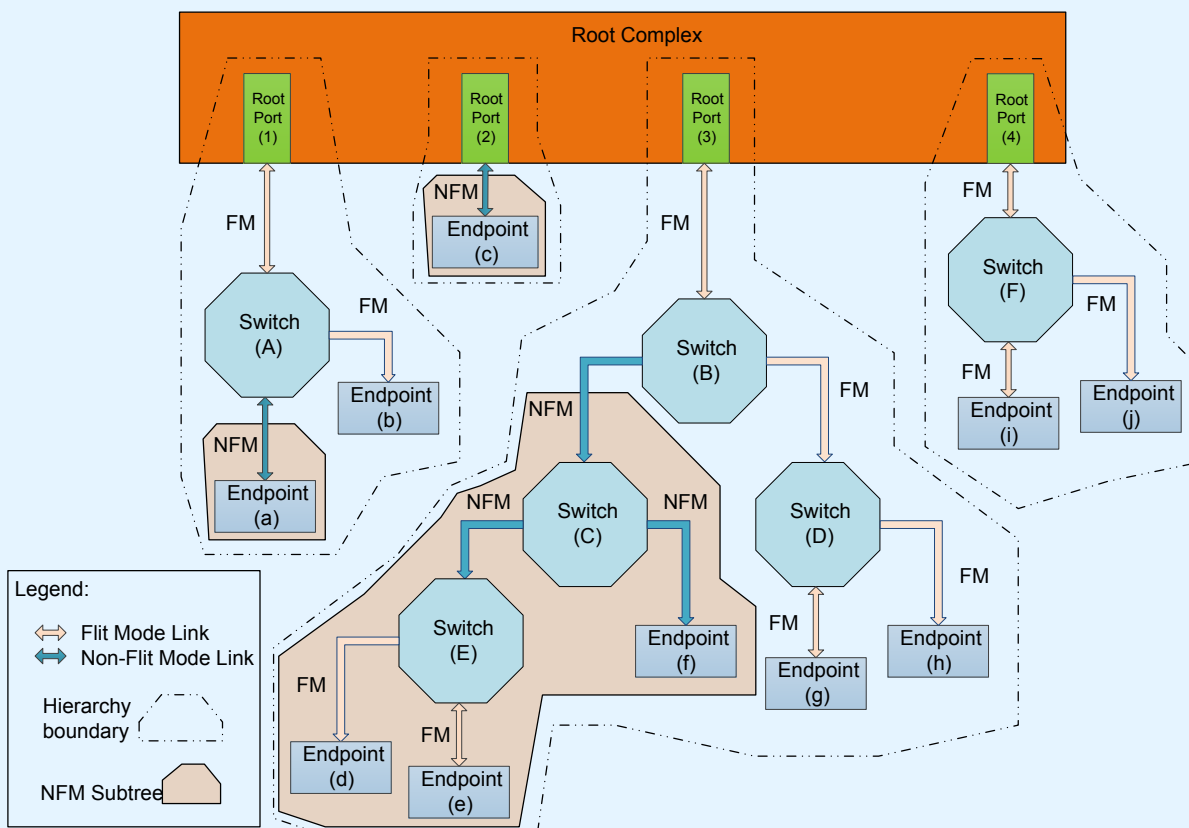


Figure 2-14 Example Topology Illustrating Multiple Segments and NFM Subtrees

The No NFM Subtree Below This Root Port bit defaults to Clear to indicate that one or more NFM subtree(s) may exist below a Root Port. Referring to the example shown in § Figure 2-14, each Root Port is in a unique Segment/Hierarchy, and Root Ports 1 through 3 have NFM subtrees below the Root Port. For RP 2, the Link immediately below the RP is in NFM, but for RP 1 and 3 the Root Port cannot directly determine the existence of a NFM subtree within its hierarchy domain, and so the default value of the No NFM Subtree Below This Root Port bit ensures that the Root Complex will take ownership of NP Requests Egressing from those Root Ports. In all cases, it is necessary that system software ensure the No NFM Subtree Below This Root Port bit for a Root Port is Clear in cases where the Root Port has one or more NFM Links or subtrees below it.

However, Root Port 4 does not have any NFM Links below it, and therefore it is not necessary for the Root Complex to take ownership of NP Requests Egressing that Root Port. It is strongly recommended that system software Set the No NFM Subtree Below This Root Port bit in such cases, and it is strongly recommended that Root Complex implementations use the value in the No NFM Subtree Below This Root Port bit to avoid taking ownership of NP Requests when it is not necessary to do so.

Note that for non-IDE Requests directed Upstream to the RC, the existence of a NFM Link between the original Requester and the Root Port is not a factor, because the RC inherently knows the Hierarchy of the Requester based on the Ingress RP of the Request, and can add the Requester Segment if needed.

Regardless of the value of the No NFM Subtree Below This Root Port bit, a Root Complex need not apply NP Memory Request tracking mechanisms for peer-to-peer Selective IDE Stream transactions that cross from one Hierarchy to another, and IDE TLPs cannot in any case be modified between the two IDE Partner Ports. When a Selective IDE Stream is established passing peer-to-peer between Hierarchies, software must ensure that the RC supports such routing, and that the entire path between the two Partner Ports is entirely in FM.

A NFM device could be hot-added into a subtree for which the No NFM Subtree Below This Root Port bit had previously been Set. In such cases it is necessary for system software to Clear the No NFM Subtree Below This Root Port bit prior to allowing the hot-added NFM device to act as a Completer for any NP Memory Request passing peer-to-peer through the RC.

- It is not permitted to configure a Selective IDE Stream passing peer-to-peer between different Hierarchies unless it is known that the RC supports flow-through IDE between the two Root Ports, and that all Links on the path between the two Partner Ports, including both the Root Ports, are in FM.
 - Root Complexes are not required to support Selective IDE Streams passing peer-to-peer through the RC.
 - If a condition exists that precludes the RC from passing an IDE TLP associated with a Selective IDE Stream configured to flow-through the RC, then the RC must treat the TLP as a Misrouted IDE TLP error at either the Ingress Port or the Egress Port.
- If a Message or Memory Request received at a RP includes a Requester Segment that does not match the Hierarchy associated with the receiving RP, the Request must be handled by the RP as an Unsupported Request.
- A RP is permitted to add a Requester Segment indication to a non-IDE Memory Write Request, or a non-IDE Route by Address Message Request, passing peer-to-peer through the RC if that TLP did not include a Requester Segment at the ingress RP, where the Requester Segment must correspond to the Segment of the Ingress RP.
- Route by ID Message Requests received at a RP without a Destination Segment, or received in NFM, are implied to be destined for a Completer within the same Hierarchy as the Ingress RP.
- When taking ownership of a NP Memory Request passing peer-to-peer through the RC:
 - The Requester ID in the Request must be replaced with one associated with the Root Complex.

- The Request must either use the Requester Segment value associated with the hierarchy domain of the Egress RP, or must not include a Requester Segment.
- The RC is permitted to replace the Tag in the Request, and must ensure the Transaction ID is unique among outstanding NP Requests associated with the same Requester ID used for taking ownership.
 - The RC is permitted to change the size of the Tag. If this is done, it is permitted to use implementation specific means to determine what size of tag is appropriate.
 - The Tag in the Completion(s) must be restored to the Tag from the original Request, as received at the Ingress RP, before returning those Completion(s).
- Completions associated with the Request must be identifiable solely by Transaction ID when received at the RP. Such Completions will not include a Destination Segment if the RP did not include a Requester Segment in the Request or if a NFM Link exists between the RP and the Completer.
- The Requester ID value in the Completion(s) must be restored to the Requester ID from the original Request, as received at the Ingress RP, before returning those Completion(s).
- If the RP that received the Request is in FM and OHC-A5 is returned to the Requester with the Completion(s):
 - The Destination Segment must be set to 00h and the DSV bit must be clear in OHC-A5 that is returned to the original Requester.
 - The Completer Segment field must not be modified if the RP receiving the Completion is in FM and OHC-A5 was received with the Completion. The Completer Segment in OHC-A5 returned to the Requester must be set to 00h if the RP receiving the Completion is in NFM or if OHC-A5 was not received with the Completion.
- When passing a NP Memory Request peer-to-peer through the RC without taking ownership:
 - The Requester ID and Tag in the Request must not be modified.
 - For non-IDE NP Memory Requests passing peer-to-peer through the RC that do not include a Requester Segment at the Ingress RP, the RC must add a Requester Segment indication at the Egress RP, using the Segment value associated with the Ingress RP.
 - Any Completion received with the DSV bit set and a Destination Segment not matching the value associated with the hierarchy domain of the receiving RP must be routed through the RC to the specified Hierarchy.
 - The Requester ID and Tag fields returned to the Requester must not be modified from the values received with the Completion in the destination hierarchy domain.
 - If the RP that received the Request is in FM and OHC-A5 is returned to the Requester with the Completion(s) the DSV bit, Destination Segment, and Completer Segment fields must not be modified from the values received with OHC-A5 in the destination hierarchy domain.

There are specific cases where a RP is not required to include Segment information:

- A RP is not required to include the Requester Segment field in any non-IDE Memory Request initiated by a Requester within the Root Complex.
- A RP is not required to include a Requester Segment field with Memory Write Requests passing peer-to-peer through the RC.
- A RP is not required to include a Requester Segment field with NP Memory Requests passing peer-to-peer through the RC if the Egress RP is taking ownership of the Request.
- A RP is not required to include the Completer Segment or Destination Segment fields in Completions associated with NP Memory Requests targeting system memory or another element of the Root Complex itself. OHC-A5 must be included if required as described in § Section 2.2.9.2 .

- A RP is not required to include the Completer Segment or Destination Segment fields in Completions associated with NP Memory Requests passing peer-to-peer through the RC. OHC-A5 must be included if required as described in § Section 2.2.9.2 .

Each Switch exists entirely within a single Hierarchy by definition. However, Switches are required to comprehend Segment fields in some TLP types for routing purposes. The following rules apply to Switches:

- For TLPs in FM for both the Ingress and Egress Ports, Switches must never modify, add, or remove any Segment field or the DSV/RSV bit(s) within the TLP.
- For Configuration Requests initiated in FM through the SFI Configuration Access Method on a Switch Downstream Port, the Destination Segment and DSV fields must reflect the values received in the associated Configuration Write or Read Request to the SFI CAM Data Register.
- A Switch for which Segment Captured is Set must handle as a TLP Translation Egress Blocked Error a NP Memory Request received at the Upstream Port destined for a Downstream Port in NFM that includes a Requester Segment that does not match the Switch's captured Segment value.
 - The Request must not be forwarded to the Downstream Port.
- If a condition exists that precludes the Switch from passing an IDE TLP associated with a Selective IDE Stream configured to flow-through the Switch without modification, then the Switch must handle the TLP as a Misrouted IDE TLP error at either the Ingress Port or the Egress Port.
- When a Switch must translate a TLP from NFM to FM:
 - If Segment Captured is Clear, OHC-C must not be added to a Request.
 - If Segment Captured is Set, the Switch is permitted to add OHC-C to Memory and Message Requests with the Requester Segment containing the value established when the Switch itself was configured.
 - OHC-C must not be added to Configuration Requests.
 - If any OHC-A with DSV and Destination Segment fields is added, the DSV bit must be Clear and the Destination Segment must be 00h.
 - For a Completion that requires OHC-A5 (see § Section 2.2.9.2),
 - if Segment Captured is Set, then the Switch must apply in the Completer Segment field the Segment value established when the Switch itself was configured,
 - if Segment Captured is Clear, then the Switch must apply in the Completer Segment field the value 00h.
- Switches must route Configuration Requests solely by the BDF fields (Destination BDF/BF in FM); the Destination Segment field must not be considered for routing.
- A Switch for which Segment Captured is Set must route Completions and Route by ID Message Requests Upstream if DSV Set and the Destination Segment does not match the Switch's captured Segment value.
- Completions and Route by ID Message Requests must be routed solely by Requester ID / Destination BDF / Destination Device ID if the Ingress Port is in NFM, a Destination Segment is not included (DSV bit is clear), or the included Destination Segment matches the Switch's captured Segment value.
- A Switch for which Segment Captured is Clear must signal a TLP Translation Egress Blocked error if a Completion or Route by ID Message Request is received with DSV Set, and the TLP must not be forwarded.
- A Switch for which Segment Captured is Clear must signal a TLP Translation Egress Blocked error if a received Message or Memory Request includes a Requester Segment. The Request must not be forwarded.
- A Switch for which Segment Captured is Set must signal a TLP Translation Egress Blocked error if a Message or Memory Request received on a Downstream Port includes a Requester Segment that does not match the Switch's captured Segment value. The Request must not be forwarded.

- When reordering Completions with other Completions, Switches are permitted to consider Destination Segment fields included in the Completions as effectively part of the Transaction ID. When not included, the Destination Segment is implied to be the same Segment where the Completion exists.
- When reordering TLPs based on ID Based Ordering (IDO), Switches must consider Requester Segment fields included in Requests, and Destination Segment fields included in Completions, as effectively part of the Transaction ID. When the Destination Segment is not included, for reordering purposes the Destination Segment must be considered to be the same Segment where the Completion exists. When the Requester Segment is not included in a Request, Switches must assume a matching value for IDO purposes.

The following rules apply to Requesters:

- When the Requester Segment field is included in a Request it must be set to the value captured from a Configuration Write Request as described in [§ Section 2.2.6.2](#).
- When the Segment Captured bit is Clear all Message and Memory Requests initiated by the Requester must not include OHC-C.
- When the Segment Captured bit is Set all Message Requests initiated by the Requester must include OHC-C with Requester Segment.
- When the Segment Captured bit is Set a Requester is permitted to include OHC-C with Requester Segment in Memory Requests.
- When the Segment Captured bit is Clear, Route by ID Message Requests initiated by the Requester must not include a Destination Segment (the DSV bit must be clear).
- When the Segment Captured bit is Set a Requester is required to include a Destination Segment, and set the DSV bit, in Route by ID Message Requests destined for a different Hierarchy. Requesters use implementation specific means to determine the Hierarchy to which a Route by ID Message Request should be routed. When the Segment Captured bit is Set the Destination Segment is required in ATS Invalidate Request, Invalidate Completion, and PRG Response Messages even if the target is in the same Hierarchy. For other Route by ID Message Requests the Destination Segment is optional when the Segment Captured bit is Set and the Requester knows, by definition or through programming, that the target of the Request is in the same Hierarchy.
- Requesters must not qualify the Destination Segment field in received Completions.
- A Requester is not required to include the Requester Segment field in any non-IDE Memory Request.

The following rules apply to Completers:

- Completers must capture their Segment value from Configuration Write Requests as described in [§ Section 2.2.6.2](#).
- When the Segment Captured bit is Clear, Completers must set the Completer Segment field to 00h in any [OHC-A5](#) that is included in a Completion.
- When the Segment Captured bit is Set, Completers must set the Completer Segment field in any [OHC-A5](#) that is included in a Completion to the Segment value that was captured as described in [§ Section 2.2.6.2](#).
 - If the Completion associated with the first Configuration Write Request includes [OHC-A5](#), the Completer Segment field must be set to the value captured from that Request.
- Completers must clear the DSV bit and set the Destination Segment field to 00h in any [OHC-A5](#) that is included with a Completion associated with a Configuration Request.
- For a NP Memory Request received without a Requester Segment field, Completers must clear the DSV bit and set the Destination Segment field to 00h in any [OHC-A5](#) that is included with the associated Completion(s).

- For a NP Memory Request received with a Requester Segment field, Completers must set the DSV bit and set the Destination Segment field to the value of the received Requester Segment in any OHC-A5 that is included with the associated Completion(s).
- When the Segment Captured bit is Set and a NP Memory Request is received with a Requester Segment value not matching the Completer's captured Segment value, all Completions associated with the Request must include OHC-A5.
- Completers must not qualify the acceptance of a Route by ID Message Request based on the value of the Destination Segment field in the Request.
- Completers must not include OHC-A5 with a Completion when all of the following are true:
 - Completion Status is successful.
 - Lower Address[1:0] equal to 00b.
 - The Completer's Segment Captured bit is Clear.
- Completers are permitted to not include OHC-A5 with a Completion when all of the following are true:
 - Completion Status is successful.
 - Lower Address[1:0] equal to 00b.
 - The Completer's Segment Captured bit is Set.
 - The associated Request either did not include a Requester Segment field or included a Requester Segment field matching the Completer's captured Segment value.

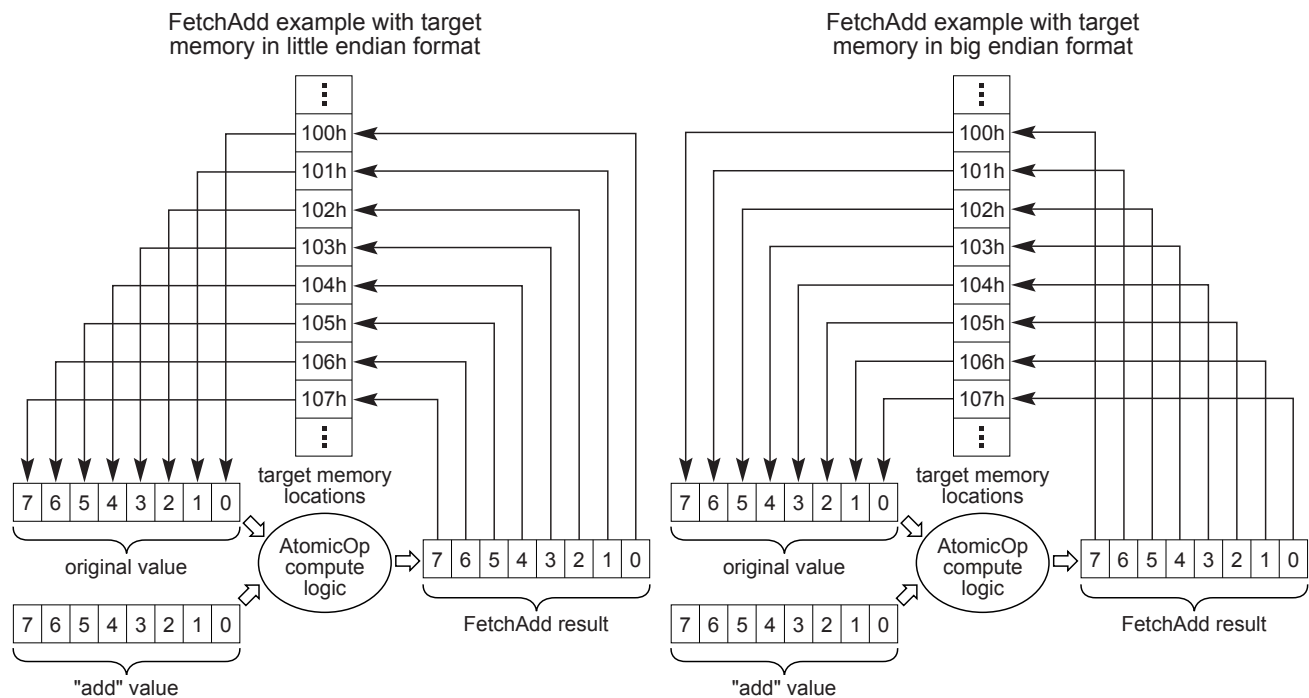
2.2.2 TLPs with Data Payloads - Rules §

- Length is specified as an integral number of DW
- Length[9:0] is Reserved for all Messages except those that explicitly refer to a data length
 - Refer to the Message Code tables in § Section 2.2.8 .
- A Function transmitting a TLP with a data payload must not allow the data payload length as indicated by the TLP's Length field to exceed the Function's applicable MPS setting. If the Function's Mixed_MPS_Supported bit is Clear or the target is host memory, the applicable MPS setting must be the Function's computed Tx_MPS_Limit, as defined below. If the Mixed_MPS_Supported bit is Set, the Function must have an implementation specific mechanism capable of supporting different MPS settings for different targets, and must handle both Request and Completion TLPs. Target-specific MPS settings are permitted to be above or below the Function's Tx_MPS_Limit, but they must never exceed the Function's Max_Payload_Size Supported field value. The Function's Tx_MPS_Limit is determined as follows:
 - For a single-Function device, the Tx_MPS_Limit must be its Max_Payload_Size field value, its "MPS setting".
 - Otherwise, for an ARI Device, the Tx_MPS_Limit must be the MPS setting in Function 0. The MPS settings in other Functions of an MFD must be ignored.
 - Otherwise, for a Function in a non-ARI MFD whose MPS settings are identical across all Functions, the Tx_MPS_Limit must be the common MPS setting.
 - Otherwise, for a Function in a non-ARI MFD whose MPS settings are not identical across all Functions, the Tx_MPS_Limit must be the MPS setting in an implementation specific Function.
 - Transmitter implementations are encouraged to use the MPS setting from the Function that generated the transaction, or else the smallest MPS setting across all Functions.
 - Software should not configure the MPS setting in different Functions to different values unless software is aware of the specific implementation.

- MPS settings apply only to TLPs with data payloads; Memory Read Requests are not restricted in length by MPS settings. The size of the Memory Read Request is controlled by the TLP's Length field.
- The data payload size in a Received TLP as indicated by the TLP's Length field must not exceed a computed Rx_MPS_Limit for the receiving Function, as determined by MPS-related parameters as indicated below.
 - Receivers must check for violations of this rule. If a Receiver determines that a TLP violates this rule, the TLP must be handled as a Malformed TLP.
 - This is a reported error associated with the Receiving Port (see § Section 6.2).
 - In the receiving Function, if the Rx_MPS_Fixed bit is Set, the Rx_MPS_Limit must be the Max_Payload_Size Supported field. Otherwise, the Rx_MPS_Limit must be determined by the Max_Payload_Size field (the "MPS setting") in one or more Functions as follows:
 - For a single-Function device, the Rx_MPS_Limit must be its MPS setting.
 - Otherwise, for an ARI Device, the Rx_MPS_Limit must be the MPS setting in Function 0. MPS settings in other Functions must be ignored.
 - Otherwise, for an Upstream Port associated with a non-ARI MFD whose MPS settings are identical across all Functions, the Rx_MPS_Limit must be the common MPS setting.
 - Otherwise, for an Upstream Port associated with a non-ARI MFD whose MPS settings are not identical across all Functions, the Rx_MPS_Limit must be the MPS setting in an implementation specific Function.
 - Receiver implementations are encouraged to use the MPS setting from the Function targeted by the transaction, or else the largest MPS setting across all Functions.
 - Software should not configure the MPS setting in different Functions to different values unless software is aware of the specific implementation.
- For TLPs that include data, the value in the Length field and the actual amount of data included in the TLP must match.
 - In NFM, Receivers must check for violations of this rule. If a Receiver determines that a TLP violates this rule, the TLP is a Malformed TLP.
 - This is a Reported Error associated with the Receiving Port (see § Section 6.2).
- The value in the Length field applies only to data - the TLP Digest is not included in the Length.
- When a data payload associated with a byte address is included in a TLP other than an AtomicOp Request or an AtomicOp Completion, the first byte of data following the header corresponds to the byte address closest to zero and the succeeding bytes are in increasing byte address sequence.
 - Example: For a 16-byte write to location 100h, the first byte following the header would be the byte to be written to location 100h, and the second byte would be written to location 101h, and so on, with the final byte written to location 10Fh.
- The data payload in AtomicOp Requests and AtomicOp Completions must be formatted such that the first byte of data following the TLP header is the least significant byte of the first data value, and subsequent bytes of data are strictly increasing in significance. With Compare And Swap (CAS) Requests, the second data value immediately follows the first data value, and must be in the same format.
 - The endian format used by AtomicOp Completers to read and write data at the target location is implementation specific, and is permitted to be whatever the Completer determines is appropriate for the target memory (e.g., little endian, big endian, etc.). Endian format capability reporting and controls for AtomicOp Completers are outside the scope of this specification.
 - Little endian example: For a 64-bit (8-byte) Swap Request targeting location 100h with the target memory in little endian format, the first byte following the header is written to location 100h, the second byte is written to location 101h, and so on, with the final byte written to location 107h. Note that before performing the writes, the Completer first reads the target memory locations so it can

return the original value in the Completion. The byte address correspondence to the data in the Completion is identical to that in the Request.

- Big endian example: For a 64-bit (8-byte) Swap Request targeting location 100h with the target memory in big endian format, the first byte following the header is written to location 107h, the second byte is written to location 106h, and so on, with the final byte written to location 100h. Note that before performing the writes, the Completer first reads the target memory locations so it can return the original value in the Completion. The byte address correspondence to the data in the Completion is identical to that in the Request.
- § Figure 2-15 shows little endian and big endian examples of Completer target memory access for a 64-bit (8-byte) FetchAdd. The bytes in the operands and results are numbered 0-7, with byte 0 being least significant and byte 7 being most significant. In each case, the Completer fetches the target memory operand using the appropriate endian format. Next, AtomicOp compute logic in the Completer performs the FetchAdd operation using the original target memory value and the “add” value from the FetchAdd Request. Finally, the Completer stores the FetchAdd result back to target memory using the same endian format used for the fetch.



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Figure 2-15 Examples of Completer Target Memory Access for FetchAdd

IMPLEMENTATION NOTE: ENDIAN FORMAT SUPPORT BY RC ATOMICOP COMPLETERS §

One key reason for permitting an AtomicOp Completer to access target memory using an endian format of its choice is so that PCI Express devices targeting host memory with AtomicOps can interoperate with host software that uses atomic operation instructions (or instruction sequences). Some host environments have limited endian format support with atomic operations, and by supporting the “right” endian format(s), an RC AtomicOp Completer may significantly improve interoperability.

For an RC with AtomicOp Completer capability on a platform supporting little-endian-only processors, there is little envisioned benefit for the RC AtomicOp Completer to support any endian format other than little endian. For an RC with AtomicOp Completer capability on a platform supporting bi-endian processors, there may be benefit in supporting both big endian and little endian formats, and perhaps having the endian format configurable for different regions of host memory.

There is no PCI Express requirement that an RC AtomicOp Completer support the host processor's “native” format (if there is one), nor is there necessarily significant benefit to doing so. For example, some processors can use load-link/store-conditional or similar instruction sequences to do atomic operations in non-native endian formats and thus not need the RC AtomicOp Completer to support alternative endian formats.

IMPLEMENTATION NOTE: MAINTAINING ALIGNMENT IN DATA PAYLOADS §

§ Section 2.3.1.1 discusses rules for forming Read Completions respecting certain natural address boundaries. Memory Write performance can be significantly improved by respecting similar address boundaries in the formation of the Write Request. Specifically, forming Write Requests such that natural address boundaries of 64 or 128 bytes are respected will help to improve system performance.

2.2.3 TLP Digest Rules - Non-Flit Mode Only §

- For any TLP, a value of 1b in the TD bit indicates the presence of the TLP Digest field including an end-to-end CRC (ECRC) value at the end of the TLP.
 - A TLP where the TD bit value does not correspond with the observed size (accounting for the data payload, if present) is a Malformed TLP.
 - This is a reported error associated with the Receiving Port (see § Section 6.2).
- If an intermediate or ultimate PCI Express Receiver of the TLP does not support ECRC checking, the Receiver must ignore the TLP Digest⁸.
 - If the Receiver of the TLP supports ECRC checking, the Receiver interprets the value in the TLP Digest field as an ECRC value, according to the rules in § Section 2.7.1 .

8. An exception is an Intermediate Receiver forwarding a Multicast TLP out an Egress Port with MC_Overlay enabled. See § Section 6.14.5 .

2.2.4 Routing and Addressing Rules §

There are three principal mechanisms for TLP routing: address, ID, and implicit. This section defines the rules for the address and ID routing mechanisms. Implicit routing is used only with Message Requests, and is covered in § Section 2.2.8.

2.2.4.1 Address-Based Routing Rules §

- Address routing is used with Memory and I/O Requests.
- Two address formats are specified, a 64-bit format used with a 4 DW header (see § Figure 2-16 and § Figure 2-18) and a 32-bit format used with a 3 DW header (see § Figure 2-17 and § Figure 2-19).

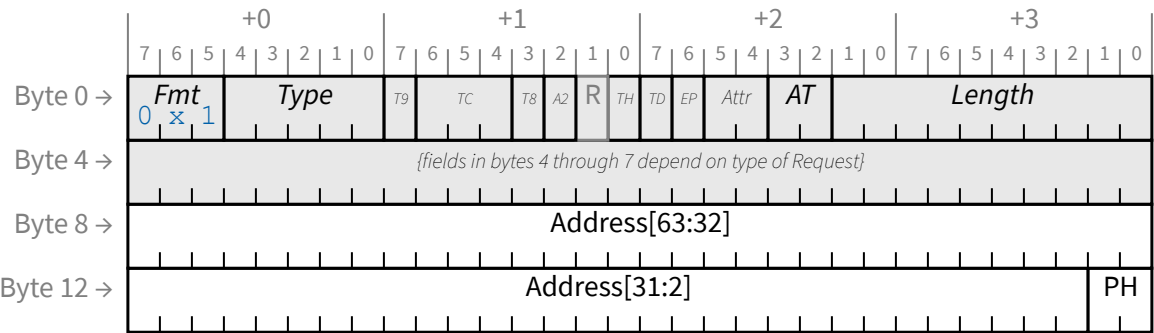


Figure 2-16 64-bit Address Routing - Non-Flit Mode

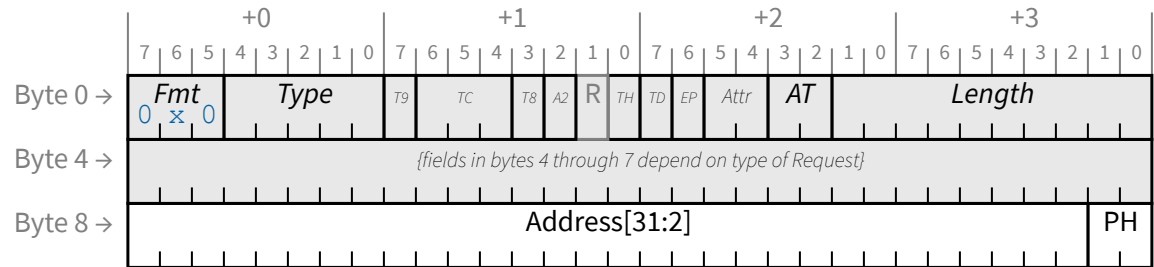


Figure 2-17 32-bit Address Routing - Non-Flit Mode

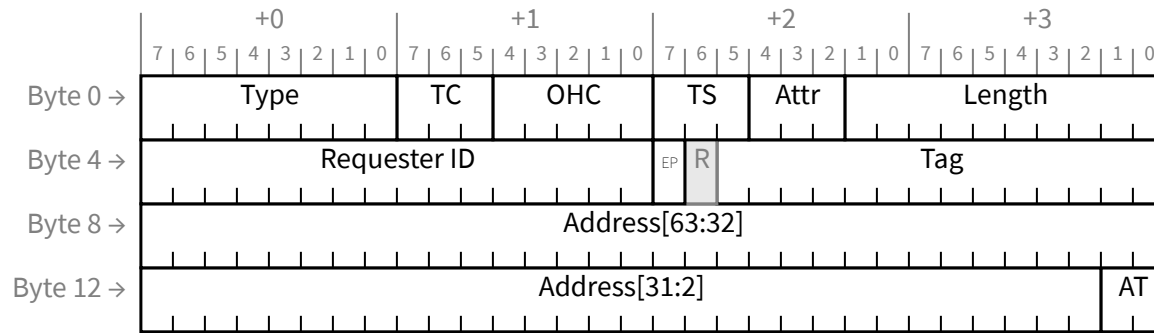


Figure 2-18 64-bit Address Routing - Flit Mode

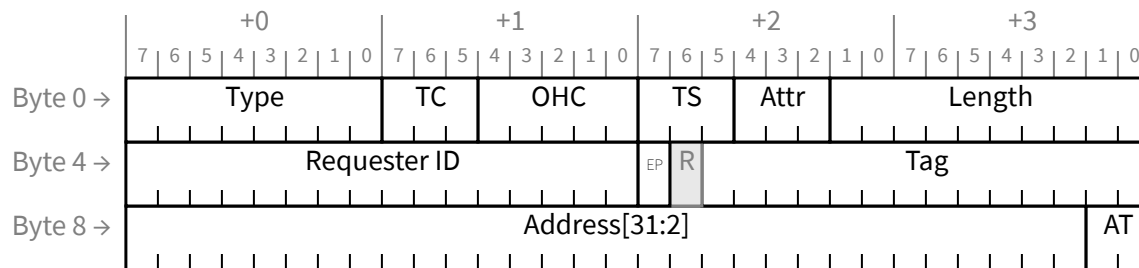


Figure 2-19 32-bit Address Routing - Flit Mode

- For Memory Read, Memory Write, DMWr, and AtomicOp Requests, the Address Type (AT) field is encoded as shown in § Table 10-1. For all other Requests, the AT field is Reserved unless explicitly stated otherwise.
- If TH is Set, the PH field is encoded as shown in § Table 2-18. If TH is Clear, the PH field is Reserved.
- Address mapping to the TLP header is shown in § Table 2-7.

Table 2-7 Address Field Mapping

Address Bits	32-bit Addressing	64-bit Addressing
63:56	Not Applicable	Bits 7:0 of Byte 8
55:48	Not Applicable	Bits 7:0 of Byte 9
47:40	Not Applicable	Bits 7:0 of Byte 10
39:32	Not Applicable	Bits 7:0 of Byte 11
31:24	Bits 7:0 of Byte 8	Bits 7:0 of Byte 12
23:16	Bits 7:0 of Byte 9	Bits 7:0 of Byte 13
15:8	Bits 7:0 of Byte 10	Bits 7:0 of Byte 14
7:2	Bits 7:2 of Byte 11	Bits 7:2 of Byte 15

- Memory Read, Memory Write, DMWr, and AtomicOp Requests use both formats.

- For Addresses below 4 GB, Requesters must use the 32-bit format. The behavior of the Receiver is not specified if a 64-bit format Request addressing below 4 GB (i.e., with the upper 32 bits of address all 0) is received.
- All other address routed requests must use 64-bit addressing.
 - When addressing below 4 GB the upper 32 address bits must be to 0000 0000h.
 - This *MUST@FLIT* include Address Routed Messages. See § Table 2-20.⁹
- I/O Read Requests and I/O Write Requests use the 32-bit format.
- All agents must decode all address bits in the header - address aliasing is not allowed.

IMPLEMENTATION NOTE: PREVENTION OF ADDRESS ALIASING §

For correct software operation, full address decoding is required even in systems where it may be known to the system hardware architect/designer that fewer than 64 bits of address are actually meaningful in the system.

2.2.4.2 ID Based Routing Rules §

- ID routing is used with Configuration Requests, with ID Routed Messages, and with Completions. This specification defines several Messages that are ID Routed (see § Table F-1). Other specifications are permitted to define additional ID Routed Messages.
- ID routing uses the Bus, Device, and Function Numbers (as applicable) to specify the destination for the TLP:
 - For non-ARI Routing IDs, Bus, Device, and (3-bit) Function Number to TLP header mapping is shown in § Table 2-8, § Figure 2-20, and § Figure 2-22.
 - For ARI Routing IDs, the Bus and (8-bit) Function Number to TLP header mapping is shown in § Table 2-9, § Figure 2-21, and § Figure 2-23.
- In FM, Completions and ID Routed Messages with a different destination Hierarchy than the Hierarchy in which they originate must be routed to the destination Hierarchy using the Destination Segment field and then routed within the destination Hierarchy by the destination Bus, Device, and Function Numbers.
- Two ID routing formats are specified, one used with a 4 DW header (see § Figure 2-20 and § Figure 2-21) and one used with a 3 DW header (see § Figure 2-23 and § Figure 2-21).
 - Header field locations are the same for both formats (see § Figure 2-5).

*Table 2-8 Header Field Locations for
non-ARI ID Routing - Non-Flit Mode*

Field	Header Location
Bus Number[7:0]	Bits 7:0 of Byte 8
Device Number[4:0]	Bits 7:3 of Byte 9
Function Number[2:0]	Bits 2:0 of Byte 9

9. Earlier versions of this specification did not specify address routed message behavior when the address was below 4 GB.

*Table 2-9 Header Field Locations for ARI
§ ID Routing*

Field	Header Location
Bus Number[7:0]	Bits 7:0 of Byte 8
Function Number[7:0]	Bits 7:0 of Byte 9

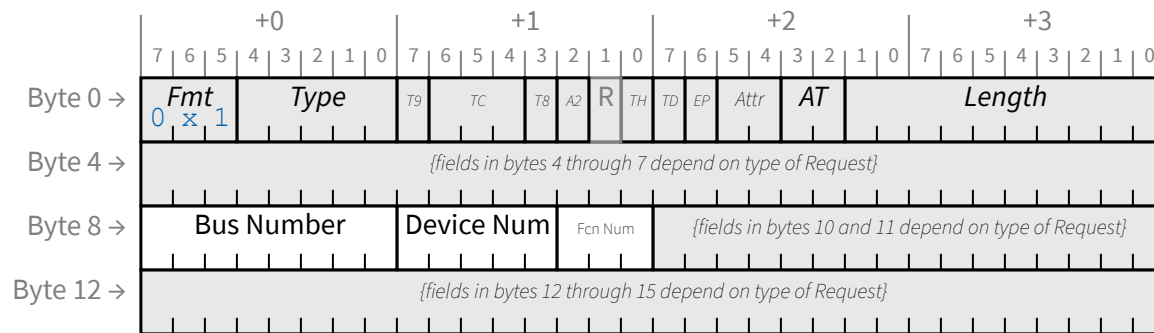


Figure 2-20 Non-ARI ID Routing with 4 DW Header - Non-Flit Mode

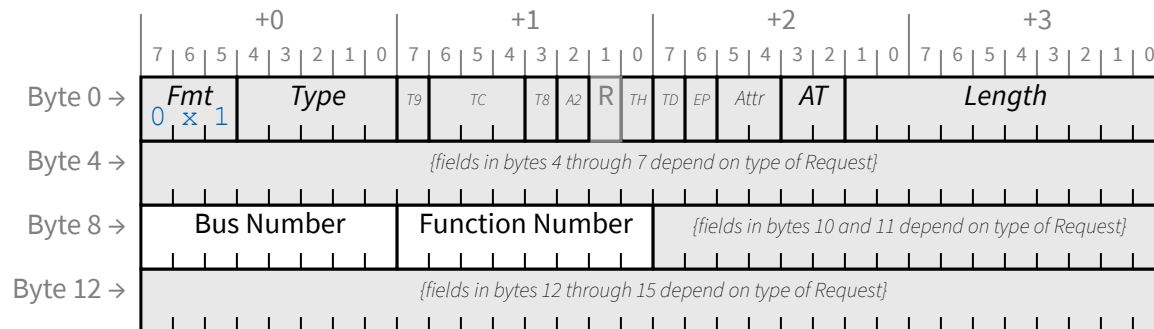


Figure 2-21 ARI ID Routing with 4 DW Header - Non-Flit Mode

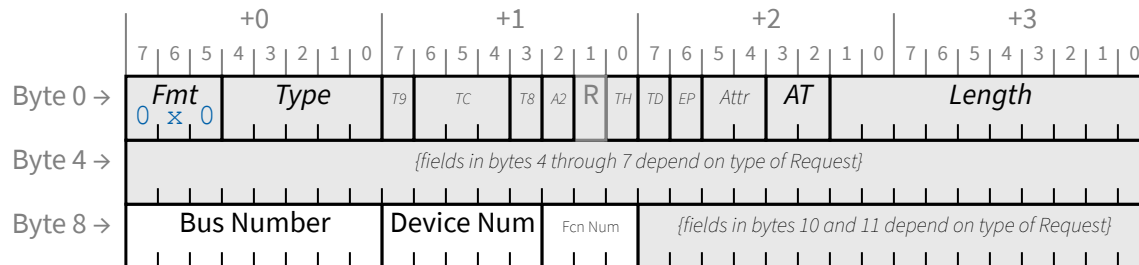


Figure 2-22 Non-ARI ID Routing with 3 DW Header - Non-Flit Mode

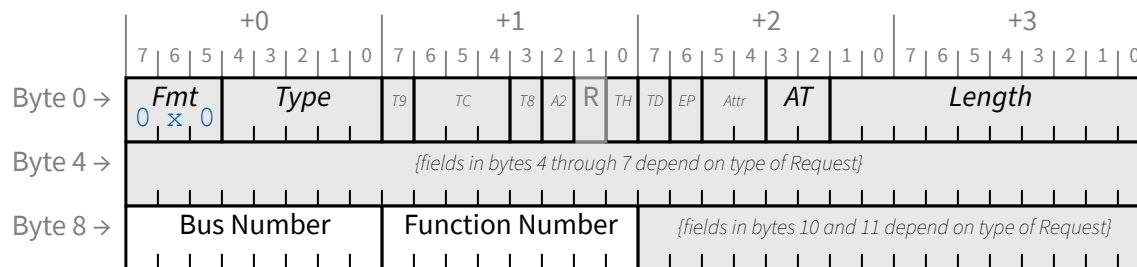


Figure 2-23 ARI ID Routing with 3 DW Header - Non-Flit Mode

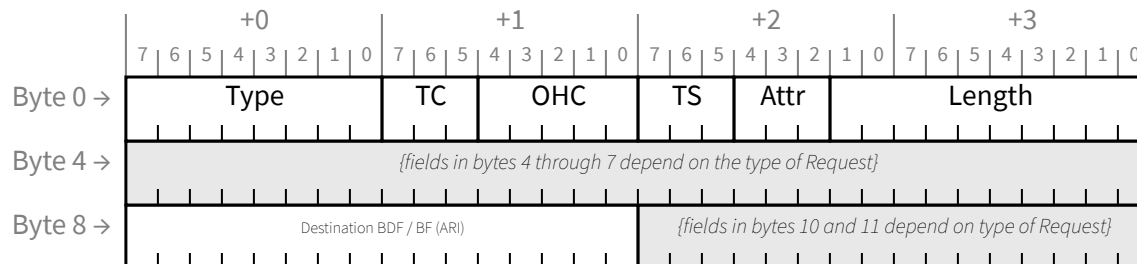


Figure 2-24 ID Routing with 3 DW Header - Flit Mode

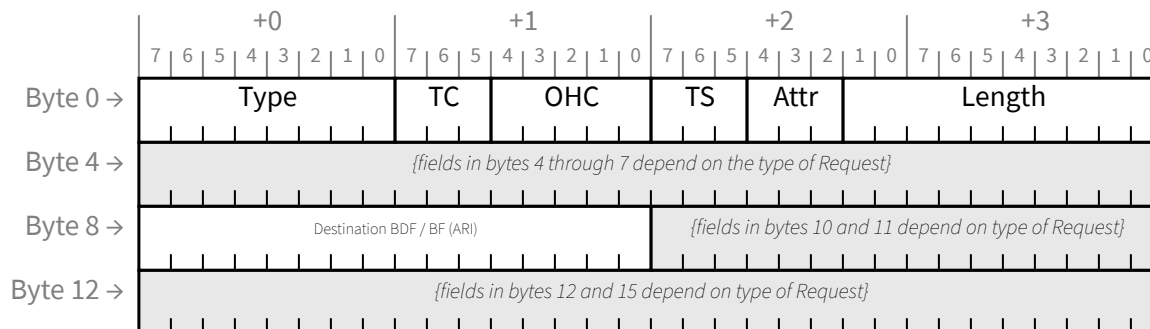


Figure 2-25 ID Routing with 4 DW Header - Flit Mode

2.2.5 First/Last DW Byte Enables Rules §

The general function of TLP Byte Enables is similar in Non-Flit Mode and Flit Mode, however the detailed rules differ.

2.2.5.1 Byte Enable Rules for Non-Flit Mode §

Byte Enables are included with Memory, I/O, and Configuration Requests. This section defines the corresponding rules. Byte Enables, when present in the Request header, are located in byte 7 of the header (see § Figure 2-26). For Memory Read Requests and DMWr Requests that have the TH bit Set, the Byte Enable fields are repurposed to carry the ST[7:0] field (refer to § Section 2.2.7.1.1 for details), and values for the Byte Enables are implied as defined below. The TH bit

must only be Set in Memory Read Requests and DMWr Requests when it is acceptable to complete those Requests as if all bytes for the requested data were enabled.

- For Memory Read Requests and DMWr Requests that have the TH bit Set, the following values are implied for the Byte Enables. See § Section 2.2.7 for additional requirements.
 - If the Length field for this Request indicates a length of 1 DW, then the value for the First DW Byte Enables is implied to be 1111b and the value for the Last DW Byte Enables is implied to be 0000b.
 - If the Length field for this Request indicates a length of greater than 1 DW, then the value for the First DW Byte Enables and the Last DW Byte Enables is implied to be 1111b.

IMPLEMENTATION NOTE: READ REQUEST WITH TPH TO NON-PREFETCHABLE SPACE §

Memory Read Requests with the TH bit Set and that target Non-Prefetchable Memory Space should only be issued when it can be guaranteed that completion of such reads will not create undesirable side effects. See § Section 7.5.1.2.1 for consideration of certain BARs that may have the Prefetchable bit Set even though they map some locations with read side-effects.

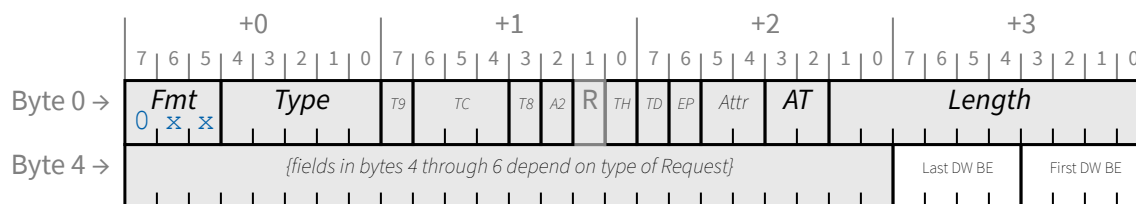


Figure 2-26 Location of Byte Enables in TLP Header - Non-Flit Mode

- The First DW BE[3:0] field contains Byte Enables for the first (or only) DW referenced by a Request.
 - If the Length field for a Request indicates a length of greater than 1 DW, this field must not equal 0000b.
- The Last DW BE[3:0] field contains Byte Enables for the last DW of a Request.
 - If the Length field for a Request indicates a length of 1 DW, this field must equal 0000b.
 - If the Length field for a Request indicates a length of greater than 1 DW, this field must not equal 0000b.
- For each bit of the Byte Enables fields:
 - a value of 0b indicates that the corresponding byte of data must not be written or, if non-prefetchable, must not be read at the Completer.
 - a value of 1b indicates that the corresponding byte of data must be written or read at the Completer.
- Non-contiguous Byte Enables (enabled bytes separated by non-enabled bytes) are permitted in the First DW BE field for all Requests with length of 1 DW.
 - Non-contiguous Byte Enable examples: 1010b, 0101b, 1001b, 1011b, 1101b
- Non-contiguous Byte Enables are permitted in both Byte Enables fields for Quad Word (QW) aligned Memory Requests with length of 2 DW (1 QW).

- All non-QW aligned Memory Requests with length of 2 DW (1 QW) and Memory Requests with length of 3 DW or more must enable only bytes that are contiguous with the data between the first and last DW of the Request.
 - Contiguous Byte Enables examples:
First DW BE: 1100b, Last DW BE: 0011b

First DW BE: 1000b, Last DW BE: 0111b
- § Table 2-10 shows the correspondence between the bits of the Byte Enables fields, their location in the Request header, and the corresponding bytes of the referenced data.

Table 2-10 Byte Enables Location and Correspondence §

Byte Enables	Header Location	Affected Data Byte ¹⁰
First DW BE[0]	Bit 0 of Byte 7	Byte 0
First DW BE[1]	Bit 1 of Byte 7	Byte 1
First DW BE[2]	Bit 2 of Byte 7	Byte 2
First DW BE[3]	Bit 3 of Byte 7	Byte 3
Last DW BE[0]	Bit 4 of Byte 7	Byte N-4
Last DW BE[1]	Bit 5 of Byte 7	Byte N-3
Last DW BE[2]	Bit 6 of Byte 7	Byte N-2
Last DW BE[3]	Bit 7 of Byte 7	Byte N-1

- A Write Request with a length of 1 DW with no bytes enabled is permitted, and has no effect at the Completer unless otherwise specified.

IMPLEMENTATION NOTE: ZERO-LENGTH WRITE §

A Memory Write Request of 1 DW with no bytes enabled, or “zero-length Write,” may be used by devices under certain protocols, in order to achieve an intended side effect.

- If a Read Request of 1 DW specifies that no bytes are enabled to be read (First DW BE[3:0] field = 0000b), the corresponding Completion must specify a length of 1 DW, and include a data payload of 1 DW.
The contents of the data payload within the Completion packet is unspecified and may be any value.
- Receiver/Completer behavior is undefined for a TLP violating the Byte Enables rules specified in this section.
- Receivers may optionally check for violations of the Byte Enables rules specified in this section. If a Receiver implementing such checks determines that a TLP violates one or more Byte Enables rules, the TLP is a Malformed TLP. These checks are independently optional (see § Section 6.2.3.4).
 - If Byte Enables rules are checked, a violation is a reported error associated with the Receiving Port (see § Section 6.2).

10. Assuming the data referenced is N bytes in length (Byte 0 to Byte N-1). Note that last DW Byte Enables are used only if the data length is greater than one DW.

IMPLEMENTATION NOTE:

ZERO-LENGTH READ §

A Memory Read Request of 1 DW with no bytes enabled, or “zero-length Read,” may be used by devices as a type of flush Request. For a Requester, the flush semantic allows a device to ensure that previously issued Posted Writes have been completed at their PCI Express destination. To be effective in all cases, the address for the zero-length Read must target the same device as the Posted Writes that are being flushed. One recommended approach is using the same address as one of the Posted Writes being flushed.

The flush semantic has wide application, and all Completers must implement the functionality associated with this semantic. Since a Requester may use the flush semantic without comprehending the characteristics of the Completer, Completers must ensure that zero-length reads do not have side-effects. This is really just a specific case of the rule that in a non-prefetchable space, non-enabled bytes must not be read at the Completer. Note that the flush applies only to traffic in the same Traffic Class as the zero-length Read.

2.2.5.2 Byte Enable Rules for Flit Mode §

Except as defined in this section, all Byte Enable Rules in Flit Mode are the same as in Non-Flit Mode.

For all Memory Requests, it is permitted for OHC-A1 (see § Figure 2-7) to be present. OHC-A1 must be included for Requests that require any of the fields included in OHC-A1. For Memory Requests, when OHC-A1 is not present, the value of the Last DW Byte Enable field must be treated as 1111b for Requests with Length greater than or equal to 2 DW, and the value of the 1st DW Byte Enable field must be treated as 1111b. If a Request requires Byte Enables field values other than these, then OHC-A1 must be present. When OHC-A1 is present, the PASID, PMR and ER fields are valid if and only if the PV bit is Set.

OHC-A2 must be included for all IO Requests.

OHC-A3 must be included for all Configuration Requests.

In all cases where OHC-A is present, the Byte Enable fields must be handled as defined in § Section 2.2.5.1 .

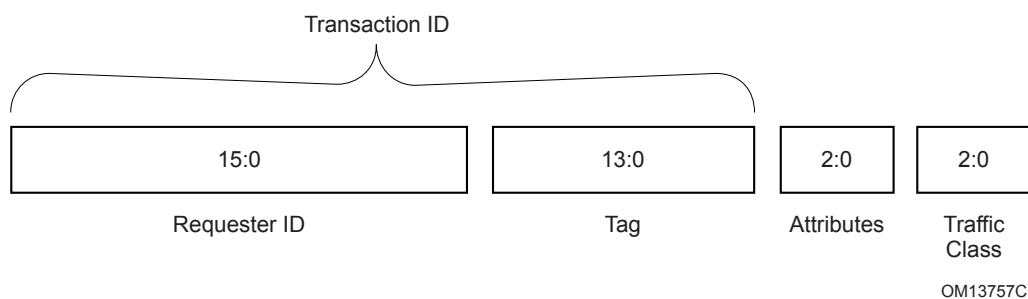
2.2.6 Transaction Descriptor §

2.2.6.1 Overview §

The Transaction Descriptor is a mechanism for carrying Transaction information between the Requester and the Completer. Transaction Descriptors are composed of three fields:

- Transaction ID - identifies outstanding Transactions
- Attributes field - specifies characteristics of the Transaction
- Traffic Class (TC) field - associates Transaction with type of required service

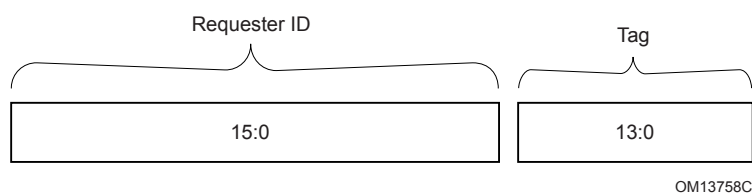
§ Figure 2-27 shows the fields of the Transaction Descriptor. Note that these fields are shown together to highlight their relationship as parts of a single logical entity. The fields are not contiguous in the packet header.



§ *Figure 2-27 Transaction Descriptor*

2.2.6.2 Transaction Descriptor - Transaction ID Field §

The Transaction ID field consists of two major sub-fields: Requester ID and Tag as shown in § Figure 2-28.



§ *Figure 2-28 Transaction ID*

Architecturally, Tag[13:0] is a 14-bit field generated by each Requester, and it must be unique for all outstanding Requests that require a Completion for that Requester ID, without regard to TC or any other field. Four Tag sizes are architected for operation: 14-bit, 10-bit, 8-Bit and 5-bit. A given Function may support different Tag sizes when operating as a Requester versus operating as a Completer. Below are the rules regarding operational Tag sizes. Also see the “Considerations for Implementing Larger-Tag Capabilities” Implementation Note later in this section.

- 14-Bit Tags and 10-Bit Tags are referred to as “larger” Tags.
- 8-Bit Tags and 5-Bit Tags are referred to as “smaller” Tags.
- All Functions must support 8-Bit Tag Completer capability.
- A Function that supports Flit Mode must support 14-Bit Tag Completer capability, and thus it automatically supports 10-Bit Tag Completer capability.
- Functions¹¹ (including those in Switches) that support 16.0 GT/s data rates or greater must support 10-Bit Tag Completer capability.
- A Function must not support 14-Bit Tag Requester capability unless it supports 14-Bit Tag Completer capability.
- A Function must not support 10-Bit Tag Requester capability unless it supports 10-Bit Tag Completer capability.

11. An exception is PCI Express to PCI/PCI-X Bridges, since 10-Bit Tag capability is not architected for these Functions.

- In Non-Flit Mode, Tag[8] and Tag[9], are not contiguous with other Tag field bits in the TLP Header. These bits were Reserved prior to 10-Bit Tags being architected. Requesters in Non-Flit Mode that do not support 10-Bit Tag Requester capability must set Tag[9:8] to 00b.
- RCs containing elements that indicate support for 14-Bit Tag Completer capability or 10-Bit Tag Completer capability must handle supported Tag-sized Requests correctly by all registers and memory regions supported as targets of PCIe Requesters; e.g., host memory targeted by DMA Requests or MMIO regions in RCiEPs.
 - Each RP indicating support must handle such Requests received by its Ingress Port.
 - Each RCiEP indicating support must handle such Requests coming from supported internal paths, including those coming through RPs.
- If an RC contains RCiEPs that indicate support for 14-Bit Tag Requester capability or 10-Bit Tag Requester capability, the RC must handle Requests from those RCiEPs correctly by all registers and memory regions supported as targets of those RCiEPs; e.g., host memory targeted by DMA Requests or MMIO regions in RCiEPs.
- Receivers/Completers must handle 8-bit Tag values correctly regardless of the setting of their Extended Tag Field Enable bit (see § Section 7.5.3.4). Refer to the *PCI Express to PCI/PCI-X Bridge Specification* for details on the bridge handling of Extended Tags.
- Receivers/Completers that support 14-Bit Tag Completer capability or 10-Bit Tag Completer capability must handle the supported Tag-size values correctly, regardless of their corresponding Tag Requester Enable bit setting. See § Section 7.5.3.16 .
- 14-Bit Tag capability and 10-Bit Tag capability are not architected for PCI Express to PCI/PCI-X Bridges, and they must not indicate the associated Tag Requester capability or Tag Completer capability.
- If one or both larger-Tag Requester Enable bits are Set, the following rules apply.
 - If both larger-Tag Requester Enable bits are Set in an Endpoint ¹² , then 14-Bit Tags are permitted for Requests that target host memory. An implementation specific hardware mechanism in the Endpoint is permitted to limit those Requests to 10-Bit Tags or smaller Tags, but generic software or firmware should not Set the 14-Bit Tag Requester Enable bit unless the host supports 14-Bit Tag Completer capability for host memory.
 - If an Endpoint ¹³ supports sending Requests to other Endpoints (as opposed to host memory), the Endpoint must not send larger-Tag Requests to another given Endpoint unless an implementation specific mechanism determines that the Endpoint supports the corresponding larger Tag Completer capability. Not sending larger-Tag Requests to other Endpoints at all may be acceptable for some implementations. More sophisticated mechanisms are outside the scope of this specification.
 - If a PIO Requester has larger-Tag Requester capability, how the Requester determines when to use larger Tags versus smaller Tags is outside the scope of this specification.
 - With 14-Bit Tags, all Tag[13:8] values except 00 0000b are valid. 14-Bit Tag values with Tag[13:8] equal to 00 0000b are invalid, and must not be generated by the Requester. This enables a Requester to determine if a Completion it receives that should have a 14-Bit Tag contains an invalid Tag value, usually caused by the Completer not supporting 14-Bit Tag Completer capability.
 - With 10-Bit Tags, all Tag[9:8] values except 00b are valid. 10-Bit Tag values with Tag[9:8] equal to 00b are invalid, and must not be generated by the Requester. This enables a Requester to determine if a Completion it receives that should have a 10-Bit Tag contains an invalid Tag value, usually caused by the Completer not supporting 10-Bit Tag Completer capability.
 - If a Requester sends a larger-Tag Request to a Completer that lacks the associated larger-Tag Completer capability, the returned Completion(s) will have Tags with invalid Tag values. Such Completions will be

12. This includes PCI Express Endpoints, Legacy PCI Express Endpoints, and Root Complex Integrated Endpoints.

13. This includes PCI Express Endpoints, Legacy PCI Express Endpoints, and Root Complex Integrated Endpoints.

handled as Unexpected Completions¹⁴, which by default are Advisory Non-Fatal Errors. The Requester must follow standard PCI Express error handling requirements.

- When a Requester handles a Completion with an invalid Tag as an Unexpected Completion, the original Request will likely incur a Completion Timeout. If the Requester handles the Completion Timeout condition in some device-specific manner that avoids data corruption, the Requester is permitted to suppress handling the Completion Timeout by standard PCI Express error handling mechanisms as required otherwise.
- If a Requester supports sending larger-Tag Requests to some Completers and smaller-Tag Requests to other Completers concurrently, the Requester must honor the Extended Tag Field Enable bit setting for the smaller-Tag Requests. That is, if the bit is Clear, only the lower 5 bits of the Tag field may be non-Zero; if the bit is Set, only the lower 8 bits of the Tag field may be non-Zero.
- If a Requester supports sending larger-Tag Requests to some Completers and smaller-Tag Requests to other Completers concurrently, the Requester must ensure that no outstanding larger Tags can alias to an outstanding smaller Tag if any larger-Tag Request is completed by a Completer that lacks larger-Tag Completer capability. See the "Using Larger Tags and Smaller Tags Concurrently" Implementation Note later in this section.
- The default value of the Extended Tag Field Enable bit is implementation specific. The default value of the 14-Bit Tag Requester Enable bit and the 10-Bit Tag Requester Enable bit is 0b.
- Receiver/Completer behavior is undefined if multiple uncompleted Requests are issued with non-unique Transaction ID values.
- If Phantom Function Numbers are used to extend the number of outstanding Requests, the combination of the Phantom Function Number and the Tag field must be unique for all outstanding Requests that require a Completion for that Requester, without regard to TC or any other field.
- If Shadow Functions are used to extend the number of outstanding Requests, the combination of the Shadow Function Number and the Tag field must be unique for all outstanding Requests that require a Completion for that Requester, without regard to TC or any other field.
- § Table 2-11 indicates how the three tag enable bits determine the maximum tag size and permitted tag value ranges for different Completers and their associated paths. For a given combination of Tag enable settings, a Requester must use a Tag value within the permitted range corresponding to Tag capabilities of the Completer and its associated path.

Table 2-11 Tag Enables, Sizes, and Permitted Ranges

14-bit Tag Requester Enable	10-bit Tag Requester Enable	Extended Tag Field Enable	Maximum Tag size	Permitted range for an 8-bit Tag Completer/path	Permitted range for a 10-bit Tag Completer/path	Permitted range for a 14-bit Tag Completer/path
0	0	0	5 bits	0 to 31	0 to 31	0 to 31
0	0	1	8 bits	0 to 255	0 to 255	0 to 255
0	1	0	10 bits	0 to 31	256 to 1023	256 to 1023
0	1	1	10 bits	0 to 255	256 to 1023	256 to 1023
1	0	0	14 bits	0 to 31	0 to 31	1024 to 16383
1	0	1	14 bits	0 to 255	0 to 255	1024 to 16383
1	1	0	14 bits	0 to 31	256 to 1023	1024 to 16383

14. If a Completion has a higher precedence error, that error should be reported instead.

14-bit Tag Requester Enable	10-bit Tag Requester Enable	Extended Tag Field Enable	Maximum Tag size	Permitted range for an 8-bit Tag Completer/path	Permitted range for a 10-bit Tag Completer/path	Permitted range for a 14-bit Tag Completer/path
1	1	1	14 bits	0 to 255	256 to 1023	1024 to 16383

Notes:

1. The permitted range for a 5-bit Tag Completer/path is always 0 to 31, so there is no column in the table to indicate this.
 2. The "X-bit Tag Completer/path" is the greatest common Tag size capability of the Completer and all routing elements along the path between the Requester and the targeted Completer. If a routing element is not the targeted Completer, but detects an Uncorrectable Error with a Request, the routing element may serve as the Completer for the Request.
 3. If a Requester supports sending larger-Tag Requests to some Completers and smaller-Tag Requests to other Completers concurrently, the Requester must ensure that no outstanding larger Tags can alias to an outstanding smaller Tag if any larger-Tag Request is completed by a Completer that lacks larger-Tag Completer capability.
- For Posted Requests, the Tag[13:8] field is Reserved in Non-Flit Mode, and Tag[13:0] is Reserved in Flit Mode.
 - An exception to this rule is allowed for the uses defined in [MCTP-VDM].
 - In Non-Flit Mode, for Posted Requests with the TH bit Set, the Tag[7:0] field is repurposed for the ST[7:0] field (refer to § Section 2.2.7.1.1 for details). For Posted Requests with the TH bit Clear, the Tag[7:0] field is undefined and may contain any value. (Refer to § Table F-1 for exceptions to this rule for certain Vendor_Defined Messages.)
 - For Posted Requests with the TH field Clear, the value in the Tag[7:0] field must not affect Receiver processing of the Request.
 - For Posted Requests with the TH bit Set, the value in the ST[7:0] field may affect Completer processing of the Request (refer to 2.2.7.1 for details).
 - Requester ID and Tag combined form a global identifier, i.e., Transaction ID for each Transaction within a Hierarchy.
 - Transaction ID is included with all Requests and Completions.
 - The Requester ID is a 16-bit value that is unique for every PCI Express Function within a Hierarchy.
 - Functions must capture the Bus and Device Numbers¹⁵ supplied with all Type 0 Configuration Write Requests completed by the Function and supply these numbers in the Bus and Device Number fields of the Requester ID¹⁶ for all Requests initiated without the use of Shadow Functions by the Device/Function. See § Section 7.9.25, for details of how the Requester ID may be modified by the use of Shadow Functions. It is recommended that Numbers are captured for successfully completed Requests only.

Exception: The assignment of Bus and Device Numbers to the Devices within a Root Complex, and Device Numbers to the Downstream Ports within a Switch, may be done in an implementation specific way.

Note that the Bus Number and Device Number¹⁷ may be changed at run time, and so it is necessary to re-capture this information with each and every Type 0 Configuration Write Request to the Device.

Configuration Write Requests addressed to unimplemented Functions *MUST@FLIT* not affect captured Bus and Device Numbers for implemented Functions.

15. In ARI Devices, Functions are only required to capture the Bus Number. ARI Devices are permitted to retain the captured Bus Number on either a per-Device or a per-Function basis. If the captured Bus Number is retained on a per-Device basis, all Functions are required to update and use the common Bus Number.

16. An ARI Requester ID does not contain a Device Number field. See § Section 2.2.4.2.

17. With ARI Devices, only the Bus Number can change.

- When generating Requests on their own behalf (for example, for error reporting), Switches must use the Requester ID associated with the primary side of the bridge logically associated with the Port (see § Section 7.1) causing the Request generation.
- Prior to the initial Configuration Write to a Function, the Function is not permitted to initiate Non-Posted Requests. (A valid Requester ID is required to properly route the resulting completions.)
 - Exception: Functions within a Root Complex are permitted to initiate Requests prior to software-initiated configuration for accesses to system boot device(s).
Note that this rule and the exception are consistent with the existing PCI model for system initialization and configuration.
- Each Function associated with a Device must be designed to respond to a unique Function Number for Configuration Requests addressing that Device. Note: Each non-ARI Device may contain up to eight Functions. Each ARI Device may contain up to 256 Functions.
- A Switch must forward Requests without modifying the Transaction ID.
- In some circumstances, a PCI Express to PCI/PCI-X Bridge is required to generate Transaction IDs for Requests it forwards from a PCI or PCI-X bus.
- In Flit Mode, Functions must capture the value of the Destination Segment supplied with all Type 0 Configuration Write Requests successfully completed by the Function. It is permitted for each Function of a Device to independently capture the Destination Segment value, or for all Functions of a Device to use the value captured by Function 0. All Functions within a Switch share a common Segment value that is captured by Functions associated with the Upstream Port. Functions also must capture the DSV bit in Type 0 Configuration Write Requests as described in the Segment Captured bit description in § Section 7.7.9.4 .
 - The Segment is effectively an extension of the Requester ID, but is formally defined as a distinct field to avoid confusion with the use of the term Transaction ID in Non-Flit Mode operation.
 - In systems that support multiple Segments, each Hierarchy must be associated with a single Segment. It is permitted for multiple hierarchy domains to be associated with a single Segment.
- In Flit-Mode, in some circumstances, the captured Segment is also explicitly indicated in a TLP, which enables the Transaction ID to be unique between Hierarchies.

IMPLEMENTATION NOTE: INCREASING THE NUMBER OF OUTSTANDING REQUESTS USING PHANTOM FUNCTIONS OR SHADOW FUNCTIONS §

To increase the maximum possible number of outstanding Requests requiring Completion beyond that possible using Tag bits alone, a device may, if the Phantom Functions Enable bit is Set (see § Section 7.5.3.4), or the Shadow Functions Enable bit is Set (see § Section 7.9.25.3), use Function Numbers not assigned to implemented Functions to logically extend the Tag identifier. For a single-Function Device, this can allow a significant increase in the maximum number of outstanding Requests.

When the Phantom Functions Enable bit is Set, unclaimed Function Numbers are referred to as Phantom Function Numbers.

Phantom Functions have a number of architectural limitations, including a lack of support by ARI Devices, Virtual Functions (VFs), and Physical Functions (PFs) when VFs are enabled. In addition, Address Translation Services (ATS) and ID-Based Ordering (IDO) do not comprehend Phantom Functions. Shadow Functions have fewer limitations. Thus, for many implementations, the use of larger Tags and Shadow Functions are better ways to increase the number of outstanding Non-Posted Requests.

IMPLEMENTATION NOTE: CONSIDERATIONS FOR IMPLEMENTING LARGER-TAG CAPABILITIES

The use of "larger" (i.e., 10-bit or 14-bit) Tags enables a Requester to increase its number of outstanding Non-Posted Requests (NPRs) substantially, which for very high rates of NPRs or very large round-trip times can avoid Tag availability from becoming a bottleneck. The following formula gives the basic relationship between payload bandwidth, number of outstanding NPRs, and other factors:

$BW = S * N / RTT$, where

BW = payload bandwidth

S = transaction payload size

N = number of outstanding NPRs

RTT = transaction round-trip time

Generally only high-speed Requesters on high-speed Links using relatively small transactions will benefit from increasing their number of outstanding NPRs beyond 256, although this can also help maintain performance in configurations where the transaction round-trip time is high.

In configurations where a Requester with larger-Tag Requester capability needs to target multiple Completers, one needs to ensure that the Requester sends larger-Tag Requests only to Completers that have sufficient larger-Tag Completer capability. This is greatly simplified if all Completers have larger-Tag capability.

For general industry enablement of larger Tags, it is strongly recommended that all Functions¹⁸ support larger-Tag Completer capability. With new implementations, Completers that don't need to operate on higher numbers of NPRs concurrently themselves can generally track larger Tags internally and return them in Completions with modest incremental investment.

Completers that actually process higher numbers of NPRs concurrently may require substantial additional hardware resources, but the full performance benefits of larger Tags generally can't be realized unless Completers actually do process higher numbers of NPRs concurrently.

For platforms where the RC supports larger-Tag Completer capability, it is strongly recommended for platform firmware or operating system software that configures PCIe hierarchies to Set one of the larger-Tag Requester Enable bits automatically in Endpoints with larger-Tag Requester capability. This enables the important class of larger-Tag capable adapters that send Memory Read Requests only to host memory.

For Endpoints other than RCiEPs, one can determine if the RC supports larger-Tag Completer capability for each one by checking the larger-Tag Completer Supported bits in its associated RP. RCiEPs have no associated RP, so for this reason they are not permitted to have one of their larger-Tag Requester Supported bits Set unless the RC supports sufficient larger-Tag Completer capability for them. Thus, software does not need to perform a separate check for RCiEPs.

Non-Flit Mode Switches that lack 10-bit Tag Completer capability are still able to forward NPRs and Completions carrying 10-bit Tags correctly, since the two new Tag bits are in TLP Header bits that were formerly Reserved, and Switches are required to forward Reserved TLP Header bits without modification. However, if such a Switch detects an error with an NPR carrying a 10-bit Tag, and that Switch handles the error by acting as the Completer for the NPR, the resulting Completion will have an invalid 10-bit Tag. Thus, it is strongly recommended that Non-Flit Mode Switches between any components using 10-bit Tags support 10-bit Completer capability. Note that Switches supporting 16.0 GT/s data rates or greater must support 10-bit Tag Completer capability.

18. An exception is PCI Express to PCI/PCI-X Bridges, since larger-Tag capability is not architected for these Functions.

For configurations where a Requester with larger-Tag Requester capability targets Completers where some do and some do not have sufficient larger-Tag Completer capability, how the Requester determines which NPRs include larger Tags is outside the scope of this specification.

IMPLEMENTATION NOTE: USING LARGER TAGS AND SMALLER TAGS CONCURRENTLY §

As stated earlier in this section, if a Requester supports sending larger-Tag Requests to some Completers and smaller-Tag Requests to other Completers concurrently, the Requester must ensure that no outstanding larger Tags can alias to an outstanding smaller Tag if any larger-Tag Request is completed by a Completer that lacks sufficient larger-Tag Completer capability.

For 10-bit Tags, one implementation approach is to have the Requester partition its 8-bit Tag space into 2 regions: one that will only be used for smaller Tags (8-bit or 5-bit Tags), and one that will only be used for the lower 8 bits of 10-bit Tags. Note that this forces a tradeoff between the Tag space available for 10-bit Tags and smaller Tags.

For example, if a Requester partitions its 8-bit Tag space to use only the lowest 4 bits for smaller Tags, this supports up to 16 outstanding smaller Tags, and it reduces the 10-bit Tag space by 3×16 values, supporting $768 - 48 = 720$ outstanding 10-bit Tags. Many other partitioning options are possible, all of which reduce the total number of outstanding Requests. In general, reserving N values for smaller Tags reduces 10-bit Tag space by $3 \times N$ values, and the total for smaller Tags plus 10-bit Tags ends up being $768 - 2 \times N$.

Similar implementation approaches for 14-Bit Tags are possible, and they are straight-forward if only 14-Bit and 8-Bit/5-Bit Tags are supported. If a Requester implementation needs to handle 14-Bit, 10-Bit, and 8-Bit/5-Bit Tag sizes concurrently, the general approach of partitioning the Requester's Tag spaces still works, but the complexity increases significantly.

2.2.6.3 Transaction Descriptor - Attributes Field §

The Attributes field is used to provide additional information that allows modification of the default handling of Transactions. These modifications apply to different aspects of handling the Transactions within the system, such as:

- Ordering
- Hardware coherency management (snoop)

Attributes are hints that allow, but do not require, optimizations in the handling of traffic. The level of optimization support is dependent on the target applications of particular PCI Express peripherals and platform building blocks. In Flit Mode the Attributes Field is contiguous in the TLP Header. In Non-Flit Mode, attribute bit 2 is sometimes labeled A2 and is not adjacent to bits 1 and 0 (see § Figure 2-30 and § Figure 2-31).

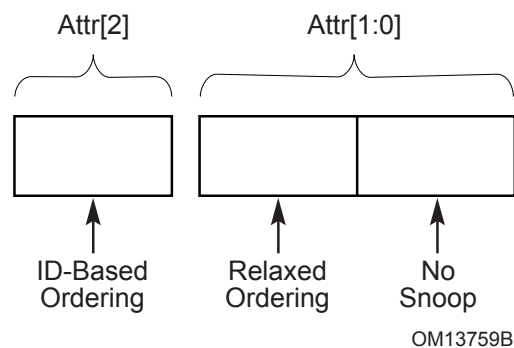


Figure 2-29 Attributes Field of Transaction Descriptor

2.2.6.4 Relaxed Ordering and ID-Based Ordering Attributes §

§ Table 2-12 defines the states of the Relaxed Ordering and ID-Based Ordering attribute fields. These attributes are discussed in § Section 2.4 . Note that Relaxed Ordering and ID-Based Ordering attributes are not adjacent in location (see § Figure 2-5).

§ Table 2-12 Ordering Attributes

Attribute Bit [2]	Attribute Bit [1]	Ordering Type	Ordering Model
0	0	Default Ordering	PCI Strongly Ordered Model
0	1	Relaxed Ordering	PCI-X Relaxed Ordering Model
1	0	ID-Based Ordering	Independent ordering based on Requester/Completer ID
1	1	Relaxed Ordering plus ID-Based Ordering	Logical “OR” of Relaxed Ordering and IDO

Attribute bit [1] is not applicable and must be Clear for Configuration Requests, I/O Requests, Memory Requests that are Message Signaled Interrupts, and Message Requests (except where specifically permitted).

Attribute bit [2], IDO, is Reserved for Configuration Requests and I/O Requests. IDO is not Reserved for all Memory Requests, including Message Signaled Interrupts (MSI/MSI-X). IDO is not Reserved for Message Requests unless specifically prohibited. A Requester is permitted to Set IDO only if the IDO Request Enable bit in the Device Control 2 register is Set.

The value of the IDO bit must not be considered by Receivers when determining if a TLP is a Malformed Packet.

A Completer is permitted to Set IDO only if the IDO Completion Enable bit in the Device Control 2 register is Set. It is not required to copy the value of IDO from the Request into the Completion(s) for that Request. If the Completer has IDO enabled, it is recommended that the Completer set IDO for all Completions, unless there is a specific reason not to (see § Appendix E.).

A Root Complex that supports forwarding TLPs peer-to-peer between Root Ports is not required to preserve the IDO bit from the Ingress to Egress Port.

2.2.6.5 No Snoop Attribute §

§ Table 2-13 defines the states of the No Snoop attribute field. Note that the No Snoop attribute does not alter Transaction ordering.

Table 2-13 Cache Coherency Management Attribute

No Snoop Attribute (b)	Cache Coherency Management Type	Coherency Model
0	Default	Hardware enforced cache coherency expected
1	No Snoop	Hardware enforced cache coherency not expected

This attribute is not applicable and must be Clear for Configuration Requests, I/O Requests, Memory Requests that are Message Signaled Interrupts, and Message Requests (except where specifically permitted).

2.2.6.6 Transaction Descriptor - Traffic Class Field §

The Traffic Class (TC) is a 3-bit field that allows differentiation of transactions into eight traffic classes.

Together with the PCI Express Virtual Channel support, the TC mechanism is a fundamental element for enabling differentiated traffic servicing. Every PCI Express Transaction Layer Packet uses TC information as an Invariant label that is carried end to end within the PCI Express fabric. As the packet traverses across the fabric, this information is used at every Link and within each Switch element to make decisions with regards to proper servicing of the traffic. A key aspect of servicing is the routing of the packets based on their TC labels through corresponding Virtual Channels. § Section 2.5 covers the details of the VC mechanism.

§ Table 2-14 defines the TC encodings.

Table 2-14 Definition of TC Field Encodings

TC Field Value (b)	Definition
000	TC0: Best Effort service class (General Purpose I/O) (Default TC - must be supported by every PCI Express device)
001 to 111	TC1 to TC7: Differentiated service classes (Differentiation based on Weighted-Round-Robin (WRR) and/or priority)

It is up to the system software to determine TC labeling and TC/VC mapping in order to provide differentiated services that meet target platform requirements.

The concept of Traffic Class applies only within the PCI Express interconnect fabric. Specific requirements of how PCI Express TC service policies are translated into policies on non-PCI Express interconnects is outside of the scope of this specification.

2.2.7 Memory, I/O, and Configuration Request Rules §

The general requirements for Memory, I/O, and Configuration Requests similar in Non-Flit Mode and Flit Mode, however some specific rules differ. Rules that are common between Non-Flit Mode and Flit-Mode follow, with rules that are specific to each in subsequent sub-sections.

2.2.7.1 Non-Flit Mode §

The following rule applies to all Memory, I/O, and Configuration Requests. Additional rules specific to each type of Request follow.

- All Memory, I/O, and Configuration Requests include the following fields in addition to the common header fields:
 - Requester ID[15:0] and Tag[9:0], forming the Transaction ID. In Non-Flit Mode, the Tag field is 10 bits.
 - Last DW BE[3:0] and First DW BE[3:0]. For Memory Read Requests, DMWr Requests, and AtomicOp Requests with the TH bit Set, the byte location for the Last DW BE[3:0] and First DW BE [3:0] fields in the header are repurposed to carry ST[7:0] field.
 - For Memory Read Requests and DMWr Requests with the TH bit Clear, see § Section 2.2.5 for First/Last DW Byte Enable Rules.
 - For AtomicOp Requests and DMWr Requests with TH bit Set, the values for the DW BE fields are implied to be Reserved. For AtomicOp Requests with TH bit Clear, the DW BE fields are Reserved.

For Memory Requests, the following rules apply:

- Memory Requests route by address, using either 64-bit or 32-bit Addressing (see § Figure 2-30 and § Figure 2-31).
- For Memory Read Requests, Length must not exceed the value specified by Max_Read_Request_Size (see § Section 7.5.3.4).
- For AtomicOp Requests, architected operand sizes and their associated Length field values are specified in § Table 2-15. If a Completer supports AtomicOps, the following rules apply. The Completer must check the Length field value. If the value does not match an architected value, the Completer must handle the TLP as a Malformed TLP. Otherwise, if the value does not match an operand size that the Completer supports, the Completer must handle the TLP as an Unsupported Request (UR). This is a reported error associated with the Receiving Port (see § Section 6.2).

Table 2-15 Length Field Values for AtomicOp Requests

AtomicOp Request	Length Field Value for Architected Operand Sizes		
	32 Bits	64 Bits	128 Bits
FetchAdd, Swap	1 DW	2 DW	N/A
CAS	2 DW	4 DW	8 DW

- A FetchAdd Request contains one operand, the “add” value.
- A Swap Request contains one operand, the “swap” value.
- A CAS Request contains two operands. The first in the data area is the “compare” value, and the second is the “swap” value.

- For AtomicOp Requests, the Address must be naturally aligned with the operand size. The Completer must check for violations of this rule. If a TLP violates this rule, the TLP is a Malformed TLP. This is a reported error associated with the Receiving Port (see § Section 6.2).
- Requests must not specify an Address/Length combination that causes a Memory Space access to cross a 4-KB boundary.
 - Receivers may optionally check for violations of this rule. If a Receiver implementing this check determines that a TLP violates this rule, the TLP is a Malformed TLP.
 - If checked, this is a reported error associated with the Receiving Port (see § Section 6.2).
 - For AtomicOp Requests, the mandatory Completer check for natural alignment of the Address (see above) already guarantees that the access will not cross a 4-KB boundary, so a separate 4-KB boundary check is not necessary.
 - If a 4-KB boundary check is performed for AtomicOp CAS Requests, this check must comprehend that the TLP Length value is based on the size of two operands, whereas the access to Memory Space is based on the size of one operand.

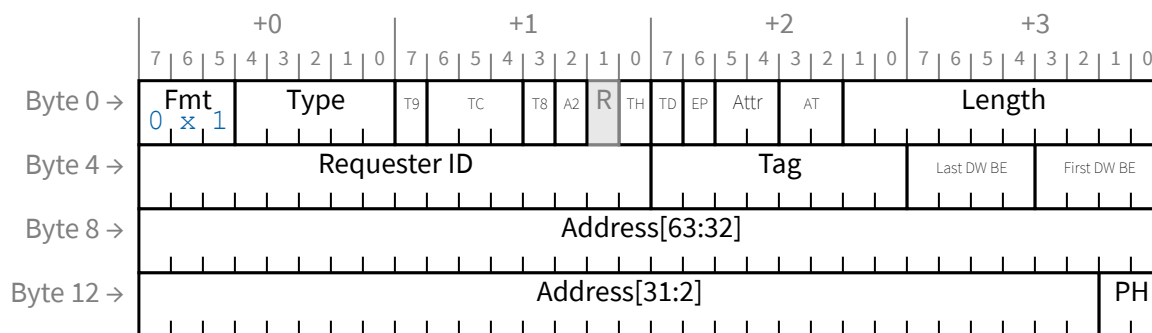


Figure 2-30 Request Header Format for 64-bit Addressing of Memory

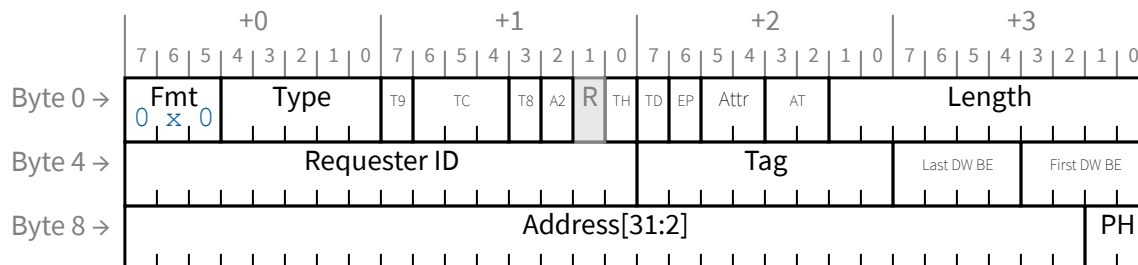


Figure 2-31 Request Header Format for 32-bit Addressing of Memory

IMPLEMENTATION NOTE: GENERATION OF 64-BIT ADDRESSES §

It is strongly recommended that PCI Express Endpoints be capable of generating the full range of 64-bit addresses. However, if a PCI Express Endpoint supports a smaller address range, and is unable to reach the full address range required by a given platform environment, the corresponding device driver must ensure that all Memory Transaction target buffers fall within the address range supported by the Endpoint. The exact means of ensuring this is platform and operating system specific, and beyond the scope of this specification.

For I/O Requests, the following rules apply:

- I/O Requests route by address, using 32-bit Addressing (see § Figure 2-32)
- I/O Requests have the following restrictions:
 - TC[2:0] must be 000b
 - TH is not applicable to I/O Request and the bit is Reserved
 - Attr[2] is Reserved
 - Attr[1:0] must be 00b
 - AT[1:0] must be 00b. Receivers are not required or encouraged to check this.
 - Length[9:0] must be 00 0000 0001b
 - Last DW BE[3:0] must be 0000b

Receivers may optionally check for violations of these rules (but must not check Reserved bits). These checks are independently optional (see § Section 6.2.3.4). If a Receiver implementing these checks determines that a TLP violates these rules, the TLP is a Malformed TLP.

- If checked, this is a reported error associated with the Receiving Port (see § Section 6.2).

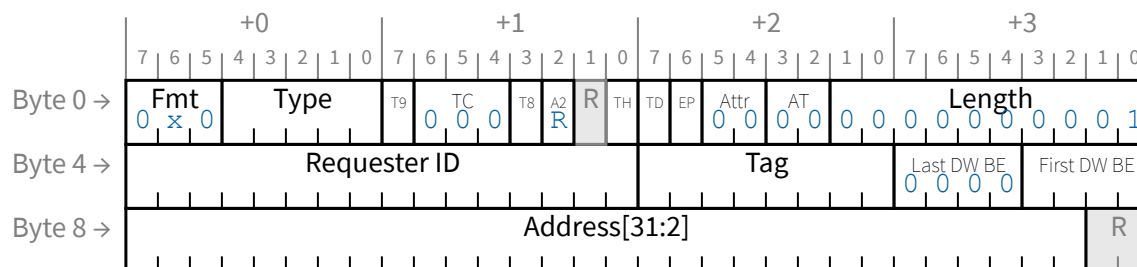


Figure 2-32 Request Header Format for I/O Transactions - Non-Flit Mode

For Configuration Requests, the following rules apply:

- Configuration Requests route by ID, and use a 3 DW header.
- In addition to the header fields included in all Memory, I/O, and Configuration Requests and the ID routing fields, Configuration Requests contain the following additional fields (see § Figure 2-33).
 - Register Number[5:0]

- Extended Register Number[3:0]
- Configuration Requests have the following restrictions:
 - TC[2:0] must be 000b
 - TH is not applicable to Configuration Requests and the bit is Reserved
 - Attr[2] is Reserved
 - Attr[1:0] must be 00b
 - AT[1:0] must be 00b. Receivers are not required or encouraged to check this.
 - Length[9:0] must be 00 0000 0001b
 - Last DW BE[3:0] must be 0000b

Receivers may optionally check for violations of these rules (but must not check reserved bits). These checks are independently optional (see § Section 6.2.3.4). If a Receiver implementing these checks determines that a TLP violates these rules, the TLP is a Malformed TLP.

- If checked, this is a reported error associated with the Receiving Port (see § Section 6.2).

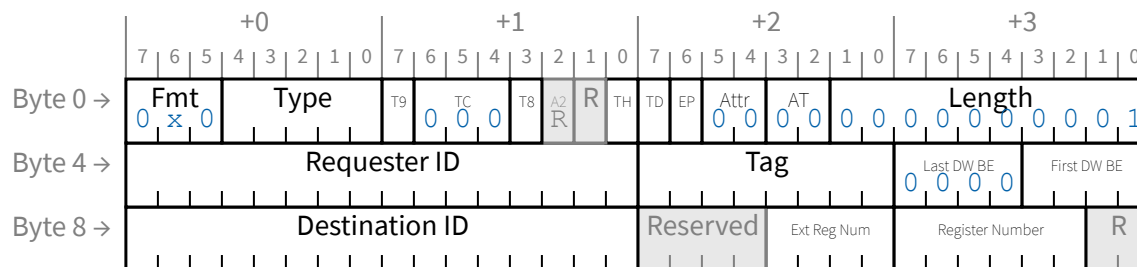
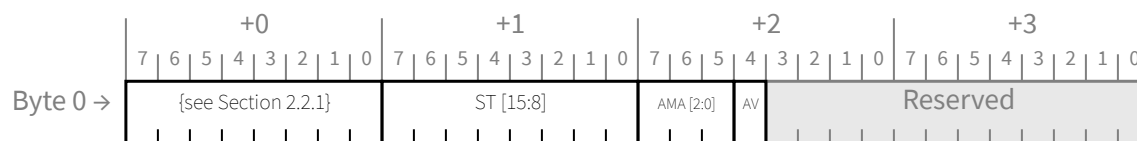


Figure 2-33 Request Header Format for Configuration Transactions - Non-Flit Mode

MSI/MSI-X mechanisms use Memory Write Requests to represent interrupt Messages (see § Section 6.1.4). The Request format used for MSI/MSI-X transactions is identical to the Memory Write Request format defined above, and MSI/MSI-X Requests are indistinguishable from memory writes with regard to ordering, Flow Control, and data integrity.

2.2.7.1.1 TPH Rules §

- Two formats are specified for TPH. The Baseline TPH format (see § Figure 2-35 and § Figure 2-36) must be used for all Requests that provide TPH. The format with the optional TPH TLP Prefix extends the TPH fields (see § Figure 2-34) to provide additional bits for the Steering Tag (ST) field.



§

Figure 2-34 TPH TLP Prefix

- The optional TPH TLP Prefix is used to provide additional TPH information.
 - The presence of a TPH TLP Prefix is determined by decoding byte 0.

*Table 2-16 TPH TLP Prefix
Bit Mapping*

Fields	TPH TLP Prefix
ST[15:8]	Bits 7:0 of byte 1
AMA[2:0]	Bits 7:5 of byte 2
AV	Bit 4 of byte 2
Reserved	Bits 3:0 of byte 2
Reserved	Bits 7:0 of byte 3

- The TPH TLP Prefix is used to send a non-Zero value for any of:
 - AMA
 - ST[15:8]
- For Requests that target Memory Space, a value of 1b in the TH bit indicates the presence of TPH in the TLP header and optional TPH TLP Prefix (if present).
 - The TH bit must be Set for Requests that provide TPH.
 - The TH bit is permitted to be Set for Requests with a TPH TLP Prefix. When the TH bit is 1b, then ST[15:8] is present and meaningful in the TPH TLP Prefix.
 - When the TH bit is Clear, the PH field is Reserved.
 - The TH bit and the PH field are not applicable and are Reserved for all other Requests.
- For Requests that target Memory Space, the TPH TLP Prefix may be present if the value of the TH bit is 0b. When the AMA Valid (AV) bit is 1b and the TPH TLP Prefix is present, AMA is present and meaningful in the TPH TLP Prefix.
- For Requests that target Memory Space with the AT field not set to 10b, the AMA field in the TPH TLP Prefix is Reserved.
- The Processing Hints (PH) fields mapping is shown in § Figure 2-35, § Figure 2-36 and § Table 2-17.

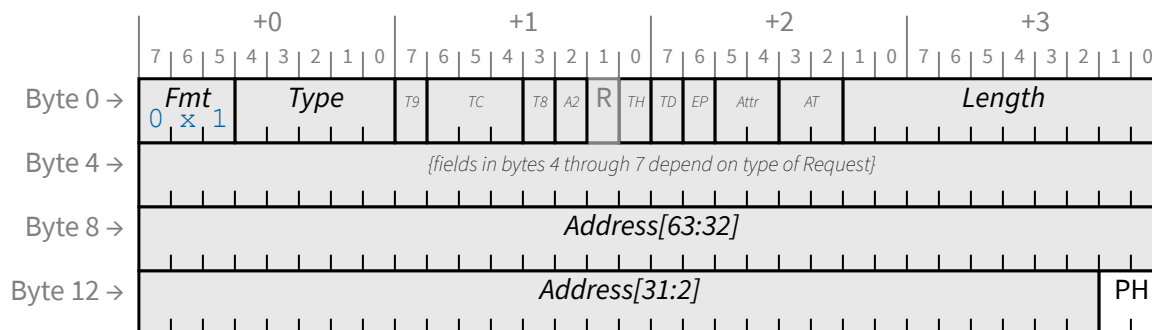


Figure 2-35 Location of PH[1:0] in a 4 DW Request Header - Non-Flit Mode

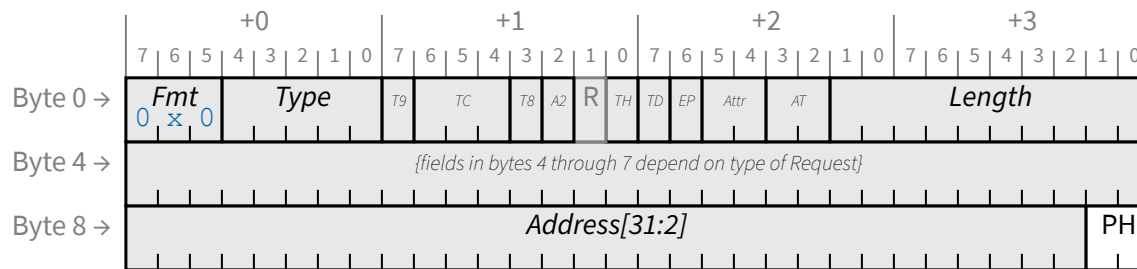


Figure 2-36 Location of PH[1:0] in a 3 DW Request Header - Non-Flit Mode

Table 2-17 Location of PH[1:0] in TLP Header

PH	32-bit Addressing	64-bit Addressing
1:0	Bits 1:0 of Byte 11	Bits 1:0 of Byte 15

- The PH[1:0] field provides information about the data access patterns and is defined as described in [§ Table 2-18](#).

§ Table 2-18 Processing Hint Encoding

PH[1:0] (b)	Processing Hint	Description
00	Bi-directional data structure	Indicates frequent read and/or write access to data by Host and device
01	Requester	Indicates frequent read and/or write access to data by device
10	Target	Indicates frequent read and/or write access to data by Host
11	Target with Priority	Indicates frequent read and/or write access by Host and indicates high temporal locality for accessed data

The Steering Tag (ST) fields are mapped to the TLP header as shown in [§ Figure 2-37](#), [§ Figure 2-38](#) and [§ Table 2-19](#).

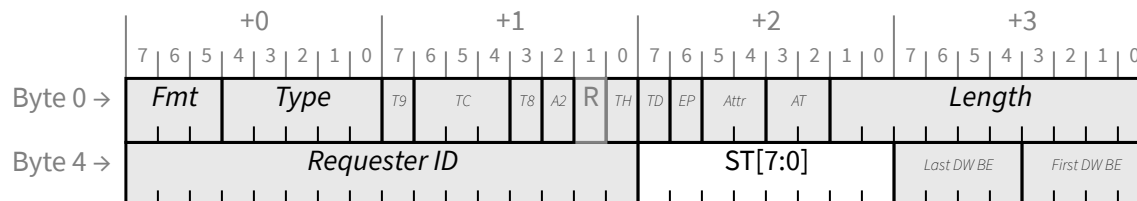


Figure 2-37 Location of ST[7:0] in the Memory Write Request Header - Non-Flit Mode

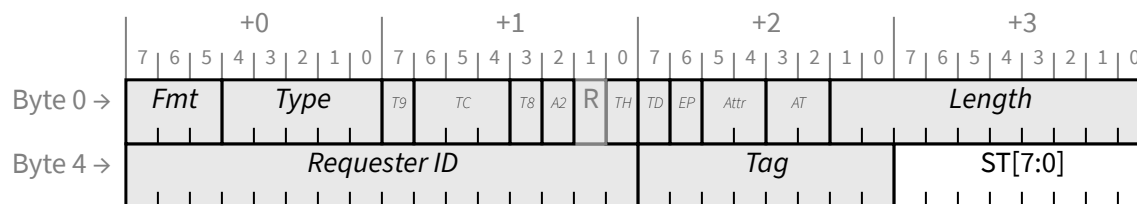


Figure 2-38 Location of ST[7:0] in Memory Read, DMWr, and AtomicOp Request Headers - Non-Flit Mode

Table 2-19 Location of ST[7:0] in TLP Headers

ST Bits	Memory Write Request	Memory Read Request or AtomicOp Request
7:0	Bits 7:0 of Byte 6	Bits 7:0 of Byte 7

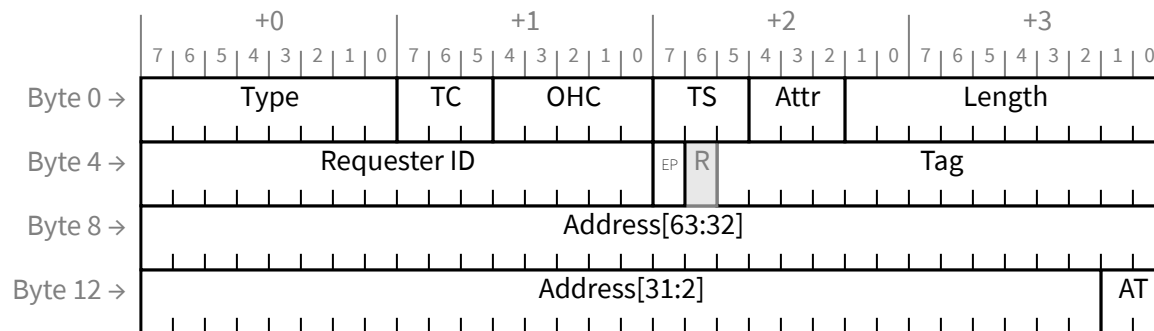
- ST[7:0] field carries the Steering Tag value
 - A value of Zero indicates no Steering Tag preference
 - A total of 255 unique Steering Tag values are provided
- A Function that does not support the TPH Completer or Routing capability and receives a transaction with the TH bit Set is required to ignore the TH bit and handle the Request in the same way as Requests of the same transaction type without the TH bit Set.

2.2.7.2 Flit Mode §

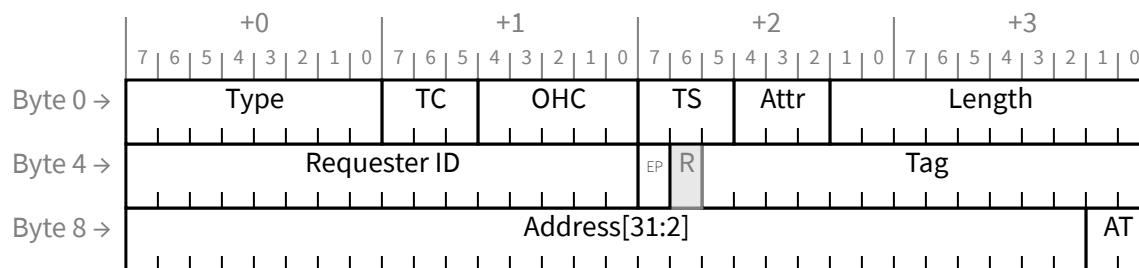
Except as stated, rules that apply in Non-Flit Mode also apply in Flit Mode.

- All Memory, I/O, and Configuration Requests include the following fields in addition to the common header fields:
 - A Transaction ID, consisting of Requester ID[15:0] and Tag[13:0], and, for Memory Requests, sometimes also including the Requester Segment[7:0]
- Byte Enable rules are in § Section 2.2.5.2 .
- For Memory Requests, including AtomicOp and DMWr, the rules for the formation and processing of Header Fields are the same as in Non-Flit Mode.
- For IO Requests, the rules for the formation and processing of Header Fields are the same as in Non-Flit Mode.
- Configuration Requests must include OHC-A3.
- Configuration Requests must only include OHC-C when they are associated with an IDE stream.

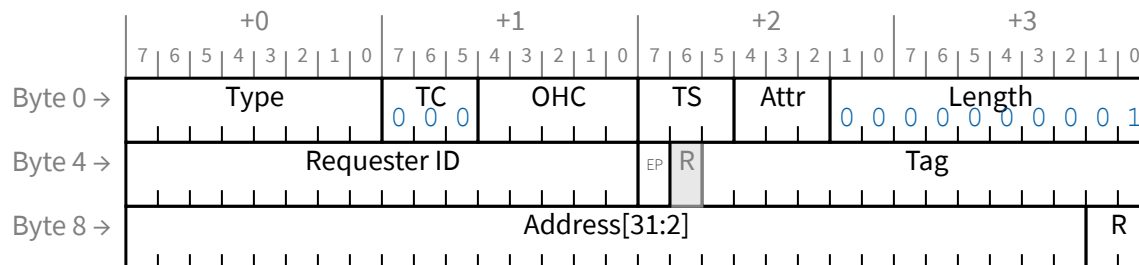
The following figures illustrate Flit Mode Request Headers.



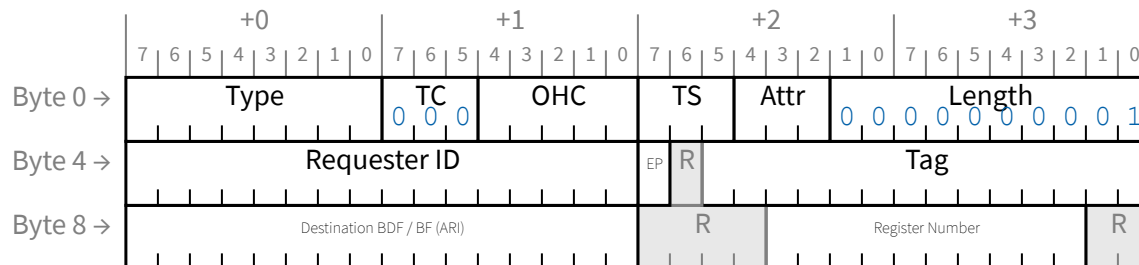
§ Figure 2-39 Flit Mode Mem64 Request



§ Figure 2-40 Flit Mode Mem32 Request



§ Figure 2-41 Flit Mode IO Request



§ Figure 2-42 Flit Mode Configuration Request

2.2.8 Message Request Rules §

This document defines the following groups of Messages:

- INTx Interrupt Signaling
- Power Management
- Error Signaling
- Locked Transaction Support
- Slot Power Limit Support
- Vendor-Defined Messages
- Latency Tolerance Reporting (LTR) Messages
- Optimized Buffer Flush/Fill (OBFF) Messages
- Device Readiness Status (DRS) Messages
- Function Readiness Status (FRS) Messages
- Hierarchy ID Messages
- Precision Time Measurement (PTM) Messages
- Integrity and Data Encryption (IDE) Messages

The following rules apply to all Message Requests. Additional rules specific to each type of Message follow.

- All Message Requests include the following fields in addition to the common header fields (see § Figure 2-43 and § Figure 2-44):
 - Requester ID[15:0]
 - Message Code[7:0] - Indicates the particular Message embodied in the Request.
 - EP - For Messages with data only, indicates data payload is poisoned (see § Section 2.7); Reserved for Messages without data.
- All Message Requests use the Msg or MsgD TLP Type.
- The Message Code field must be fully decoded (Message aliasing is not permitted).
- The Attr[2] field is not Reserved unless specifically indicated as Reserved.
- Except as noted, the Attr[1:0] field is Reserved.
- Except as noted, TH is not applicable to Message Requests and the bit is Reserved.
- AT[1:0] must be 00b. Receivers are not required or encouraged to check this.
- Bytes 8 through 15 are Reserved unless specifically defined.
- Message Requests are posted and do not require Completion.
- Message Requests follow the same ordering rules as Memory Write Requests.

Many types of Messages, including Vendor-Defined Messages, are potentially usable in non-D0 states, and it is strongly recommended that the handling of Messages by Ports be the same when the Port's Bridge Function is in D1, D2, and D3_{Hot} as it is in D0. It is strongly recommended that Type 0 Functions support the generation and reception of Messages in non-D0 states.

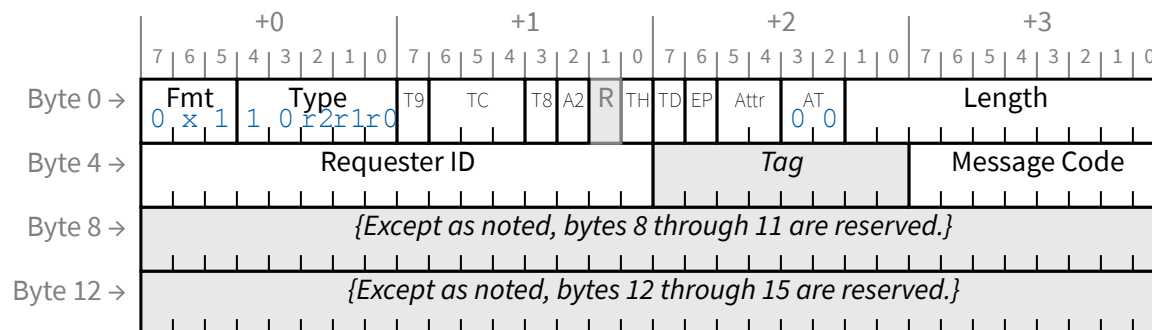


Figure 2-43 Message Request Header - Non-Flit Mode

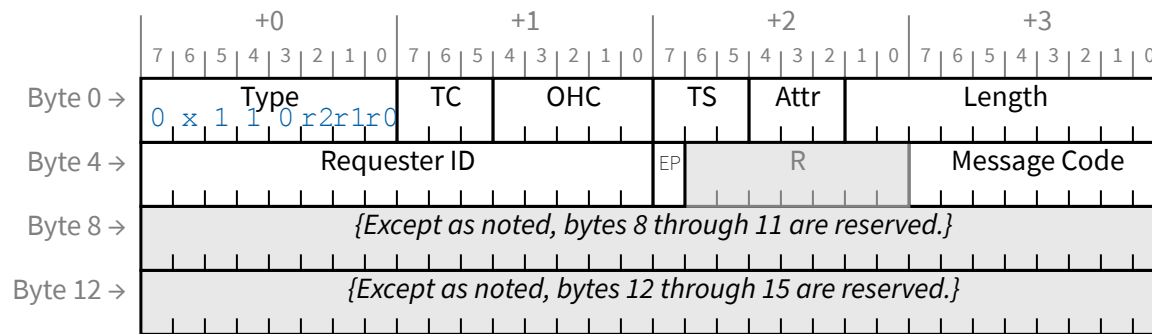


Figure 2-44 Message Request Header - Flit Mode

In addition to address and ID routing, Messages support several other routing mechanisms. These mechanisms are referred to as “implicit” because no address or ID specifies the destination, but rather the destination is implied by the routing type. The following rules cover Message routing mechanisms:

- Message routing is determined using the r[2:0] sub-field of the Type field
 - Message Routing r[2:0] values are defined in § Table 2-20
 - Permitted values are defined in the following sections for each Message

Table 2-20 Message Routing

r[2:0] (b)	Description	Bytes 8 to 15 ¹⁹
000	Routed to Root Complex	Reserved
001	Routed by Address ²⁰	Address
010	Routed by ID	See § Section 2.2.4.2
011	Broadcast from Root Complex	Reserved
100	Local - Terminate at Receiver	Reserved

19. Except as noted, e.g., Vendor_Defined Messages.

20. Note that no Messages defined in this document use Address routing.

r[2:0] (b)	Description	Bytes 8 to 15
101	Gathered and routed to Root Complex ²¹	Reserved
110 to 111	Reserved - Terminate at Receiver	Reserved

In Flit Mode, when Route by ID is used and the Destination Segment is different from the Requester Segment, OHC-A4 must be present and include the Destination Segment in byte 0 and DSV must be Set. DSV is permitted to be Set when the Destination Segment is the same as the Requester Segment. DSV must be Clear when Route by ID is not used. When DSV is clear, the Destination Segment field must be set to 00h. OHC-A4 must be present for Route by ID Messages that require PASID.

2.2.8.1 INTx Interrupt Signaling - Rules §

A Message Signaled Interrupt (MSI or MSI-X) is the preferred interrupt signaling mechanism in PCI Express (see § Section 6.1). However, in some systems, there may be Functions that cannot support the MSI or MSI-X mechanisms. The INTx virtual wire interrupt signaling mechanism is used to support Legacy Endpoints and PCI Express/PCI(-X) Bridges in cases where the MSI or MSI-X mechanisms cannot be used. Switches must support this mechanism. The following rules apply to the INTx Interrupt Signaling mechanism:

- The INTx mechanism uses eight distinct Messages (see § Table 2-21).
- Assert_INTx/Deassert_INTx Messages do not include a data payload (TLP Type is Msg).
- The Length field is Reserved.
- With Assert_INTx/Deassert_INTx Messages, the Function Number field in the Requester ID must be 0. Note that the Function Number field is a different size for non-ARI and ARI Requester IDs.
- Assert_INTx/Deassert_INTx Messages are only issued by Upstream Ports.
 - Receivers may optionally check for violations of this rule. If a Receiver implementing this check determines that an Assert_INTx/Deassert_INTx violates this rule, it must handle the TLP as a Malformed TLP.
 - This is a reported error associated with the Receiving Port (see § Section 6.2).
- Assert_INTx and Deassert_INTx interrupt Messages must use the default Traffic Class designator (TC0). Receivers must check for violations of this rule. If a Receiver determines that a TLP violates this rule, it must handle the TLP as a Malformed TLP.
 - This is a reported error associated with the Receiving Port (see § Section 6.2).

§ Table 2-21 INTx Mechanism Messages

Name	Code[7:0] (b)	Routing r[2:0] (b)	Support ²²				Description/Comments
			RC	Ep	Sw	Br	
Assert_INTA	0010 0000	100	All:				Assert INTA virtual wire Note: These Messages are used for Conventional PCI-compatible INTx emulation.
			r		tr		
			As Required:				
				t		t	

21. This routing type is used only for PME_TO_Ack, and is described in § Section 5.3.3.2.1 .

22. Abbreviations: RC = Root Complex Sw = Switch (only used with “Link” routing) Ep = Endpoint Br = PCI Express (primary) to PCI/PCI-X (secondary) Bridge r = Supports as Receiver t = Supports as Transmitter

Name	Code[7:0] (b)	Routing r[2:0] (b)	Support				Description/Comments
			RC	Ep	Sw	Br	
Assert_INTB	0010 0001	100	All:				Assert INTB virtual wire
			r		tr		
			As Required:				
				t		t	
Assert_INTC	0010 0010	100	All:				Assert INTC virtual wire
			r		tr		
			As Required:				
				t		t	
Assert_INTD	0010 0011	100	All:				Assert INTD virtual wire
			r		tr		
			As Required:				
				t		t	
Deassert_INTA	0010 0100	100	All:				Deassert INTA virtual wire
			r		tr		
			As Required:				
				t		t	
Deassert_INTB	0010 0101	100	All:				Deassert INTB virtual wire
			r		tr		
			As Required:				
				t		t	
Deassert_INTC	0010 0110	100	All:				Deassert INTC virtual wire
			r		tr		
			As Required:				
				t		t	
Deassert_INTD	0010 0111	100	All:				Deassert INTD virtual wire
			r		tr		
			As Required:				
				t		t	

The Assert_INTx/Deassert_INTx Message pairs constitute four “virtual wires” for each of the legacy PCI interrupts designated A, B, C, and D. The following rules describe the operation of these virtual wires:

- The components at both ends of each Link must track the logical state of the four virtual wires using the Assert/Deassert Messages to represent the active and inactive transitions (respectively) of each corresponding virtual wire.
 - An Assert_INTx represents the active going transition of the INTx (x = A, B, C, or D) virtual wire
 - A Deassert_INTx represents the inactive going transition of the INTx (x = A, B, C, or D) virtual wire
- When the local logical state of an INTx virtual wire changes at an Upstream Port, the Port must communicate this change in state to the Downstream Port on the other side of the same Link using the appropriate Assert_INTx or Deassert_INTx Message.

Note: Duplicate Assert_INTx/Deassert_INTx Messages have no effect, but are not errors.

- INTx Interrupt Signaling is disabled when the Interrupt Disable bit of the Command register (see § Section 7.5.1.1.3) is Set.
 - Any INTx virtual wires that are active when the Interrupt Disable bit is Set must be deasserted by transmitting the appropriate Deassert_INTx Message(s).
- Virtual and actual PCI to PCI Bridges must map the virtual wires tracked on the secondary side of the Bridge according to the Device Number of the device on the secondary side of the Bridge, as shown in § Table 2-22.
- Switches must track the state of the four virtual wires independently for each Downstream Port, and present a “collapsed” set of virtual wires on its Upstream Port.
- If a Switch Downstream Port goes to DL_Down status, the INTx virtual wires associated with that Port must be deasserted, and the Switch Upstream Port virtual wire state updated accordingly.
 - If this results in deassertion of any Upstream INTx virtual wires, the appropriate Deassert_INTx Message(s) must be sent by the Upstream Port.
- The Root Complex must track the state of the four INTx virtual wires independently for each of its Downstream Ports, and map these virtual signals to system interrupt resources.
 - Details of this mapping are system implementation specific.
- If a Downstream Port of the Root Complex goes to DL_Down status, the INTx virtual wires associated with that Port must be deasserted, and any associated system interrupt resource request(s) must be discarded.

Table 2-22 Bridge Mapping for INTx Virtual Wires

Requester ID[7:3] from the Assert_INTx/Deassert_INTx Message received on Secondary Side of Bridge (Interrupt Source ²³) If ARI Forwarding is enabled, the value 0 must be used instead of Requester ID[7:3].	INTx Virtual Wire on Secondary Side of Bridge	Mapping to INTx Virtual Wire on Primary Side of Bridge
0,4,8,12,16,20,24,28	INTA	INTA
	INTB	INTB
	INTC	INTC
	INTD	INTD
1,5,9,13,17,21,25,29	INTA	INTB
	INTB	INTC
	INTC	INTD
	INTD	INTA

23. The Requester ID of an Assert_INTx/Deassert_INTx Message will correspond to the Transmitter of the Message on that Link, and not necessarily to the original source of the interrupt.

Requester ID[7:3] from the Assert_INTx/Deassert_INTx Message received on Secondary Side of Bridge (Interrupt Source) If ARI Forwarding is enabled, the value 0 must be used instead of Requester ID[7:3].	INTx Virtual Wire on Secondary Side of Bridge	Mapping to INTx Virtual Wire on Primary Side of Bridge
2,6,10,14,18,22,26,30	INTA	INTC
	INTB	INTD
	INTC	INTA
	INTD	INTB
3,7,11,15,19,23,27,31	INTA	INTD
	INTB	INTA
	INTC	INTB
	INTD	INTC

IMPLEMENTATION NOTE: SYSTEM INTERRUPT MAPPING §

Note that system software (including BIOS and operating system) needs to comprehend the remapping of legacy interrupts (INTx mechanism) in the entire topology of the system (including hierarchically connected Switches and subordinate PCI Express/PCI Bridges) to establish proper correlation between PCI Express device interrupt and associated interrupt resources in the system interrupt controller. The remapping described by § Table 2-22 is applied hierarchically at every Switch. In addition, PCI Express/PCI and PCI/PCI Bridges perform a similar mapping function.

IMPLEMENTATION NOTE: VIRTUAL WIRE MAPPING FOR INTX INTERRUPTS FROM ARI DEVICES §

The implied Device Number for an ARI Device is 0. When ARI-aware software (including BIOS and operating system) enables ARI Forwarding in the Downstream Port immediately above an ARI Device in order to access its Extended Functions, software must comprehend that the Downstream Port will use Device Number 0 for the virtual wire mappings of INTx interrupts coming from all Functions of the ARI Device. If non-ARI-aware software attempts to determine the virtual wire mappings for Extended Functions, it can come up with incorrect mappings by examining the traditional Device Number field and finding it to be non-0.

2.2.8.2 Power Management Messages §

These Messages are used to support PCI Express power management, which is described in detail in § Chapter 5. . The following rules define the Power Management Messages:

- § Table 2-23 defines the Power Management Messages.

- Power Management Messages do not include a data payload (TLP Type is Msg).
- The Length field is Reserved.
- With PM_Active_State_Nak Messages, the Function Number field in the Requester ID must contain the Function Number of the Downstream Port that sent the Message, or else 000b for compatibility with earlier revisions of this specification.
- With PME_TO_Ack Messages, the Function Number field in the Requester ID must be Reserved, or else for compatibility with earlier revisions of this specification must contain the Function Number of one of the Functions associated with the Upstream Port. Note that the Function Number field is a different size for non-ARI and ARI Requester IDs.
- Power Management Messages must use the default Traffic Class designator (TC0). Receivers must check for violations of this rule. If a Receiver determines that a TLP violates this rule, it must handle the TLP as a Malformed TLP.
 - This is a reported error associated with the Receiving Port (see [§ Section 6.2](#)).

Table 2-23 Power Management Messages

Name	Code[7:0] (b)	Routing r[2:0] (b)	Support				Description/Comments
			RC	Ep	Sw	Br	
<i>PM_Active_State_Nak</i>	0001 0100	100	t	r	tr	r	Terminate at Receiver
<i>PM_PME</i>	0001 1000	000	All:				Sent Upstream by PME-requesting component. Propagates Upstream.
			r		tr	t	
			If PME supported:				
				t			
<i>PME_Turn_Off</i>	0001 1001	011	t	r		r	Broadcast Downstream
<i>PME_TO_Ack</i>	0001 1011	101	r	t		t	Sent Upstream by Upstream Port. See § <u>Section 5.3.3.2.1</u> .
			(Note: Switch handling is special)				

2.2.8.3 Error Signaling Messages §

Error Signaling Messages are used to signal errors that occur on specific transactions and errors that are not necessarily associated with a particular transaction. These Messages are initiated by the agent that detected the error.

- [§ Table 2-24](#) defines the Error Signaling Messages.
- Error Signaling Messages do not include a data payload (TLP Type is Msg).
- The Length field is Reserved.
- With Error Signaling Messages, the Function Number field in the Requester ID must indicate which Function is signaling the error. Note that the Function Number field is a different size for non-ARI and ARI Requester IDs.
- Error Signaling Messages must use the default Traffic Class designator (TC0) Receivers must check for violations of this rule. If a Receiver determines that a TLP violates this rule, it must handle the TLP as a Malformed TLP.

- This is a reported error associated with the Receiving Port (see § Section 6.2).

§ Table 2-24 Error Signaling Messages

Name	Code[7:0] (b)	Routing r[2:0] (b)	Support				Description/Comments
			RC	Ep	Sw	Br	
ERR_COR	0011 0000	000	r	t	tr	t	This Message is issued when the Function or Device detects a correctable error on the PCI Express interface.
ERR_NONFATAL	0011 0001	000	r	t	tr	t	This Message is issued when the Function or Device detects a Non-Fatal, uncorrectable error on the PCI Express interface.
ERR_FATAL	0011 0011	000	r	t	tr	t	This Message is issued when the Function or Device detects a Fatal, uncorrectable error on the PCI Express interface.

The initiator of the Message is identified with the Requester ID of the Message header. The Root Complex translates these error Messages into platform level events. Refer to § Section 6.2 for details on uses for these Messages.

- **ERR_COR** Messages have an **ERR_COR Subclass (ECS)** field in the Message header that enables different subclasses to be distinguished from each other. See § Figure 2-45. **ERR_NONFATAL** and **ERR_FATAL** Messages do not have the ECS field.

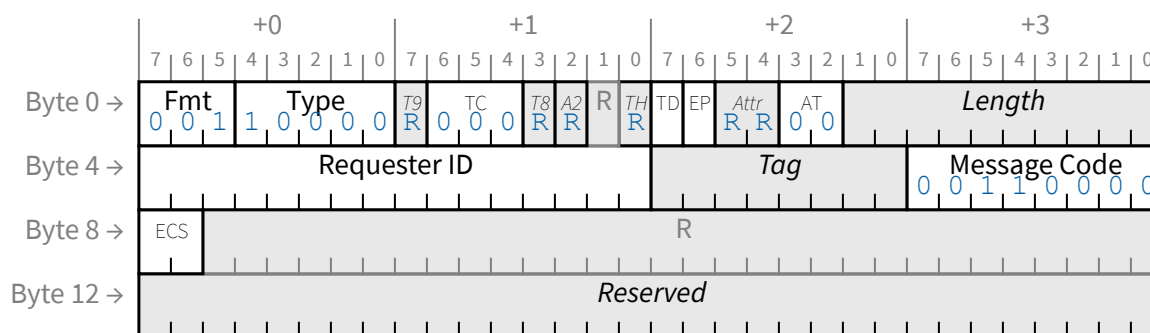


Figure 2-45 **ERR_COR** Message - Non-Flit Mode

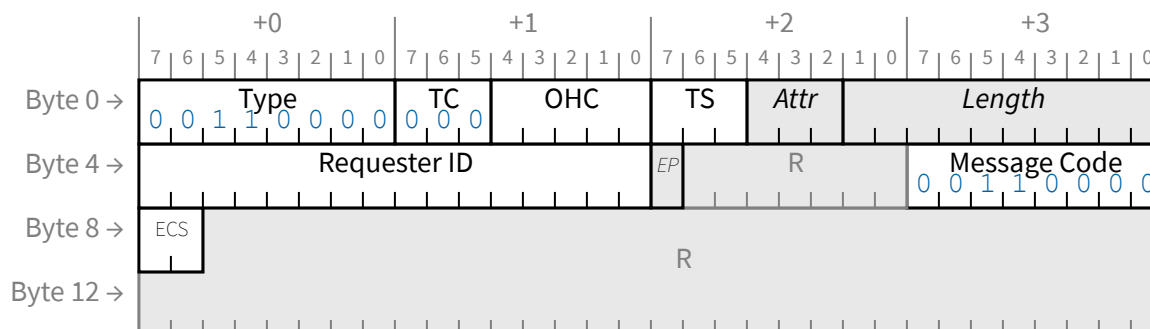


Figure 2-46 **ERR_COR** Message - Flit Mode

- The ERR_COR Subclass (ECS) field is encoded as shown in § Table 2-25, indicating the ERR_COR Message subclass.

Table 2-25 ERR_COR Subclass (ECS) Field Encodings

ECS Coding	Description
00	ECS Legacy - The value inherently used if a Requester does not support ECS capability. ECS-capable Requesters must not use this value. See <u>§ Section 7.5.3.3</u> .
01	ECS SIG_SFW - Must be used by an ECS-capable Requester when signaling a <u>DPC</u> or <u>SFI</u> event with an <u>ERR_COR Message</u> .
10	ECS SIG_OS - Must be used by an ECS-capable Requester when signaling an <u>AER</u> or <u>RP PIO</u> event with an <u>ERR_COR Message</u> .
11	ECS Extended - Intended for possible future use. Requesters must not use this value. Receivers must handle the signal internally the same as <u>ECS SIG_OS</u> .

2.2.8.4 Locked Transactions Support §

The Unlock Message is used to support Lock Transaction sequences. Refer to § Section 6.5 for details on Lock Transaction sequences. The following rules apply to the formation of the Unlock Message:

- § Table 2-26 defines the Unlock Messages.
- The Unlock Message does not include a data payload (TLP Type is Msg).
- The Length field is Reserved.
- With Unlock Messages, the Function Number field in the Requester ID is Reserved.
- The Unlock Message must use the default Traffic Class designator (TC0). Receivers must check for violations of this rule. If a Receiver determines that a TLP violates this rule, it must handle the TLP as a Malformed TLP.
 - This is a reported error associated with the Receiving Port (see § Section 6.2).

Table 2-26 Unlock Message

Name	Code[7:0] (b)	Routing r[2:0] (b)	Support				Description/Comments
			RC	Ep	Sw	Br	
Unlock	0000 0000	011	t	r	tr	r	Unlock Completer

2.2.8.5 Slot Power Limit Support §

This Message is used to convey a slot power limitation value from a Downstream Port (of a Root Complex or a Switch) to an Upstream Port of a component (with Endpoint, Switch, or PCI Express-PCI Bridge Functions) attached to the same Link.

- § Table 2-27 defines the Set_Slot_Power_Limit Message.
- The Set_Slot_Power_Limit Message includes a 1 DW data payload (TLP Type is MsgD).
- The Set_Slot_Power_Limit Message must use the default Traffic Class designator (TC0). Receivers must check for violations of this rule. If a Receiver determines that a TLP violates this rule, it must handle the TLP as a Malformed TLP.

- This is a reported error associated with the Receiving Port (see § Section 6.2).

Table 2-27 **Set_Slot_Power_Limit Message**

Name	Code[7:0] (b)	Routing r[2:0] (b)	Support				Description/Comments
			RC	Ep	Sw	Br	
Set_Slot_Power_Limit	0101 0000	100	t	r	tr	r	Set Slot Power Limit in Upstream Port

The Set_Slot_Power_Limit Message includes a one DW data payload. The data payload is copied from the Slot Capabilities register of the Downstream Port and is written into the Device Capabilities register of the Upstream Port on the other side of the Link. Bits 1:0 of Byte 1 of the data payload map to the Slot Power Limit Scale field and bits 7:0 of Byte 0 map to the Slot Power Limit Value field. Bits 7:0 of Byte 3, 7:0 of Byte 2, and 7:2 of Byte 1 of the data payload must all be set to zero by the Transmitter and ignored by the Receiver. This Message must be sent automatically by the Downstream Port (of a Root Complex or a Switch) when one of the following events occurs:

- On a Configuration Write to the Slot Capabilities register (see § Section 7.5.3.9) when the Data Link Layer reports DL_Up status.
- Any time when a Link transitions from a non-DL_Up status to a DL_Up status (see § Section 2.9.2) and the Auto Slot Power Limit Disable bit is Clear in the Slot Control Register. This transmission is optional if the Slot Capabilities register has not yet been initialized.

The component on the other side of the Link (with Endpoint, Switch, or Bridge Functions) that receives Set_Slot_Power_Limit Message must copy the values in the data payload into the Device Capabilities register associated with the component's Upstream Port. PCI Express components that are targeted exclusively for integration on the system planar (e.g., system board) as well as components that are targeted for integration on an adapter where power consumption of the entire adapter is below the lowest power limit specified for the adapter form factor (as defined in the corresponding form factor specification) are permitted to hardwire the value of all 0's in the Captured Slot Power Limit Scale and Captured Slot Power Limit Value fields of the Device Capabilities Register, and are not required to copy the Set_Slot_Power_Limit Message payload into that register.

For more details on Power Limit control mechanism see § Section 6.9.

2.2.8.6 Vendor_Defined Messages §

The Vendor_Defined Messages allow expansion of PCI Express messaging capabilities, either as a general extension to [PCIe] or a vendor-specific extension. This section defines the rules associated with these Messages generically.

- The Vendor_Defined Messages (see § Table 2-28) use the header format shown in § Figure 2-47.
 - The Requester ID is implementation specific. The Requester ID field *MUST*²⁴ contain the value associated with the Requester.
 - If the Route by ID routing is used, bytes 8 and 9 form a 16-bit field for the destination ID
 - otherwise these bytes are Reserved.
 - Bytes 10 and 11 form a 16-bit field for the Vendor ID, as defined by PCI-SIG®, of the vendor defining the Message.
 - Bytes 12 through 15 are available for vendor definition.

24. ACS Source Validation (see § Section 6.12.1.1) checks the Requester ID on all Requests, including Vendor_Defined Messages. This validation depends on the Requester ID properly identifying the Requester.

§ Table 2-28 Vendor_Defined Messages

Name	Code[7:0] (b)	Routing r[2:0] (b)	Support				Description/Comments
			RC	Ep	Sw	Br	
Vendor_Defined Type 0	0111 1110	000, 010, 011, 100	See Note 1.				Triggers detection of UR by Completer if not implemented.
Vendor_Defined Type 1	0111 1111	000, 010, 011, 100	See Note 1.				Silently discarded by Completer if not implemented.

1. Note 1: Transmission by Endpoint/Root Complex/Bridge is implementation specific. Switches must forward received Messages using Routing r[2:0] field values of 000b, 010b, and 011b.

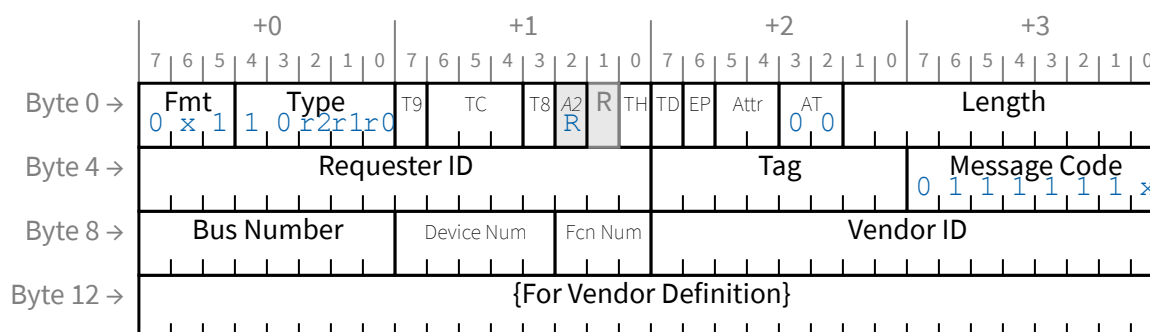


Figure 2-47 Header for Vendor-Defined Messages - Non-Flit Mode

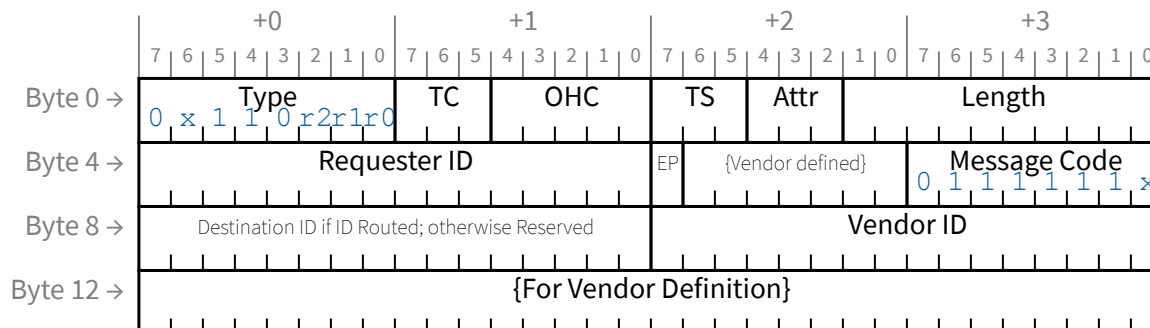


Figure 2-48 Header for Vendor-Defined Messages - Flit Mode

- A data payload may be included with either type of Vendor_Defined Message (TLP type is Msg if no data payload is included or MsgD if a data payload is included).
- For both types of Vendor_Defined Messages, the Attr[1:0] and Attr[2] fields are not Reserved.
- Messages defined by different vendors or by PCI-SIG are distinguished by the value in the Vendor ID field.
 - The further differentiation of Messages defined by a particular vendor is beyond the scope of this document.

- Support for Messages defined by a particular vendor is implementation specific, and beyond the scope of this document.
- Completers silently discard Vendor_Defined Type 1 Messages that they are not designed to receive - this is not an error condition.
- Completers handle the receipt of an unsupported Vendor_Defined Type 0 Message as an Unsupported Request, and the error is reported according to [§ Section 6.2](#).

[PCIe-to-PCI-PCI-X-Bridge] defines additional requirements for Vendor_Defined Messages that are designed to be interoperable with PCI-X Device ID Messages. This includes restrictions on the contents of the Tag[7:0] field and the Length[9:0] field as well as specific use of Bytes 12 through 15 of the message header. Vendor_Defined Messages intended for use solely within a PCI Express environment (i.e., not intended to address targets behind a PCI Express to PCI/PCI-X Bridge) are not subject to the additional rules. Refer to [\[PCIe-to-PCI-PCI-X-Bridge\]](#) for details. Refer to [§ Section 2.2.6.2](#) for considerations regarding larger-Tag capabilities.

2.2.8.6.1 PCI-SIG Defined VDMs [§](#)

PCI-SIG-Defined VDMs are Vendor_Defined Type 1 Messages that use the PCI-SIG® Vendor ID (0001h). As a Vendor_Defined Type 1 Message, each is silently discarded by a Completer if the Completer does not implement it.

Beyond the rules for other Vendor_Defined Type 1 Messages, the following rules apply to the formation of the PCI-SIG-Defined VDMs:

- PCI-SIG-Defined VDMs use the Header format shown in [§ Figure 2-49](#).
- The Requester ID field must contain the value associated with the Requester.
- The Message Code must be 01111111b.
- The Vendor ID must be 0001h, which is assigned to the PCI-SIG.
- The Subtype field distinguishes the specific PCI-SIG-Defined VDMs. See [§ Appendix F](#) for a list of PCI-SIG-Defined VDMs.

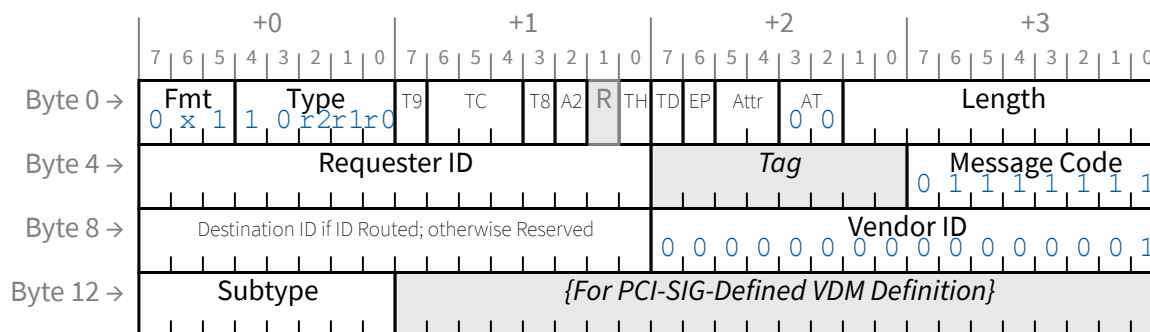


Figure 2-49 Header for PCI-SIG-Defined VDMs - Non-Flit Mode

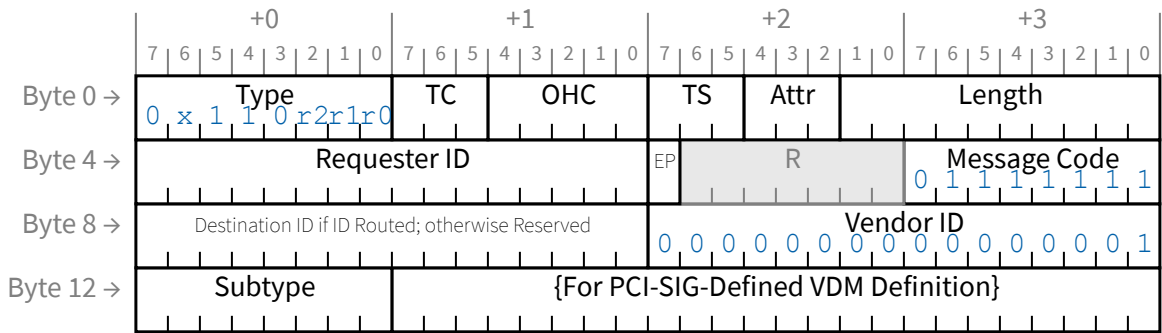


Figure 2-50 Header for PCI-SIG-Defined VDMs - Flit Mode

2.2.8.6.2 Device Readiness Status (DRS) Message §

The Device Readiness Status (DRS) protocol (see § Section 6.22.1) uses the PCI-SIG-Defined VDM mechanism (see § Section 2.2.8.6.1). The DRS Message is a PCI-SIG-Defined VDM (Vendor-Defined Type 1 Message) with no payload.

Beyond the rules for other PCI-SIG-Defined VDMs, the following rules apply to the formation of DRS Messages:

- § Table 2-29 and § Figure 2-51 illustrate and define the DRS Message.
- The TLP Type must be Msg.
- The TC[2:0] field must be 000b.
- The Attr[2:0] field is Reserved.
- The Tag field is Reserved.
- The Subtype field must be 08h.
- The Message Routing field must be set to 100b - Local - Terminate at Receiver.

Receivers may optionally check for violations of these rules (but must not check reserved bits). These checks are independently optional (see § Section 6.2.3.4). If a Receiver implementing these checks determines that a TLP violates these rules, the TLP is a Malformed TLP.

- If checked, this is a reported error associated with the Receiving Port (see § Section 6.2).

§ Table 2-29 DRS Message

Name	Code[7:0] (b)	Routing r[2:0] (b)	Support				Description/Comments
			RC	Ep	Sw	Br	
DRS Message	0111 1111	100	r	t	tr		Device Readiness Status

The format of the DRS Message is shown in § Figure 2-51 below:

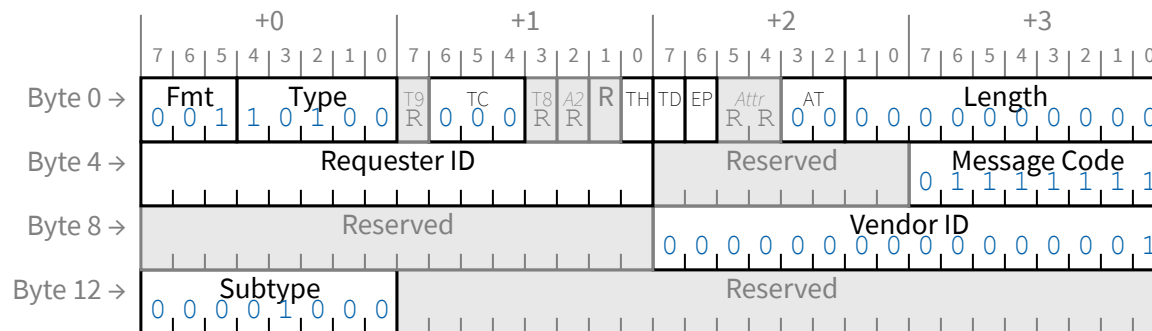
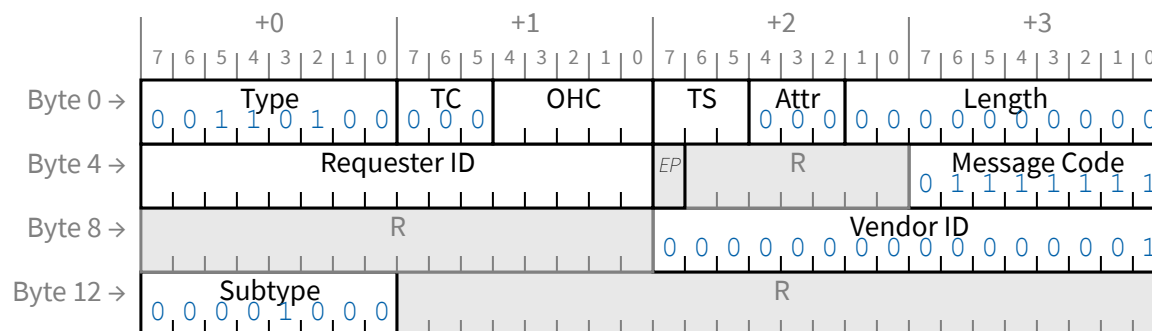


Figure 2-51 DRS Message - Non-Flit Mode



§ Figure 2-52 DRS Message - Flit Mode

2.2.8.6.3 Function Readiness Status Message (FRS Message) §

The Function Readiness Status (FRS) protocol (see § Section 6.22.2) uses the PCI-SIG-Defined VDM mechanism (see § Section 2.2.8.6.1). The FRS message is a PCI-SIG-Defined VDM (Vendor_Defined Type 1 Message) with no payload.

Beyond the rules for other PCI-SIG-Defined VDMs, the following rules apply to the formation of FRS Messages:

§ Table 2-30 and § Figure 2-53 illustrate and define the FRS Message.

- The TLP Type must be Msg.
- The TC[2:0] field must be 000b.
- The Attr[2:0] field is Reserved.
- The Tag field is Reserved.
- The Subtype field must be 09h.
- The FRS Reason[3:0] field indicates why the FRS Message was generated:

0001b: DRS Message Received

The Downstream Port indicated by the Message Requester ID received a DRS Message and has the DRS Signaling Control field in the Link Control Register set to DRS to FRS Signaling Enabled

0010b: D3_{Hot} to D0 Transition Completed

A D3_{Hot} to D0 transition has completed, and the Function indicated by the Message Requester ID is now Configuration-Ready and has returned to the D0_{uninitialized} or D0_{active} state depending on the setting of the No_Soft_Reset bit (see § Section 7.5.2.2)

0011b: FLR Completed

An FLR has completed, and the Function indicated by the Message Requester ID is now Configuration-Ready

1000b: VF Enabled

The Message Requester ID indicates a Physical Function (PF) - All Virtual Functions (VFs) associated with that PF are now Configuration-Ready

1001b: VF Disabled

The Message Requester ID indicates a PF - All VFs associated with that PF have been disabled and the Single Root I/O Virtualization (SR-IOV) data structures in that PF may now be accessed.

Others:

All other values Reserved

- The Message Routing field must be Cleared to 000b - Routed to Root Complex

Receivers may optionally check for violations of these rules (but must not check reserved bits). These checks are independently optional (see § Section 6.2.3.4). If a Receiver implementing these checks determines that a TLP violates these rules, the TLP is a Malformed TLP.

- If checked, this is a reported error associated with the Receiving Port (see § Section 6.2).

§ Table 2-30 FRS Message

Name	Code[7:0] (b)	Routing r[2:0] (b)	Support				Description/Comments
			RC	Ep	Sw	Br	
FRS Message	0111 1111	000	r	t	tr		Function Readiness Status

The format of the FRS Message is shown in § Figure 2-53 and § Figure 2-54 below:

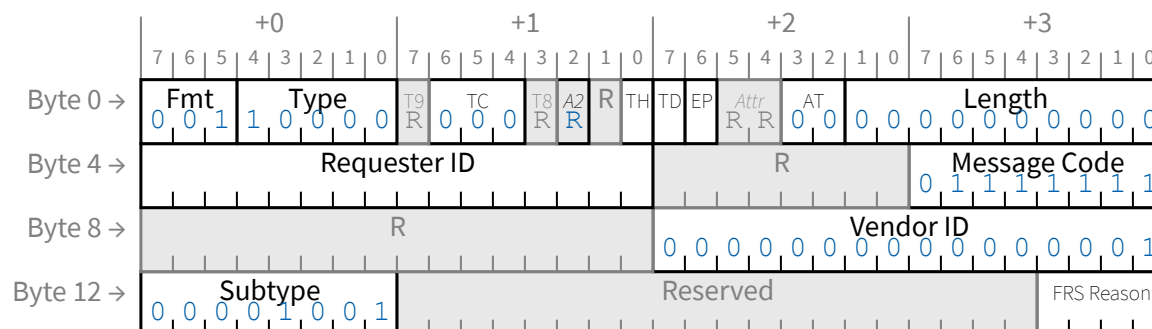
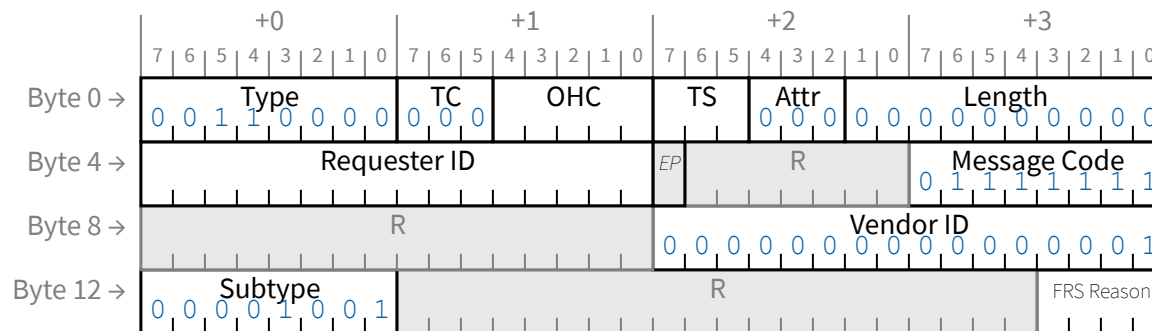


Figure 2-53 FRS Message - Non-Flit Mode



§ Figure 2-54 FRS Message - Flit Mode

2.2.8.6.4 Hierarchy ID Message §

Hierarchy ID uses the PCI-SIG-Defined VDM mechanism (see § Section 2.2.8.6.1). The Hierarchy ID Message is a PCI-SIG-Defined VDM (Vendor_Defined Type 1 Message) with payload (MsgD).

Beyond the rules for other PCI-SIG-Defined VDMs, the following rules apply to the formation of Hierarchy ID Messages:

- § Table 2-31 and § Figure 2-55 illustrate and define the Hierarchy ID Message.
- The TLP Type must be MsgD.
- Each Message must include a 4-DWORD data payload.
- The Length field must be 4.
- The TC[2:0] field must be 000b.
- The Attr[2:0] field is Reserved.
- The Tag field is Reserved.
- The Subtype field is 01h.
- The Message Routing field must be 011b - Broadcast from Root Complex.

Receivers may optionally check for violations of these rules (but must not check reserved bits). These checks are independently optional (see § Section 6.2.3.4). If a Receiver implementing these checks determines that a TLP violates these rules, the TLP is a Malformed TLP.

- If checked, this is a reported error associated with the Receiving Port (see § Section 6.2).

The payload of each Hierarchy ID Message contains the lower 128-bits of the System GUID.

For details of the Hierarchy ID, GUID Authority ID, and System GUID fields see § Section 6.25 .

Table 2-31 Hierarchy ID Message

Name	Code[7:0] (b)	Routing r[2:0] (b)	Support				Description/Comments
			RC	Ep	Sw	Br	
Hierarchy ID Message	0111 1111	011	t	r	tr		Hierarchy ID

The format of the Hierarchy ID Message is shown in § Figure 2-55 and § Figure 2-56 below:

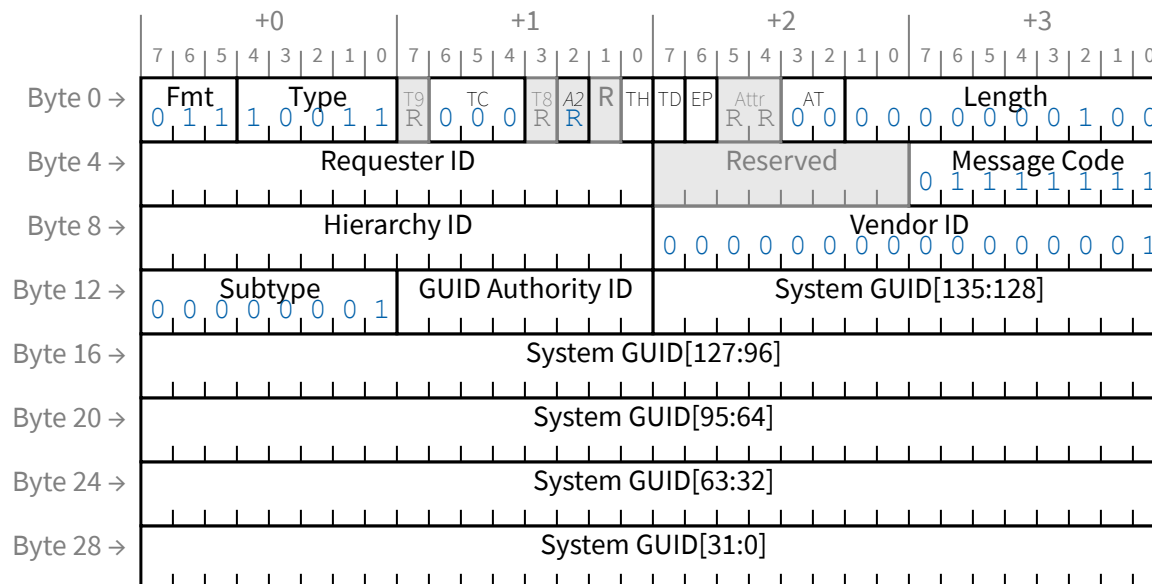


Figure 2-55 Hierarchy ID Message - Non-Flit Mode

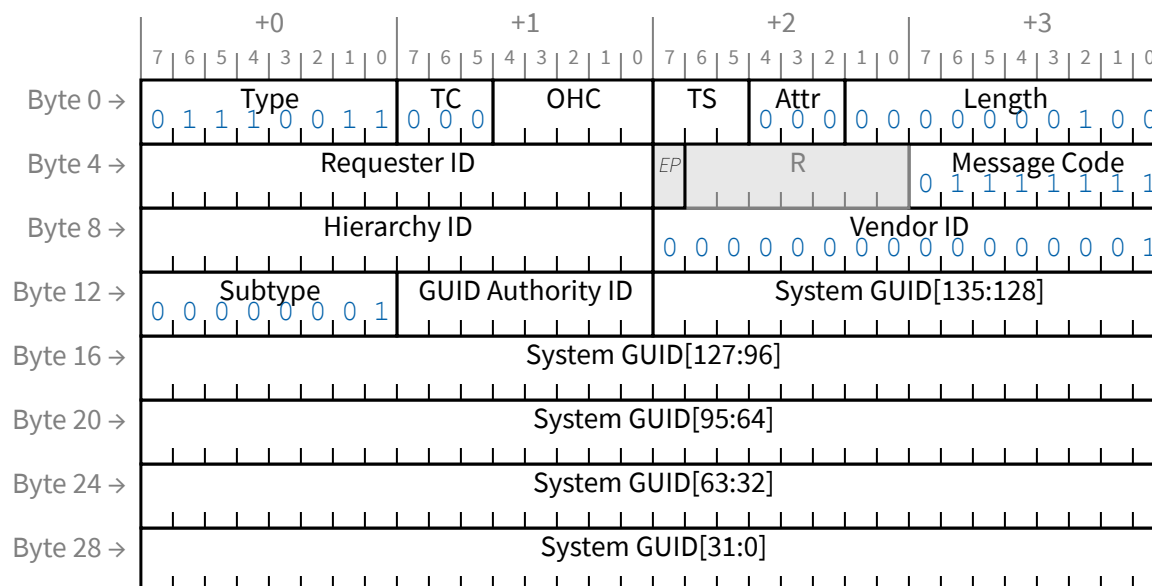


Figure 2-56 Hierarchy ID Message - Flit Mode

2.2.8.7 Ignored Messages §

The messages listed in § Table 2-32 were previously used for a mechanism (Hot-Plug Signaling) that is no longer supported. Transmitters *MUST@FLIT* not transmit these messages. If message transmission is implemented, it must conform to the requirements of [PCIe-1.0a].

Beyond normal Link-Layer processing and mandatory checking for properly-formed TLPs, Receivers *MUST@FLIT* not process these messages further (i.e., carry out their originally architected Transaction-Layer functionality). If complete processing of these messages is implemented, Receivers must process these messages in conformance with the requirements [PCIe-1.0a].

Ignored messages listed in § Table 2-32 are handled by the Receiver as follows:

- The Physical and Data Link Layers must handle these messages identical to handling any other TLP.
- The Transaction Layer must account for flow control credit but take no other action in response to these messages.

§ Table 2-32 Ignored Messages

Name	Code[7:0] (b)	Routing r[2:0] (b)	Support				Description/Comments
			RC	Ep	Sw	Br	
Ignored Message	0100 0001	100					
Ignored Message	0100 0011	100					
Ignored Message	0100 0000	100					
Ignored Message	0100 0101	100					
Ignored Message	0100 0111	100					
Ignored Message	0100 0100	100					
Ignored Message	0100 1000	100					

2.2.8.8 Latency Tolerance Reporting (LTR) Message §

The LTR Message is optionally used to report device behaviors regarding its tolerance of Read/Write service latencies. Refer to § Section 6.18 for details on LTR. The following rules apply to the formation of the LTR Message:

- § Table 2-33 defines the LTR Message.
- The LTR Message does not include a data payload (the TLP Type is Msg).
- The Length field is Reserved.
- The LTR Message must use the default Traffic Class designator (TC0). Receivers that implement LTR support must check for violations of this rule. If a Receiver determines that a TLP violates this rule, it must handle the TLP as a Malformed TLP.
 - This is a reported error associated with the Receiving Port (see § Section 6.2).

§

Table 2-33 LTR Message

Name	Code[7:0] (b)	Routing r[2:0] (b)	Support ¹				Description/Comments
			RC	Ep	Sw	Br	
LTR	0001 0000	100	r	t	tr		Latency Tolerance Reporting

Notes:

Name	Code[7:0] (b)	Routing r[2:0] (b)	Support ¹				Description/Comments
			RC	Ep	Sw	Br	

1. Support for LTR is optional. Functions that support LTR must implement the reporting and enable mechanisms described in § Chapter 7. .

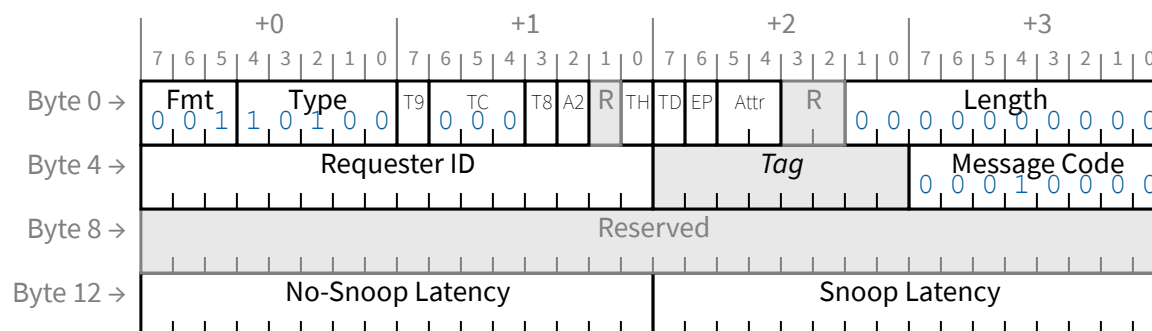
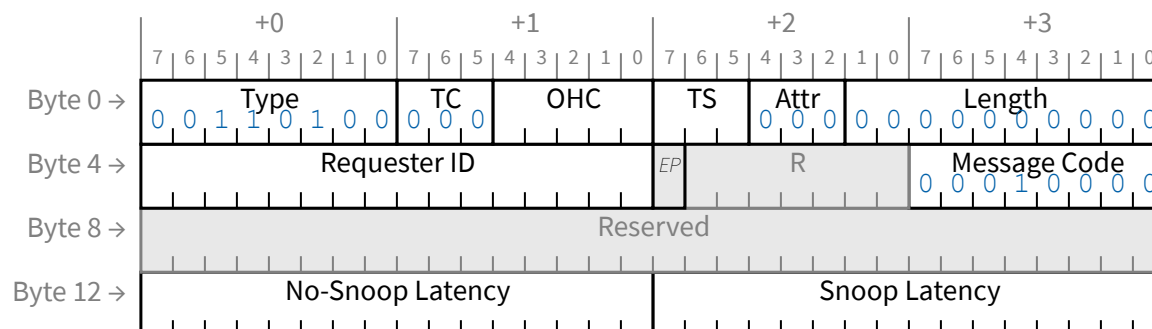


Figure 2-57 LTR Message - Non-Flit Mode



§ Figure 2-58 LTR Message - Flit Mode

2.2.8.9 Optimized Buffer Flush/Fill (OBFF) Message §

The OBFF Message is optionally used to report platform central resource states to Endpoints. This mechanism is described in detail in § Section 6.19 .

The following rules apply to the formation of the OBFF Message:

- § Table 2-34 defines the OBFF Message.
- The OBFF Message does not include a data payload (TLP Type is Msg).
- The Length field is Reserved.
- The Requester ID must be set to the Transmitting Port's ID.

- The OBFF Message must use the default Traffic Class designator (TC0). Receivers that implement OBFF support must check for violations of this rule. If a Receiver determines that a TLP violates this rule, it must handle the TLP as a Malformed TLP.
 - This is a reported error associated with the Receiving Port (see § Section 6.2).

§ *Table 2-34 OBFF Message*

Name	Code[7:0] (b)	Routing r[2:0] (b)	Support ¹				Description/Comments
			RC	Ep	Sw	Br	
OBFF	0001 0010	100	t	r	tr		Optimized Buffer Flush/Fill

Notes:

- Support for OBFF is optional. Functions that support OBFF must implement the reporting and enable mechanisms described in § Chapter 7., Software Initialization and Configuration.

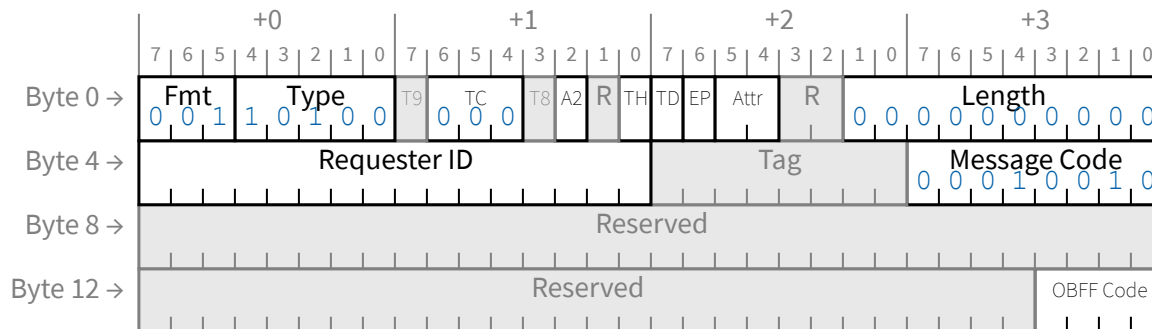
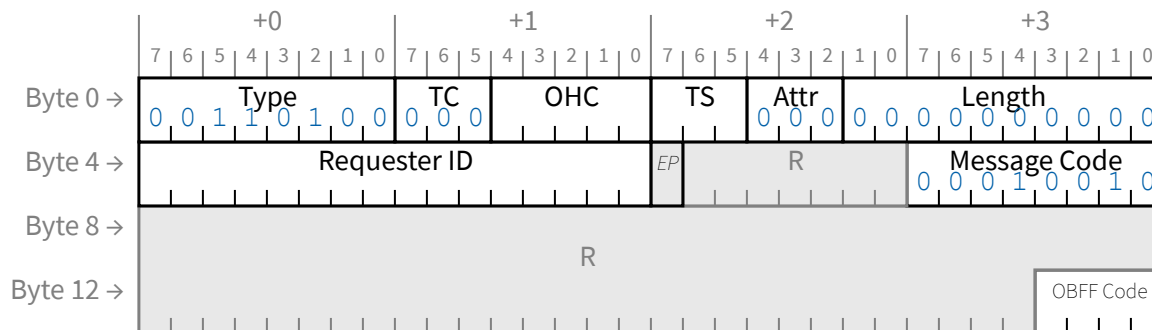


Figure 2-59 OBFF Message - Non-Flit Mode



§ *Figure 2-60 OBFF Message - Flit Mode*

2.2.8.10 Precision Time Measurement (PTM) Messages §

§ Table 2-35 defines the PTM Messages.

- The PTM Request and PTM Response Messages must use a TLP Type of Msg, and must not include a data payload. The Length field is reserved.
 - § Figure 2-61 illustrates the format of the PTM Request and Response Messages.
- The PTM ResponseD Message must use a TLP Type of MsgD, and must include a 64 bit PTM Master Time field in bytes 8 through 15 of the TLP header and a 1 DW data payload containing the 32 bit Propagation Delay field.
 - § Figure 2-62 illustrates the format of the PTM ResponseD Message.
 - Refer to § Section 6.21.3.2 for details regarding how to populate the PTM ResponseD Message.
- The Requester ID must be set to the Transmitting Port's ID.
- A PTM dialog is defined as a matched pair of messages consisting of a PTM Request and the corresponding PTM Response or PTM ResponseD message.
- The PTM Messages must use the default Traffic Class designator (TC0). Receivers implementing PTM must check for violations of this rule. If a Receiver determines that a TLP violates this rule, it must handle the TLP as a Malformed TLP.
 - This is a reported error associated with the Receiving Port (see § Section 6.2).

Table 2-35 Precision Time Measurement Messages

Name	TLP Type	Code[7:0] (b)	Routing r[2:0] (b)	Support				Description/Comments
				RC	EP	Sw	Br	
PTM Request	Msg	0101 0010	100	r	t	tr		Initiates PTM dialog
PTM Response	Msg	0101 0011	100	t	r	tr		Completes current PTM dialog - does not carry timing information
PTM ResponseD	MsgD	0101 0011	100	t	r	tr		Completes current PTM dialog - carries timing information

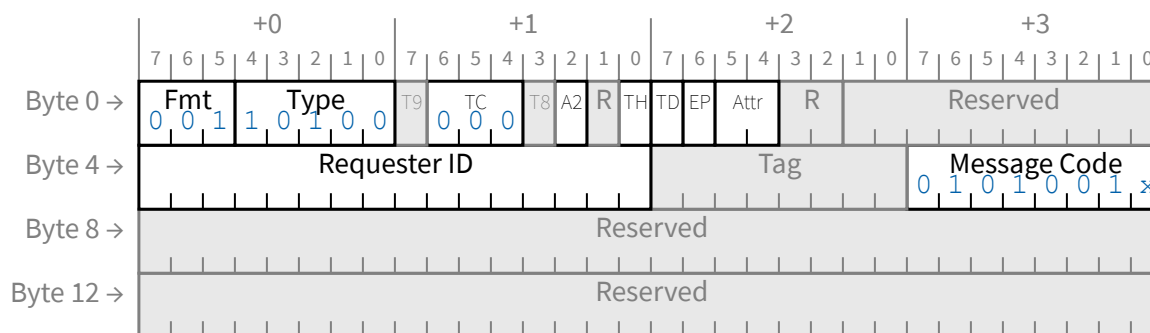


Figure 2-61 PTM Request/Response Message - Non-Flit Mode

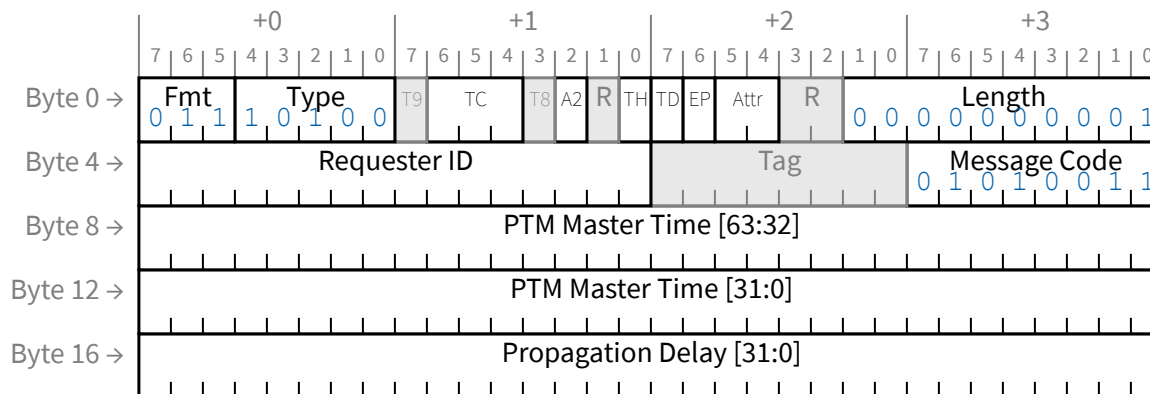


Figure 2-62 PTM ResponseD Message - Non-Flit Mode

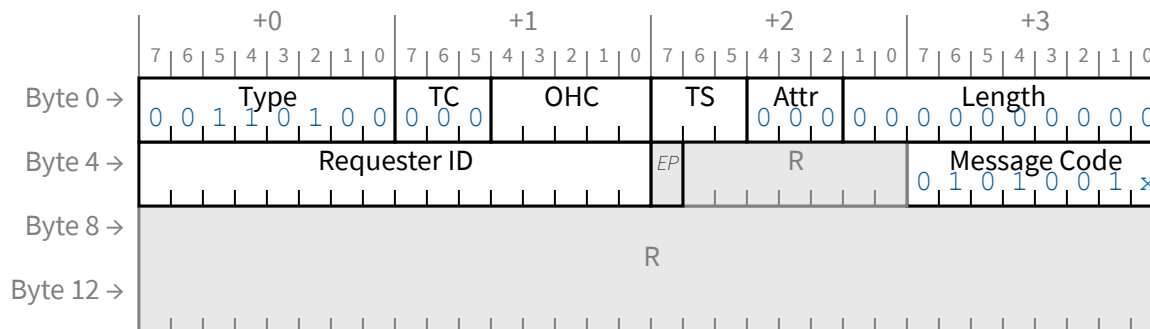


Figure 2-63 PTM Request/Response Message - Flit Mode

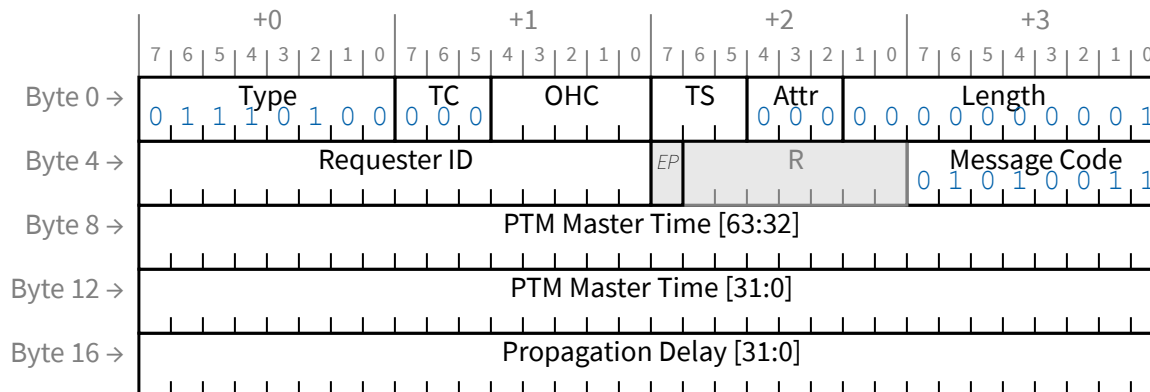


Figure 2-64 PTM ResponseD Message - Flit Mode

IMPLEMENTATION NOTE: PROPAGATION DELAY[31:0] ENDIANNESS §

The bytes within the Propagation Delay[31:0] field (shown in § Figure 2-62) are such that:

- Data Byte 0 contains Propagation Delay [31:24]
- Data Byte 1 contains Propagation Delay [23:16]
- Data Byte 2 contains Propagation Delay [15:8]
- Data Byte 3 contains Propagation Delay [7:0]

Due to ambiguity in previous versions of this document, some implementations made this interpretation:

- Data Byte 0 contains Propagation Delay [7:0]
- Data Byte 1 contains Propagation Delay [15:8]
- Data Byte 2 contains Propagation Delay [23:16]
- Data Byte 3 contains Propagation Delay [31:24]

As a result, it is recommended that implementations provide mechanisms for adapting to either byte interpretation. One such mechanism is the optional PTM Propagation Delay Adaptation Capability.

2.2.8.11 Integrity and Data Encryption (IDE) Messages §

IDE Messages are used with the optional Integrity and Data Encryption (IDE) mechanism (see § Section 6.33). The following rules apply to the formation of IDE Messages:

- § Table 2-36 defines the IDE Messages.
- The IDE Messages do not include a data payload (TLP Type is Msg).
- The Length field is Reserved.
- The Requester ID must be set to the RID of the Function implementing IDE at the Transmitting Port.
- IDE Sync and IDE Fail Messages associated with a Link IDE Stream must use Local routing (100b).
- IDE Sync and IDE Fail Messages associated with a Selective IDE Stream must use Route by ID (010b), and the Destination ID must contain the value in the RID Base field of the Selective IDE RID Association Register Block.
- IDE Sync and IDE Fail Messages must use the same Traffic Class designator as the associated IDE Stream.
- IDE Sync Messages are implicitly associated with the same IDE Stream as indicated in the IDE Prefix applied to the IDE Sync Message.

§ *Table 2-36 IDE Messages*

Name	TLP Type	Code[7:0] (b)	Routing r[2:0] (b)	Support ¹				Description/Comments
				RC	EP	Sw	Br	
IDE Sync	Msg	0101 0100	100 / 100	tr	tr	tr		Synchronization of IDE PR Count for the associated IDE Stream

Name	TLP Type	Code[7:0] (b)	Routing r[2:0] (b)	Support ¹				Description/Comments
				RC	EP	Sw	Br	
IDE Fail	Msg	0101 0101	100 / 100	tr	tr	tr		Notification of IDE failure for a specific IDE Stream from the detecting Port to the IDE Partner Port

Notes:

- Support for these messages is required when the optional IDE mechanism is implemented

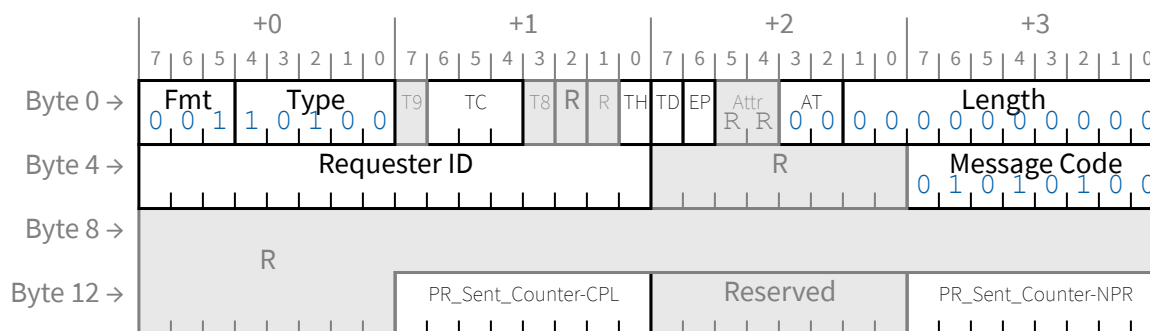


Figure 2-65 IDE Sync Message for Link IDE Stream - Non-Flit Mode

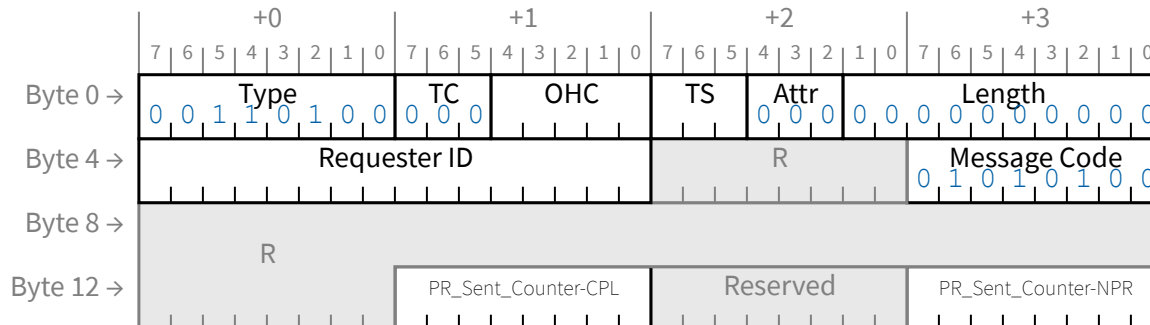


Figure 2-66 IDE Sync Message for Link IDE Stream - Flit Mode

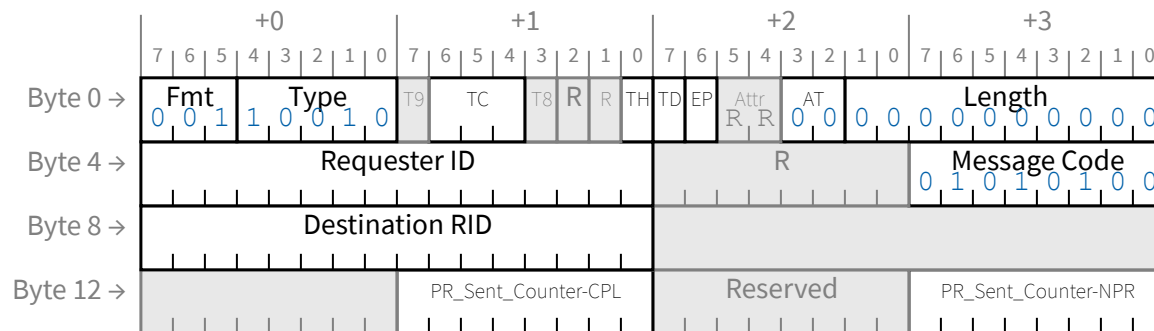


Figure 2-67 IDE Sync Message for Selective IDE Stream - Non-Flit Mode

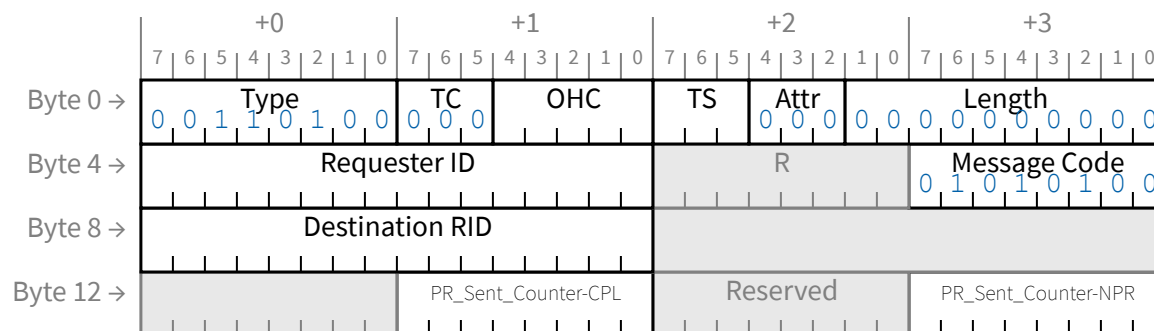


Figure 2-68 IDE Sync Message for Selective IDE Stream - Flit Mode

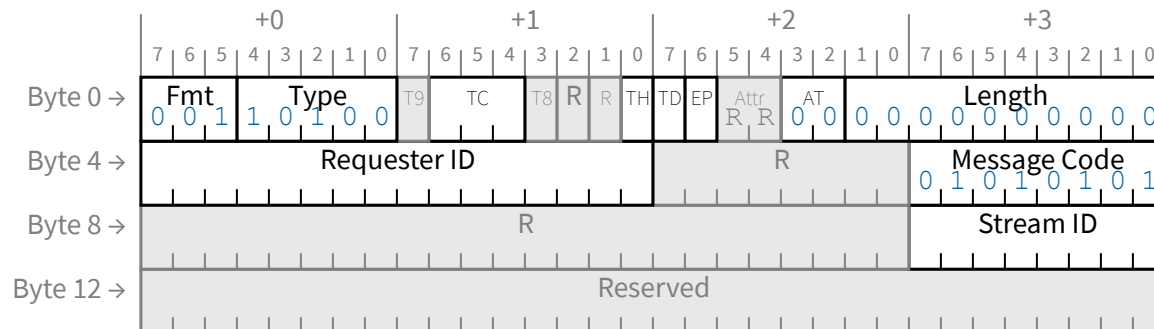


Figure 2-69 IDE Fail Message for Link IDE Stream - Non-Flit Mode

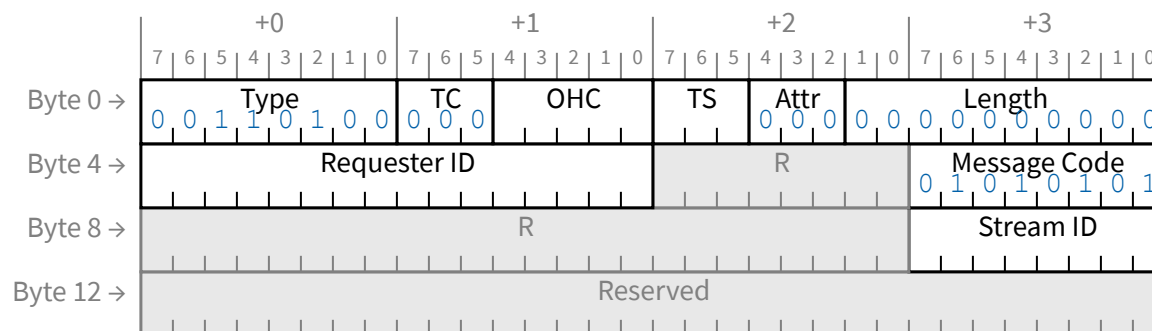


Figure 2-70 IDE Fail Message for Link IDE Stream - Flit Mode

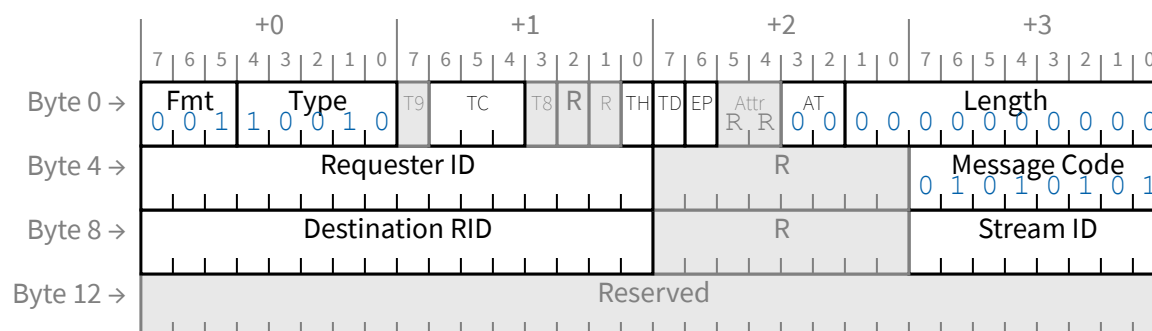


Figure 2-71 IDE Fail Message for Selective IDE Stream - Non-Flit Mode

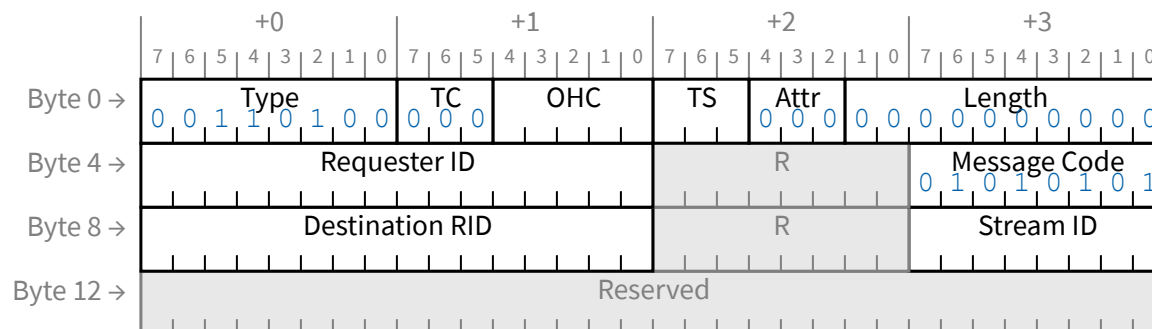


Figure 2-72 IDE Fail Message for Selective IDE Stream - Flit Mode

2.2.9 Completion Rules §

All Read, Non-Posted Write, DMWR, and AtomicOp Requests require Completion. Completions include a Completion header that, for some types of Completions, will be followed by some number of DWs of data. The rules for each of the fields of the Completion header are defined in the following sections.

2.2.9.1 Completion Rules for Non-Flit Mode §

- Completions route by ID, and use a 3 DW header.
 - Note that the routing ID fields correspond directly to the Requester ID supplied with the corresponding Request. Thus, for Completions these fields will be referred to collectively as the Requester ID instead of the distinct fields used generically for ID routing.
- In addition to the header fields included in all TLPs and the ID routing fields, Completions contain the following additional fields (see § Figure 2-73):
 - Completer ID[15:0] - Identifies the Completer - described in detail below
 - Completion Status[2:0] - Indicates the status for a Completion (see § Table 2-37)
 - Rules for determining the value in the Completion Status[2:0] field are in § Section 2.3.1 .
 - BCM - Byte Count Modified - this bit must not be set by PCI Express Completers, and may only be set by PCI-X completers
 - Byte Count[11:0] - The remaining Byte Count for Request
 - The Byte Count value is specified as a binary number, with 0000 0000 0001b indicating 1 byte, 1111 1111 1111b indicating 4095 bytes, and 0000 0000 0000b indicating 4096 bytes.
 - For Memory Read Completions, Byte Count[11:0] is set according to the rules in § Section 2.3.1.1 .
 - For AtomicOp Completions, the Byte Count value must equal the associated AtomicOp operand size in bytes.
 - For all other types of Completions, the Byte Count value must be 4.
 - Tag[9:0] - in combination with the Requester ID field, corresponds to the Transaction ID. In Non-Flit Mode, the Tag field is 10 bits.
 - Lower Address[6:0] - lower byte address for starting byte of Completion
 - For Memory Read Completions, the value in this field is the byte address for the first enabled byte of data returned with the Completion (see the rules in § Section 2.3.1.1).
 - For AtomicOp Completions, the Lower Address field is Reserved.
 - This field is set to all 0's for all remaining types of Completions. Receivers may optionally check for violations of this rule. See § Section 2.3.2 , second bullet, for details.

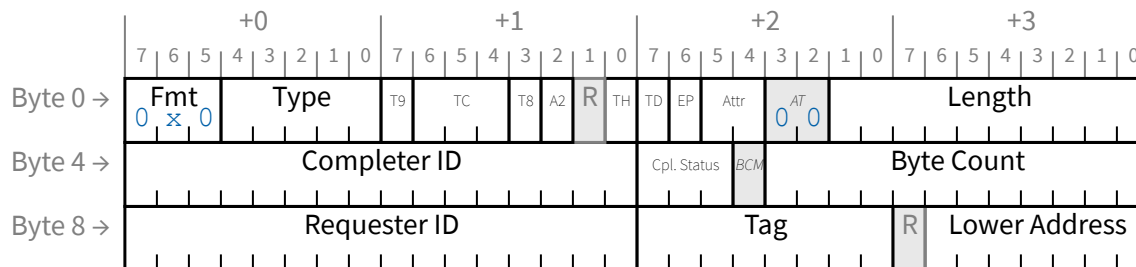
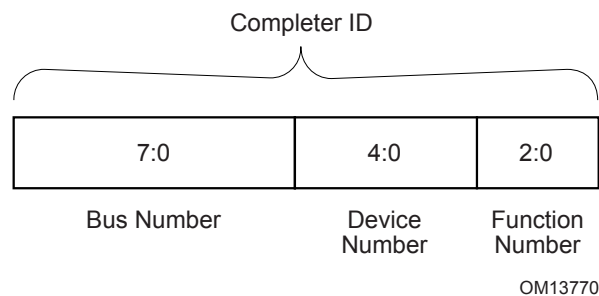
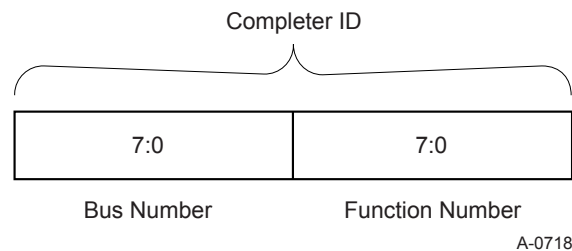


Figure 2-73 Completion Header Format - Non-Flit Mode

Table 2-37 Completion Status Field Values[§]

Cpl. Status[2:0] Field Value (b)	Completion Status
000	Successful Completion (SC)
001	Unsupported Request (UR)
010	Request Retry Status (RRS)
100	Completer Abort (CA)
all others	Reserved

- The Completer ID[15:0] is a 16-bit value that is unique for every PCI Express Function within a Hierarchy (see § Figure 2-74 and § Figure 2-75)

*§ Figure 2-74 (Non-ARI) Completer ID**§ Figure 2-75 ARI Completer ID*

- Functions must capture the Bus and Device Numbers²⁵ supplied with all Type 0 Configuration Write Requests completed by the Function, and supply these numbers in the Bus and Device Number fields of the Completer ID²⁶ for all Completions generated by the Device/Function.
 - If a Function must generate a Completion prior to the initial device Configuration Write Request, 0's must be entered into the Bus Number and Device Number fields
 - Note that Bus Number and Device Number may be changed at run time, and so it is necessary to re-capture this information with each and every Configuration Write Request.

25. With ARI Devices, Functions are only required to capture the Bus Number. ARI Devices are permitted to retain the captured Bus Number on either a per-Device or a per-Function basis. See § Section 2.2.6.2.

26. An ARI Completer ID does not contain a Device Number field. See § Section 2.2.4.2.

- Exception: The assignment of Bus Numbers to the Devices within a Root Complex may be done in an implementation specific way.
- In some cases, a Completion with UR Completion Status may be generated by an MFD without associating the Completion with a specific Function within the device - in this case, the Function Number field ²⁷ is Reserved.
 - Example: An MFD receives a Read Request that does not target any resource associated with any of the Functions of the device - the device generates a Completion with UR status and sets a value of all 0's in the Function Number field of the Completer ID.
- Completion headers must supply the same values for the Requester ID, Tag, and Traffic Class as were supplied in the header of the corresponding Request.
- Completion headers must supply the same values for the Attribute as were supplied in the header of the corresponding Request, except as explicitly allowed:
 - when IDO is used (see § Section 2.2.6.4)
 - when RO is used in a Translation Completion (see § Section 10.2.3)
- The TH bit is reserved for Completions.
- AT[1:0] must be 00b. Receivers are not required or encouraged to check this.
- The Completer ID field is not meaningful prior to the software initialization and configuration of the completing device (using at least one Configuration Write Request), and for this case the Requester must ignore the value returned in the Completer ID field.
- A Completion including a data payload must specify the actual amount of data returned in that Completion, and must include the amount of data specified.
 - It is a TLP formation error to include more or less data than specified in the Length field, and the resulting TLP is a Malformed TLP.

Note: This is simply a specific case of the general rule requiring the TLP data payload length to match the value in the Length field.

2.2.9.2 Completion Rules for Flit Mode §

In Flit Mode, the rules for Completions are the same as in Non-Flit Mode, except as defined in this section. In Flit Mode, Completions must use the Completion Header Base (see § Figure 2-76)

In Flit Mode, the Tag field is 14 bits.

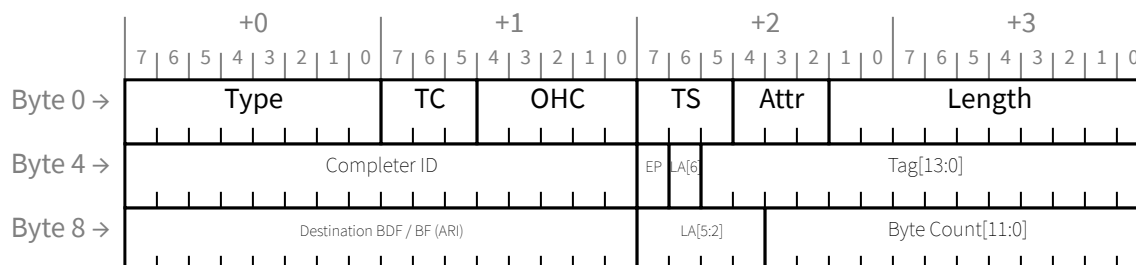


Figure 2-76 Completion Header Base Format - Flit Mode

27. With an ARI Completer ID, the Function Number field is 8 bits.

OHC-A5 (see § Figure 2-11) is required for all:

- Unsuccessful Completions
- Completions with Lower Address[1:0] not equal to 00b
- Completions that require the Destination Segment due to the associated Non-Posted Request containing a Requester Segment that does not match the Completer's captured Segment.

When OHC-A5 is not present it is implied that the Completion Status is Successful, that Lower Address[1:0] = 00b, and that Completer Segment and Destination Segment need not be explicitly indicated (see Segment rules in § Section 2.2.1.2).

When OHC-A5 is present:

- the Completion Status and Lower Address[1:0] fields must contain valid values
- the Completer Segment field must contain the Segment value captured by the Function as described in § Section 2.2.6.2 ; the Completer Segment field must be 00h if Segment Captured is Clear
- if the associated Request did not include a Requester Segment, the Destination Segment field must be 00h and the DSV bit must be clear. If the associated Request included a Requester Segment, the Destination Segment field must reflect the value of the Requester Segment and the DSV bit must be Set.

The BCM field, present in Non-Flit Mode Completions, is not supported in Flit Mode.

2.2.10 TLP Prefix Rules §

2.2.10.1 TLP Prefix General Rules - Non-Flit Mode §

In NFM, the following rules apply to any TLP that contains a TLP Prefix:

- For any TLP, a value of 100b in the Fmt[2:0] field in byte 0 of the TLP indicates the presence of a TLP Prefix and the Type[4] bit indicates the type of TLP Prefix.
 - A value of 0b in the Type[4] bit indicates the presence of a Local TLP Prefix
 - A value of 1b in the Type[4] bit indicates the presence of an End-End TLP Prefix
- The format for bytes 1 through 3 of a TLP Prefix is defined by its TLP Prefix type.
- A TLP that contains a TLP Prefix must have an underlying TLP Header. A received TLP that violates this rule is handled as a Malformed TLP. This is a reported error associated with the Receiving Port (see § Section 6.2).
- It is permitted for a TLP to contain more than one TLP Prefix of any type
 - When a combination of Local and End-End TLP Prefixes are present in TLP, it is required that all the Local TLP Prefixes precede any End-End TLP Prefixes. A received TLP that violates this rule is handled as a Malformed TLP. This is a reported error associated with the Receiving Port (see § Section 6.2).
- The size of each TLP Prefix is 1 DW. A TLP Prefix may be repeated to provide space for additional data.
- If the value in the Fmt and Type field indicates the presence of a Local TLP Prefix, handle according to the Local TLP Prefix handling (see § Section 2.2.10.2).
- If the value in the Fmt and Type field indicates the presence of an End-End TLP Prefix, handle according to the End-End TLP Prefix handling (see § Section 2.2.10.4).

2.2.10.2 Local TLP Prefix Processing §

The following rules apply to Local TLP Prefixes:

- In Flit Mode, TLP Prefix types are determined using the Type[7:0] field (see § Table 2-5)
- In Non-Flit Mode, Local TLP Prefix types are determined using the L[3:0] sub-field of the Type field
 - Type[4] must be 0b
 - Local TLP Prefix L[3:0] values are defined in § Table 2-38

§ Table 2-38 Local TLP Prefix Types

Local TLP Prefix Type	L[3:0] (b)	Description
MR-IOV	0000	MR-IOV TLP Prefix - Refer to [MR-IOV] for details.
FlitModePrefix	1101	Flit Mode Local TLP Prefix - See § Section 2.2.10.3
VendPrefixL0	1110	Vendor Defined Local TLP Prefix - Refer to § Section 2.2.10.2.1 for further details.
VendPrefixL1	1111	Vendor Defined Local TLP Prefix - Refer to § Section 2.2.10.2.1 for further details.
		All other encodings are Reserved.

- The size, routing, and flow control rules are specific to each Local TLP Prefix type.
- It is an error to receive a TLP with a Local TLP Prefix type not supported by the Receiver. If the Extended Fmt Field Supported bit is Set, TLPs in violation of this rule are handled as a Malformed TLP unless explicitly stated differently in another specification. This is a reported error associated with the Receiving Port (see § Section 6.2). If the Extended Fmt Field Supported bit is Clear, behavior is device specific.
- No Local TLP Prefixes are protected by ECRC even if the underlying TLP is protected by ECRC.

2.2.10.2.1 Vendor Defined Local TLP Prefix §

As described in § Table 2-38, Types VendPrefixL0 and VendPrefixL1 are defined for use as Vendor Defined Local TLP Prefixes. To maximize interoperability and flexibility the following rules are applied to such prefixes:

- Components must not send TLPs containing Vendor Defined Local TLP Prefixes unless this has been explicitly enabled (using vendor-specific mechanisms).
- Components that support any usage of Vendor Defined Local TLP Prefixes must support the 3-bit definition of the Fmt field and have the Extended Fmt Field Supported bit Set (see § Section 7.5.3.15).
- It is recommended that components be configurable (using vendor-specific mechanisms) so that all vendor defined prefixes can be sent using either of the two Vendor Defined Local TLP Prefix encodings. Such configuration need not be symmetric (for example each end of a Link could transmit the same Prefix using a different encoding).

2.2.10.3 Flit Mode Local TLP Prefix §

This prefix (see § Figure 2-77) is only permitted when operating in Flit Mode.